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United States Patent	12415120
Kind Code	B2
Date of Patent	September 16, 2025
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Golf club having an adjustable weight assembly

Abstract

An iron-type golf club head includes a body, a recessed channel, and a weight assembly. The body includes a striking face, a sole, and a back portion positioned between the sole and the striking face. The recessed channel is formed in the back portion. The weight assembly includes a weight at least partially disposed within the recessed channel and configured to move therein, a cover, and a fastener coupling the cover to the body and configured to retain the weight in the recessed channel indirectly by the cover.

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Family ID: 1000008821042

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Appl. No.: 17/556154

Filed: December 20, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220111270 A1	Apr. 14, 2022

Related U.S. Application Data

continuation-in-part parent-doc US 17362488 20210629 US 11497974 child-doc US 17556154
continuation-in-part parent-doc US 17222774 20210405 US 11439879 child-doc US 17362488
continuation-in-part parent-doc US 17122887 20201215 US 11229827 child-doc US 17222774
continuation-in-part parent-doc US 16843640 20200408 US 10918917 20210216 child-doc US 17122887
continuation-in-part parent-doc US 16708255 20191209 US 11090536 20210817 child-doc US

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continuation-in-part parent-doc US 16535844 20190808 US 10926143 20210223 child-doc US
16708255
continuation-in-part parent-doc US 16387859 20190418 US 10695628 20200630 child-doc US
16535844

Publication Classification

Int. Cl.: **A63B53/06** (20150101); **A63B53/04** (20150101); **A63B53/08** (20150101)

U.S. Cl.:

CPC **A63B53/08** (20130101); **A63B53/047** (20130101); **A63B53/06** (20130101);
A63B2053/0479 (20130101); A63B2053/0491 (20130101)

Field of Classification Search

CPC: A63B (53/047); A63B (53/0475); A63B (2053/0479); A63B (2053/0483); A63B
(2053/0491)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/362,488, filed Jun. 29, 2021, which is a continuation-in-part of U.S. patent application Ser. No. 17/222,774, filed Apr. 5, 2021, which is a continuation-in-part of U.S. patent application Ser. No. 17/122,887, filed Dec. 15, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/843,640, filed Apr. 8, 2020, now U.S. Pat. No. 10,918,917, which is a continuation-in-part of U.S. patent application Ser. No. 16/708,255, filed Dec. 9, 2019, now U.S. Pat. No. 11,090,536, which is a continuation-in-part of U.S. patent application Ser. No. 16/535,844, filed Aug. 8, 2019, now U.S. Pat. No. 10,926,143, which is a continuation-in-part of U.S. patent application Ser. No. 16/387,859, filed Apr. 18, 2019, now U.S. Pat. No. 10,695,628, and which are hereby incorporated by reference in their entireties. To the extent appropriate, the present application claims priority to the above-referenced applications.

BACKGROUND

(1) The flight characteristics of a golf ball after being struck by a golf club are dependent not only on the swing of the golf club but also on the golf club itself. For example, flight characteristics of the golf ball, such as fades, draws, launch angles, ball spin, and speed are impacted by the design of the golf club. By adjusting one or more design properties of the golf club, the flight characteristics of the golf ball can be improved, thereby increasing golf club performance. In some examples, adjusting a center of gravity (CG) and/or a moment of inertia (MOI) of a head of the golf club through selective weight placement impacts the flight characteristics of the golf ball. However, these adjustable weights need to be both securely attached to the golf club head and selectively moveable. As such, improvements to adjustable weight assemblies for golf club heads are desired.

SUMMARY

(2) In an aspect, the technology relates to an iron-type golf club head including: a body including: a striking face having a lower leading edge, a sole extending from the lower leading edge, and a back portion positioned rearward of the striking face and coupled to a rearward portion of the sole; a recessed channel formed in the back portion; and a weight assembly including: a weight at least partially disposed within the recessed channel and configured to move therein, a cover adapted to releasably secure the weight within the recessed channel, and a fastener coupling the cover to the body and selectively movable along a fastener axis forming a fastener angle of about 0 degrees to

about 90 degrees to a striking face plane, wherein the striking face plane is tangential to an outermost surface of the striking face, the fastener angle is formed in a cross-sectional plane, and the fastener angle is measured from the striking face plane to the fastener axis along a counter-clockwise direction when the cross-sectional plane is viewed in a toe-to-heel direction.

(3) In an example, the cover is positionable in at least an unlocked configuration whereby the cover is raised at least partially out of the recessed channel and the weight is selectively movable within the recessed channel, and a locked configuration whereby the cover is at least partially disposed within the recessed channel and the weight is secured within the recessed channel. In another example, the weight is engaged with the cover so as to move with the cover between the locked configuration and the unlocked configuration. In another example, the cover includes a flange, and the weight has a groove shaped and sized to receive at least a portion of the flange to slidably engage the weight with the cover. In another example, the weight remains in the recessed channel when the cover moves between the locked configuration and the unlocked configuration. In another example, between about 0% and about 30% of an outer surface of the weight is visible in the locked configuration. In another example, the recessed channel has a bottom track offset from the outer surface of the body, and the cover extends at least partially over the recessed channel to cover the entire bottom track. In another example, the recessed channel has a bottom track offset from the outer surface of the body, and the cover extends partially over the recessed channel and exposes at least part of the bottom track. In another example, the iron-type golf club head is a wedge-type golf club head.

(4) In another aspect, the technology relates to an iron-type golf club head including: a body including: a striking face having an upper topline edge and lower leading edge, a sole extending from the lower leading edge and having a rearward portion distal to the lower leading edge, and a first back surface coupled to the rearward portion and extending towards the upper topline edge, wherein one or more of the striking face, the sole, and the first back surface define at least part of an outer surface of the body; a recessed channel defined in the first back surface; and a weight assembly including: a weight at least partially disposed within the recessed channel, a cover extending at least partially over the recessed channel, and a fastener coupling the cover to the body, the fastener being adapted to retain the weight in the recessed channel indirectly by the cover, and wherein the cover is positionable in at least an unlocked configuration, whereby the cover is raised at least partially out of the recessed channel and the weight is selectively moveable within the recessed channel, and a locked configuration whereby the cover is at least partially disposed within the recessed channel and the weight is secured within the recessed channel.

(5) In an example, a fastener axis defined by the fastener intersects a striking face plane tangential to an outermost surface of the striking face at a point closer to the lower leading edge than to the upper topline edge. In another example, the fastener has a height defined along a direction from the lower leading edge to the upper topline edge, and the height of the fastener increases as the position of the cover is changed from the locked configuration to the unlocked configuration. In another example, the fastener defines a fastener axis forming a fastener angle of about 0 degrees to about 90 degrees to a striking face plane, wherein the striking face plane is tangential to an outermost surface of the striking face, the fastener angle is formed in a cross-sectional plane, and the fastener angle is measured from the striking face plane to the fastener axis along a counter-clockwise direction when the cross-sectional plane is viewed in a toe-to-heel direction. In another example, a fastener axis defined by the fastener intersects the striking face. In another example, the body further includes a second back surface coupled between the first back surface and the upper topline edge of the striking face, and the second back surface forms an obtuse angle with the first back surface at a junction between the first back surface and the second back surface. In another example, the body has a fastener hollow at a position between the first and second back surfaces and adapted to receive at least part of the fastener, and the recessed channel is spaced apart from the fastener hollow. In another example, the body includes at least one locating lug on a bottom

track of the recessed channel offset from the outer surface of the body, and the weight includes an indent shaped and sized to receive at least a portion of the at least one locating lug to at least partially define the position of the weight within the recessed channel. In another example, the recessed channel has a first sidewall along a toe-heel direction and adjacent to the rearward portion of the sole, the first sidewall including at least one dimple, and the weight includes a protruding position indicator configured to selectively engage with the at least one dimple. In another example, the cover includes at least one locating groove, and wherein the weight includes a protruding position indicator configured to selectively engaged with the at least one locating groove.

(6) In another aspect, the technology relates to an iron-type golf club head including: a body having an outer surface; a recessed channel formed in the outer surface; and a weight assembly including: a weight at least partially disposed within the recessed channel and configured to move therein, a cover adapted to releasably secure the weight within the recessed channel, and a fastener coupling the cover to the body and selectively movable along a fastener axis that intersects a striking face of the body.

(7) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Non-limiting and non-exhaustive examples are described with reference to the following Figures.

(2) FIG. 1 is a perspective view of a sole of a golf club head with an exemplary weight assembly.

(3) FIG. 2 is a cross-sectional view of the golf club head taken along line 2-2 in FIG. 1 where the weight assembly is in a locked configuration.

(4) FIG. 3 is a cross-sectional view of the weight assembly taken along line 3-3 in FIG. 2.

(5) FIG. 4 is a cross-sectional view of the golf club head taken along line 2-2 in FIG. 1 where the weight assembly is in an unlocked configuration.

(6) FIG. 5 is a cross-sectional view of the weight assembly taken along line 5-5 in FIG. 4.

(7) FIG. 6 is a perspective view of the sole of the golf club head with another weight assembly.

(8) FIG. 7 is a cross-sectional view of the weight assembly taken along line 7-7 in FIG. 6.

(9) FIG. 8 is a perspective view of the golf club head with another weight assembly.

(10) FIG. 9 is a perspective view of the sole of the golf club head with another weight assembly.

(11) FIG. 10 is a top view of the golf club head shown in FIG. 9 with a portion of a crown removed.

(12) FIG. 11 is a cross-sectional view of the weight assembly taken along line 11-11 in FIG. 9.

(13) FIG. 12 is a cross-sectional view of the weight assembly taken along line 12-12 in FIG. 9.

(14) FIG. 13 is a cross-sectional view of another weight assembly.

(15) FIG. 14 is a perspective view of the sole of the golf club head with another weight assembly.

(16) FIG. 15 is a cross-sectional view of the golf club head taken along line 15-15 in FIG. 14 and showing the weight assembly.

(17) FIG. 16 is a cross-sectional view of the weight assembly taken along line 16-16 in FIG. 14.

(18) FIG. 17 is a cross-sectional view of the weight assembly taken along line 17-17 in FIG. 14.

(19) FIG. 18 is an exploded perspective view the golf club head with another weight assembly.

(20) FIG. 19 is a cross-sectional view of the weight assembly taken along line 19-19 in FIG. 18.

(21) FIG. 20 is a partial cross-sectional perspective view of another weight assembly.

(22) FIG. 21 is another cross-sectional view of the weight assembly shown in FIG. 20.

(23) FIG. 22 is a perspective view of the sole of the golf club head with another weight assembly in a locked configuration.

(24) FIG. 23 is a cross-sectional view of the weight assembly taken along line 23-23 in FIG. 22.

(25) FIG. 24 is a perspective view of the sole of the golf club head with the weight assembly shown in FIG. 22 in an unlocked configuration.

(26) FIG. 25 is a cross-sectional view of the weight assembly taken along line 25-25 in FIG. 24.

(27) FIG. 26 is a perspective view of the sole of the golf club head with another weight assembly.

(28) FIG. 27 is a cross-sectional view of the weight assembly taken along line 27-27 in FIG. 26.

(29) FIG. 28 is an exploded perspective view of the sole of the golf club head with another weight assembly.

(30) FIG. 29 is a cross-sectional view of the weight assembly shown in FIG. 28.

(31) FIG. 30 is a perspective view of the sole of the golf club head with another weight assembly.

(32) FIG. 31 is a cross-sectional view of the weight assembly taken along line 31-31 in FIG. 30.

(33) FIG. 32 is a perspective view of the sole of the golf club head with another weight assembly.

(34) FIG. 33 is a perspective view of the sole of the golf club head with another weight assembly.

(35) FIG. 34 is a perspective view of the sole of the golf club head with another weight assembly.

(36) FIG. 35 is a perspective view of the sole of the golf club head with another weight assembly.

(37) FIG. 36 is a perspective view of the sole of the golf club head with another weight assembly.

(38) FIG. 37 is an exploded perspective view of the weight assembly shown in FIG. 36.

(39) FIG. 38 is a cross-sectional view of the weight assembly taken along line 38-38 in FIG. 36.

(40) FIG. 39 is an inside surface view of a cover of the weight assembly shown in FIG. 36.

(41) FIG. 40 is a cross-sectional view of the weight assembly taken along line 40-40 in FIG. 36 and in a weight sliding configuration.

(42) FIG. 41 is a cross-sectional view of the weight assembly taken along line 40-40 in FIG. 36 and in a weight removal configuration.

(43) FIG. 42 is a perspective view of a sole of another golf club head with another weight assembly in a locked configuration.

(44) FIG. 43 is a perspective view of the sole of the golf club head with the weight assembly shown in FIG. 42 in an unlocked configuration.

(45) FIG. 44 is a cross-sectional view of the golf club head with the weight assembly taken along line 44-44 in FIG. 42.

(46) FIG. 45 is a partial perspective cross-sectional view of the weight assembly taken along line 44-44 in FIG. 42.

(47) FIG. 46 is a bottom view of the golf club head with another weight assembly.

(48) FIG. 47 is a perspective cross-section view of the golf club head with weight assembly taken along line 47-47 in FIG. 46.

(49) FIG. 48 is a perspective view of another golf club head.

(50) FIG. 49 is a bottom view of the club head shown in FIG. 48 with another weight assembly.

(51) FIG. 50 is a cross-section view of another weight assembly.

(52) FIG. 51 is a schematic view of the weight assembly shown in FIG. 50.

(53) FIG. 52 is a top view of a cover of the weight assembly shown in FIG. 50.

(54) FIG. 53 is a side view of the cover of the weight assembly shown in FIG. 50.

(55) FIG. 54 is a bottom view of another golf club head with another weight assembly.

(56) FIG. 55 is a perspective, cross-sectional, view of the weight assembly taken along line 54-54 in FIG. 54.

(57) FIG. 56 is a cross-sectional view of a cover taken along line 54-54 in FIG. 54.

(58) FIG. 57 is a perspective view of a weight of the weight assembly shown in FIGS. 55 and 56.

(59) FIG. 58 is a schematic top plan view of the weight shown in FIG. 57.

(60) FIG. 59 is a schematic perspective view of an exemplary test mule with another weight

assembly.

(61) FIG. **60** is a cross-sectional view of the weight assembly taken along line **60-60** in FIG. **59**.

(62) FIG. **61** is another cross-sectional view of the weight assembly taken along line **61-61** in FIG. **59**.

(63) FIG. **62** is a schematic perspective view of another test mule with another weight assembly.

(64) FIGS. **63A-E** are cross-sectional views of the weight assembly taken along line **63-63** in FIG. **62** and with a weight in a variety of different positions.

(65) FIG. **64** is another cross-sectional view of the weight assembly taken along line **64-64** in FIG. **62**.

(66) FIG. **65** is a partial perspective view of an exemplary recessed channel within a body of a test mule.

(67) FIG. **66** is another partial perspective view of the recessed channel shown in FIG. **65**.

(68) FIG. **67** is a schematic perspective view of another test mule with another weight assembly.

(69) FIG. **68** is a cross-sectional view of the weight assembly in a first configuration taken along line **67-67** in FIG. **67**.

(70) FIG. **69** is a cross-sectional view of the weight assembly in a second configuration taken along line **67-67** in FIG. **67**.

(71) FIG. **70** is a schematic perspective view of another test mule with another weight assembly.

(72) FIG. **71** is a partial cross-sectional view of the weight assembly shown in FIG. **70** in an unlocked configuration.

(73) FIG. **72** is a partial cross-sectional view of the weight assembly shown in FIG. **70** in a locked configuration.

(74) FIG. **73** is a cross-sectional view of another weight assembly that can be used with the test mule shown in FIG. **70**.

(75) FIG. **74** is an exploded perspective view of another test mule with another weight assembly.

(76) FIG. **75** is a perspective view of a sole of another golf club head with another weight assembly.

(77) FIG. **76** is a cross-sectional view of the weight assembly taken along line **76-76** in FIG. **75**.

(78) FIG. **77** is a cross-sectional view of the weight assembly taken along line **77-77** in FIG. **75**.

(79) FIG. **78** is an exploded view of a cover of the weight assembly shown in FIG. **75**.

(80) FIG. **79** is a perspective view of the weight assembly shown in FIG. **75** in a locked configuration.

(81) FIG. **80** is a perspective view of the weight assembly shown in FIG. **75** in an unlocked configuration.

(82) FIG. **81** is a perspective view of the weight assembly shown in FIG. **75** in a weight removal configuration.

(83) FIG. **82** is a perspective view of a sole of another golf club head with another weight assembly.

(84) FIG. **83** is a cross-sectional view of the weight assembly taken along line **83-83** in FIG. **82**.

(85) FIG. **84** is a perspective view of a cover of the weight assembly shown in FIG. **82**.

(86) FIG. **85** is a perspective view of the weight assembly shown in FIG. **82** in a locked configuration.

(87) FIG. **86** is a perspective view of the weight assembly shown in FIG. **82** in an unlocked configuration.

(88) FIG. **87** is a perspective view of the weight assembly shown in FIG. **82** in a weight removal configuration.

(89) FIG. **88** is a perspective view of a sole of another golf club head with another weight assembly.

(90) FIG. **89** is a perspective view of a sole of another golf club head with another weight assembly in a locked configuration.

(91) FIG. **90** is a perspective view of the weight assembly shown in FIG. **89** in an unlocked configuration.

(92) FIG. **91** is a perspective view of the weight assembly shown in FIG. **89** in a weight adjustment configuration.

(93) FIG. **92** is a cross-sectional view of the weight assembly taken along line **92-92** in FIG. **91**.

(94) FIG. **93** is a perspective view of a sole of another golf club head with another weight assembly.

(95) FIG. **94** is a perspective view of the weight assembly shown in FIG. **93**.

(96) FIG. **95** is a cross-sectional view of the weight assembly taken along line **93-93** in FIG. **93** in a locked configuration.

(97) FIG. **96** is a cross-sectional view of the weight assembly taken along line **93-93** in FIG. **93** in an unlocked configuration.

(98) FIG. **97** is a bottom view of a sole of a golf club head with another weight assembly.

(99) FIG. **98** is a perspective cross-sectional view of the golf club head taken along line **97-97** in FIG. **97** and in an locked configuration.

(100) FIG. **99** is another perspective cross-sectional view of the golf club head taken along line **97-97** in FIG. **97** and in an unlocked configuration.

(101) FIG. **100** is another bottom view of the sole of the golf club head shown in FIG. **97**.

(102) FIG. **101** is an inside surface view of a cover and a weight of the weight assembly shown in FIGS. **97-99**.

(103) FIG. **102** is a side view of the cover and the weight shown in FIG. **101**.

(104) FIG. **103** is another inside surface view of the cover shown in FIG. **101**.

(105) FIG. **104** is a cross-sectional view of the cover taken along line **104-104** in FIG. **103**.

(106) FIG. **105** is a partially exploded, perspective view of an iron-type golf club head with another weight assembly.

(107) FIG. **106** is another partially exploded, perspective view of the golf club head of FIG. **105**.

(108) FIG. **107** is a back view of the golf club head of FIG. **105** with the weight assembly in a locked configuration.

(109) FIG. **108** is another back view of the golf club head of FIG. **105** with the weight assembly in an unlocked configuration.

(110) FIG. **109** is another partially exploded, perspective view of the golf club head of FIG. **105**.

(111) FIG. **110** is a top view of the golf club head of FIG. **105** without the weight assembly.

(112) FIG. **111** is a cross-sectional view in the toe-to-heel direction of the golf club head of FIG. **105** taken along line **110a-110a** in FIG. **110** and with the weight assembly in the locked configuration.

(113) FIG. **112** is a cross-sectional view in the toe-to-heel direction of the golf club head of FIG. **105** taken along line **110a-110a** in FIG. **110** and with the weight assembly in the unlocked configuration.

(114) FIG. **113** is a back view of a golf club head with another weight assembly in the locked configuration.

(115) FIG. **114** is a back view of a golf club head with another weight assembly.

(116) FIG. **115** is a top view of the golf club head of FIG. **114**.

(117) FIG. **116** is a back view of a golf club head with another weight assembly in the locked configuration.

(118) FIG. **117** is a back view of the golf club head of FIG. **116** when the weight assembly is in the unlocked configuration.

(119) FIG. **118** is a partially exploded, back view of the golf club head of FIG. **116**.

(120) FIG. **119** is cross-sectional view in the toe-to-heel direction of the golf club head of FIG. **116** taken along line **116a-116a** and when the weight assembly is in the locked configuration.

(121) FIG. **120** is cross-sectional view in the toe-to-heel direction of the golf club head of FIG. **116**

taken along line **116a-116a** and when the weight assembly is in the unlocked configuration.

(122) FIG. **121** is a back view of a golf club head with another weight assembly in the locked configuration.

(123) FIG. **122** is another back view of the golf club head of FIG. **121** when the weight assembly is in the unlocked configuration.

(124) FIG. **123** is a perspective view of the golf club head of FIG. **121** when the weight assembly is in the unlocked configuration.

(125) FIG. **124** is a partially exploded, perspective view of the golf club head of FIG. **121**.

(126) FIG. **125** is a cross-sectional view in the toe-to-heel direction of the golf club head of FIG. **121** along line **121a-121a** and when the weight configuration is in the locked configuration.

(127) FIG. **126** is a cross-sectional view of the golf club head of FIG. **121** along line **121a-121a** and when the weight configuration is in the unlocked configuration.

DETAILED DESCRIPTION

(128) The technologies described herein contemplate a golf club head, such as a fairway metal, driver, or other golf club head, that includes an adjustable weight assembly. Through the weight balance of the golf club head, the flight characteristics of the golf ball can be improved, thereby increasing golf club performance. In the examples described herein, the weight assembly enables for the CG and/or MOI of a head of the golf club to be adjusted through selective weight placement to impact the flight characteristics of the golf ball, such as fades, draws, launch angles, ball spin, and speed. Additionally or alternatively, the weight assembly enables for the swing weight of the golf club head to be adjustable (e.g., increasing or decreasing the weight of the club head).

(129) In examples, the present technologies provide a golf club head with a recessed channel defined therein. A slidable weight is disposed at least partially within the channel and secured therein by a cover and a fastener. The cover is configured to retain the weight within the channel indirectly so that the fastener never engages with the weight. This configuration enables for the size, shape, and/or density of the weight to be defined so that the CG and MOI of the golf club head can be finely tuned. Additionally, the cover includes additional features that increase securement of the weight within the channel and reduce undesirable rattling or movement during the golf club swing. Furthermore, the weight assemblies described herein allow for the weight to be adjusted quickly and easily without requiring any component to be fully detached from the club head. Thereby reducing lost or misplaced components during club head adjustment. In an aspect, the weight is engaged with the cover so that the two components can move together with respect to the golf club head. Additionally, the weight is restricted from tilting relative to the cover so as to reduce or prevent binding of the weight within the channel.

(130) FIG. **1** is a perspective view of a sole **102** of a golf club head **100** with an exemplary weight assembly **104**. The golf club head **100** is a metalwood-type golf club head having a body **106** that includes a striking face **108** positioned towards the front of the club head **100** and having a lower edge **110** and an upper edge **112** (e.g., shown in FIG. **8**) each extending between a toe **114** and heel **116** of the club head **100**. The sole **102** extends from the lower edge **110** on the bottom side of the club head **100** and a crown **118** extends from the upper edge **112** on the top of the club head **100**. The sole **102**, the striking face **108**, and the crown **118** are coupled together so as to define an outer surface **120** of the body **106** with an interior cavity **122** (shown in FIG. **2**) formed within. A hosel **124** is disposed at the heel **116** and is configured to couple to a shaft (not shown). In some examples, a skirt **126** (shown in FIG. **8**) may also form a portion of the club head **100** and is positioned between the crown **118** and the sole **102**. In such examples and for purposes of this application, the crown **118** may still be considered to be attached or coupled to the sole **102**, via the skirt **126**. Furthermore, the body **106** may form any type club head, such as an iron-type club head or hybrid-type club head, as required or desired.

(131) In operation, the sole **102** generally provides the lower surface of the club head **100** when the club head **100** is placed in an address position. The club head **100** defines a center of gravity (CG)

and a moment of inertia (MOI) that impact flight characteristics of a golf ball (not shown) when hit with the striking face **108**. The weight assembly **104** is coupled to the club head **100** such that the CG and/or the MOI of the club head **100** can be selectively adjusted as required or desired. In the example, the weight assembly **104** includes a movable weight **128**, a cover **130** configured to secure the weight **128** in place, and a fastener **132** for coupling the weight assembly **104** to one or more other portions of the club head **100**. In some examples, the weight **128** may be formed from tungsten. In examples, the weight **128** may be between about 2 grams to 15 grams. In some specific examples, the weight **128** may be about 9 grams.

(132) A recessed elongated channel **134** is formed in the outer surface **120** of the club head **100**. More specifically, the channel **134** is substantially linear and defined in the sole **102** of the club head **100**. In other examples, the channel **134** may be defined at any other location of the body **106** (e.g., the crown **118** or the skirt **126**) as required or desired. The channel **134** is sized and shaped to receive at least a portion of the weight **128** so that the weight **128** can be slidable therein. In the example, the channel **134** extends substantially linearly in a toe **114**-heel **116** direction so that the CG and the MOI of the club head **100** can be adjusted (by selectively moving the weight **128**) for fade or draw bias. The channel **134** can be angularly offset from the plane of the striking face **108** as illustrated in FIG. **1**. In other examples, the channel **134** may extend substantially parallel to the striking face **108**. In the example, the fastener **132** is positioned proximate to the heel side of the channel **134**. In other examples, the fastener **132** may be positioned at any other location relative to the channel **134** to enable the weight assembly **104** to function as described herein. For example, at approximately a midpoint of the channel **134** as described in reference to FIG. **26** or proximate the toe side of the channel **134**.

(133) In operation and through use of the fastener **132**, the cover **130** is coupled to the body **106** and extends at least partially over the channel **134** so as to selectively secure the weight **128** to the club head **100**. Additionally, the cover **130** covers at least a portion of the channel **134** so as to reduce dust and dirt from accumulating therein. However, the fastener **132** is separate from the weight **128** and only indirectly (e.g., via the cover **130**) secures the weight **128** to the club head **100**. In examples, the fastener **132** and the cover **130** are adapted to retain the weight **128** in the channel **134** only by contact with the cover **130** such that the fastener **132** never engages the weight **128**. As described herein, when the fastener **132** indirectly retains the weight **128**, the fastener **132** never engages the weight **128** directly and it is a separate component (e.g., the cover **130**) that directly engages the weight **128** for securement to the club head **100**.

(134) The cover **130** may be loosened or completely removed, via the fastener **132**, from the club head **100** to enable the weight **128** to slide within the channel **134** and selectively adjust the CG and the MOI as required or desired. Because the weight **128** is selectively moveable, the weight assembly **104** (e.g., the fastener **132**, the weight **128**, and the cover **130**) enables the movement of the weight **128**, while also securing the weight **128** to one or more portions of the club head **100** so that undesirable movement (e.g., during a club swing) is reduced or prevented. By separating the fastener **132** from the weight **128**, the size, shape, and/or density of the weight **128** may be configured so that the CG and the MOI of the club head **100** may be more finely tuned, thereby increases the performance of the golf club head **100**. The weight assembly **104** is described further below.

(135) FIG. **2** is a cross-sectional view of the golf club head **100** taken along line 2-2 in FIG. **1** and showing the weight assembly **104** in a locked configuration **136**. FIG. **3** is a cross-sectional view of the weight assembly **104** taken along line 3-3 in FIG. **2**. Referring concurrently to FIGS. **2** and **3**, when the weight assembly **104** is in the locked configuration **136**, the cover **130** is disposed within the channel **134** and the weight **128** is secured within the channel **134** such that movement is restricted. In the example, to lock the cover **130** to the body **106**, the fastener **132** may be a threaded bolt that threadingly engages with a nut **138** positioned within the heel end of the channel **134**. In some examples, the nut **138** may be integrally formed within the body **106**.

(136) When the cover **130** is in the locked configuration **136**, an exterior surface **140** of the cover **130** is substantially aligned (e.g., flush) with the outer surface **120** of the body **106**. Additionally, the fastener **132** defines a fastener axis **142**. In the example, the fastener axis **142** is disposed at an angle **144** relative to a plane **146** that is normal to the exterior surface **140** of the cover **130** proximate the fastener **132**. The angle **144** defines the orientation that the cover **130** may move relative to the body **106**. The angle **144** may be between about 0° (e.g., aligned with the plane **146**) and about 88°. In examples, the angle **144** may be between about 20° and 50°. In one example, the angle **144** may be about 45°.

(137) In the example, only a single fastener **132** is used to couple the cover **130** to the body **106** and the fastener **132** is positioned at the heel end of the weight assembly **104**. As such, to connect the toe end of the cover **130** to the body **106**, the cover **130** may include one or more projections **148** that extend from the toe end. The projection **148** is sized and shaped to be received within one or more corresponding chambers **150** defined at the toe end of the channel **134**. When the weight assembly **104** is in the locked configuration **136**, the projection **148** is received at least partially within the chamber **150** and engaged therewith. By engaging the cover **130** to the body **106** at a position opposite from the fastener **132**, when the weight **128** is positioned away from the fastener **132**, the cover **130** still enables securement of the weight **128** within the channel **134** and reduces or prevents movement of the weight **128** in the locked configuration **136**. In the example, the projection **148** extends in the toe-heel direction of the cover **130** and includes at least one oblique surface **152** that frictionally engages with a corresponding at least one oblique surface **154** of the chamber **150**. In some examples, the oblique surfaces **152**, **154** may be substantially parallel to the fastener axis **142**. In other examples, the oblique surfaces **152**, **154** may be oriented at a different angle than the fastener axis **142** (e.g., steeper or shallower angles). Additionally or alternatively, the projection **148** and chamber **150** may extend substantially orthogonal to the toe-heel direction (e.g., in and out of the page of FIG. 2).

(138) The cover **130** may also be engaged with the body **106** at one or more intermediate positions between the fastener **132** and the opposite end. A seat **156** may protrude into the channel **134** at a location between the toe end and the heel end, for example, proximate a midpoint location of the channel **134**. The seat **156** is sized and shaped to be received within a corresponding notch **158** defined in the cover **130**. When the weight assembly **104** is in the locked configuration **136**, the seat **156** is received at least partially within the notch **158** and engaged therewith. This engagement of the cover **130** to the body **106** at a position away from the fastener **132**, also secures the weight **128** within the channel **134** and reduces or prevents movement of the weight **128** in the locked configuration **136**. In the example, the seat **156** extends in the toe-heel direction of the channel **134** and includes at least one oblique surface **160** that frictionally engages with a corresponding at least one oblique surface **162** of the notch **158**. In some examples, the oblique surfaces **160**, **162** may be substantially parallel to the fastener axis **142**. In other examples, the oblique surfaces **160**, **162** may extend at angle relative to the bottom of the channel **134** between about 3° and 88°. In one example, the oblique surfaces **160**, **162** may extend at an angle relative to the bottom of the channel **134** of about 30°.

(139) A cam **164** may also protrude into the channel **134** at a location between the toe end and the heel end, for example, between the seat **156** and the chamber **150**. The cam **164** is sized and shaped to receive within a corresponding cutout **166** defined in the cover **130**. When the weight assembly **104** is in the locked configuration **136**, the cam **164** is received at least partially within the cutout **166**. The cam **164** and the cutout **166** are described further below in reference to FIG. 4.

(140) In the example, the cover **130** is substantially L-shaped with a long leg **168** and a short leg **170**. In the locked configuration **136**, the long leg **168** forms the exterior surface **140** and the short leg **170** extends within the channel **134**. The channel **134** is formed from two opposing sidewalls **172**, **174** and a bottom track **176** offset from the outer surface **120** of the body **106**. The long leg **168** of the cover **130** opposes the track **176** of the channel **134** and the short leg **170** of the cover

130 is adjacent to one of the sidewalls **172**. The seat **156** and the cam **164** may protrude from the sidewall **172** of the channel **134** and the corresponding notch **158** and cutout **166** may be defined in the short leg **170** of the cover **130**. When the weight **128** is secured within the channel **134** and in the locked configuration **136**, the weight **128** is compressed between cover **130** and one or more walls (e.g., the sidewall **174** and/or the track **176**) of the channel **134**. As such, the weight **128** is frictionally secured to one or more portions of the club head **100** by the weight assembly **104**.

(141) Additionally, the weight **128** may be slidably coupled to the cover **130**. The long leg **168** of the cover **130** may include a flange **178** extending therefrom. The flange **178** is sized and shaped to be received at least partially within a corresponding groove **180** defined in the weight **128**. In the locked configuration **136**, a portion of the weight **128** is not covered by the cover **130** and exposed within the channel **134** such that the portion forms part of the outer surface **120** of the body **106**. This enables for the location of the weight **128** within the channel **134** to be easily determined by visual inspection.

(142) FIG. 4 is a cross-sectional view of the club head **100** taken along line 2-2 in FIG. 1 and showing the weight assembly **104** in an unlocked configuration **182**. FIG. 5 is a cross-sectional view of the weight assembly **104** taken along line 5-5 in FIG. 4. Referring concurrently to FIGS. 4 and 5, when the weight assembly **104** is in the unlocked configuration **182**, at least a portion of the cover **130** is lifted and raised out of the channel **134** such that the weight **128** is selectively slidable (e.g., along a toe-heel direction **184**) within the channel **134**. In the example, the fastener **132** may be coupled to the cover **130** (e.g., with a lock washer **186** (shown in FIG. 16)), so that the cover **130** moves along the fastener axis **142** (shown in FIG. 2) upon rotation of the fastener **132**. The cover **130** and the fastener **132** may be completely removed from the body **106** as required or desired so as to completely remove the weight **128** from the channel **134**. However, in examples, moving the weight assembly **104** between the locked configuration **136** (shown in FIGS. 2 and 3) and the unlocked configuration **182** does not require that the weight assembly **104** be uncoupled from the body **106**. As such, in the unlocked configuration **182**, the cover **130** may remain coupled to the body **106** so that it is less likely that the components become lost or misplaced. In some examples, the fastener **132** and/or the nut **138** may include a hard stop (not shown) that prevents the fastener **132** from being completely de-threaded from the club head **100** as required or desired.

(143) Since only a single fastener **132** is used to couple the cover **130** to the body **106** and the fastener **132** is positioned at the heel end of the weight assembly **104**, the cam **164** may be used to assist the toe end of the cover **130** with lifting from the channel **134** in the unlocked configuration **182**. This enables the weight **128** to more easily slide to positions away from the fastener **132**. In the example, the cam **164** extends in the toe-heel direction of the channel **134** and includes at least one camming surface **188** that slidably engages with a corresponding camming surface **190** of the cutout **166**. As the cover **130** moves from the locked configuration **136**, where the cam **164** is received within the cutout **166**, toward the unlocked configuration **182**, the camming surfaces **188**, **190** slide against one another to lift the toe end of the cover **130**. In some examples, when the weight assembly **104** is in the unlocked configuration **182**, a portion of the cover **130** may be supported on the cam **164**. The camming surfaces **188**, **190** may be substantially parallel to the fastener axis **142**.

(144) Additionally, in the unlocked configuration **182**, the notch **158** may lift away from the seat **156** to disengage the oblique surfaces **160**, **162** (shown in FIG. 2). In the unlocked configuration **182**, the notch **158** may lift partially or completely for the seat **156**. The projection **148** may also lift away from the chamber **150**. However, the projection **148** may remain at least partially engaged with the chamber **150** so that the weight **128** cannot slide out of the toe end of the cover **130** and remain within the channel **134** in the unlocked configuration **182**. Furthermore, because the weight **128** is engaged with the cover **130** (e.g., the flange **178** and the groove **180**), the weight **128** moves with the cover **130** between the locked configuration **136** and the unlocked configuration **182**. This enables the weight **128** to be more easily slidable in the unlocked configuration **182**.

(145) In some examples, one or more of the weight **128**, the cover **130**, and the channel **134** may include complementary features (e.g., corresponding detents **192** on the cover **130** and recesses (not shown) on the weight **128**) that index the location of the weight **128** to the channel **134** and/or the cover **130**. These complementary indexing features may provide tactile and/or audible feedback when the weight **128** is moved. Additionally, the complementary indexing features may also provide increased resistance to the relative movement between the weight **128** and the channel **134** and/or cover **130** when the weight assembly **104** is in the locked configuration **136**.

(146) FIG. **6** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **200**. FIG. **7** is a cross-sectional view of the weight assembly **200** taken along line 7-7 in FIG. **6**. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **6** and **7**, the weight assembly **200** includes a recessed channel **202** defined within the sole **102** of the body **106** of the club head **100**, however, the channel **202** extends substantially linearly in a front-rear direction so that the CG and the MOI of the club head **100** can be adjusted for launch angle bias. The channel **202** can be substantially orthogonal to the striking face **108** as illustrated in FIG. **6**. In other examples, the channel **202** may extend at either an acute or obtuse angle relative to the striking face **108**. The weight assembly **200** also includes a slidable weight **204**, a cover **206**, and a fastener **208**. In this example, the fastener **208** is positioned proximate to the rear of channel **202** and opposite of the striking face **108**. In other examples, the fastener **208** may be positioned at any other location relative to the channel **202** to enable the weight assembly **200** to function as described herein. For example, at approximately a midpoint of the channel **202** or proximate the striking face **108** side of the channel **202**.

(147) In this example, the channel **202** is formed by two opposing sidewalls, a cover sidewall **210** and an undercut sidewall **212**, and a bottom track **214** offset from the outer surface **120** of the body **106**. A partial wall **216** also extends from the bottom track **214**. Here, the cover **206** is located adjacent to the cover sidewall **210** and includes an angled surface **218**. As such, when the weight assembly **200** is in a locked configuration (e.g., FIG. **7**), the cover **206** generates a compressive force **220** along the angled surface **218** that acts in both a downward direction and a transverse direction to secure the weight **204** between the cover **206** and the undercut sidewall **212**.

Accordingly, the weight **204** is frictionally secured to one or more portions of the club head **100** by the weight assembly **200** and at least partially underneath the angled surface **218** and the undercut sidewall **212**. The weight **204** is at least partially trapezoidal in cross-sectional shape so that the undercuts of the sidewall **212** and the cover **206** assist in retaining the weight **204** within the channel **202**. Additionally, the cover **206** engages with the partial wall **216** so that the portion of the cover **206** away from the fastener **208** is restricted from moving within the channel **202** (e.g., bending or flexing) towards the undercut sidewall **212**. Furthermore, the partial wall **216** is substantially parallel to the fastener axis (not shown) of the fastener **208** so that the cover **206** is guided between the locked and unlocked configuration. In some example, the weight assembly **200** may include the seat/notch interface as described above to further engage the cover **206** within the channel **202** and increase the securement of the weight **204** to one or more portions of the club head **100**.

(148) FIG. **8** is a perspective view of the golf club head **100** with another weight assembly **300**. Certain components are described above, and thus, are not necessarily described further. In this example, the club head **100** includes the skirt **126** positioned between the crown **118** and the sole **102**, opposite of the striking face **108**. The weight assembly **300** includes a recessed channel **302** defined within the skirt **126** of the body **106** of the club head **100** and extends along the rear perimeter of the club head **100** such that the channel **302** has a curved shape. The weight assembly **300** also includes a slidable weight **304**, a cover **306**, and a fastener **308**. In this example, the fastener **308** is coupled to the heel **116** side of the body **106**. In other examples, the fastener **308** may be coupled to the toe **114** side of the body **106** as required or desired. The weight assembly **300** may include one or more of the weight assembly features described herein to enable the CG

and the MOI of the club head **100** to be adjustable for fade-draw bias, while securing the weight **304** in a locked configuration (as shown in FIG. **8**).

(149) FIG. **9** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **400**. FIG. **10** is a top view of the golf club head **100** shown in FIG. **9** with a portion of the crown **118** removed. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **9** and **10**, the weight assembly **400** includes a recessed channel **402** defined within the sole **102** of the body **106** of the club head **100** that extends substantially linearly in the toe **114**-heel **116** direction. The weight assembly **400** also includes a slidable weight **404**, a cover **406**, and a fastener **408**. The channel **402** includes a bottom track **410** that the weight **404** is slidable on. In this example, the fastener **408**, and also a nut **412** that the fastener **408** couples to, are offset from the track **410** and positioned towards the rear of the body **106**. By offsetting the fastener **408** from the track **410**, the length of the track **410** can be extended in the toe-heel direction so that the weight **404** can be positioned at a greater number of locations on the sole **102**. In other examples, the fastener **408** may be offset from the track **410** and positioned towards the front and the striking face **108** of the body **106** as required or desired.

(150) In this example, one or more support ribs **414** may extend from the channel **402** and within the interior cavity **122** of the body **106**. The support ribs **414** are substantially orthogonal to the length of the channel **402**. The support ribs **414** provide structural strength to the channel **402** so that the channel **402** is resistant to deformation when the cover **406** compresses the weight **404** therein. In some examples, the support ribs **414** may extend the entire distance between the sole **102** and the crown **118** within the interior cavity **122**.

(151) FIG. **11** is a cross-sectional view of the weight assembly **400** taken along line **11-11** in FIG. **9**. FIG. **12** is a cross-sectional view of the weight assembly **400** taken along line **12-12** in FIG. **9**. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **11** and **12**, the weight assembly **400** is illustrated in a locked configuration so that the weight **404** is secured within the channel **402**. In this example, the weight **404** includes an elastomeric material **416** (e.g., a rubber-based material) that engages with the channel **402** and/or the cover **406** and further increase securement of the weight **404** in the locked configuration. Additionally, the elastomeric material **416** decreases rattling of the weight **404** within the channel **402** during the swing of the club head.

(152) In this example, the channel **402** is formed from two opposing sidewalls **418**, **420** and the track **410**. One sidewall **420** may include an elongate fin **422** extending into the channel **402**. The weight **404** is sized and shaped to be received at least partially within the channel **402** and includes a bottom surface **424** that is positioned adjacent to the track **410** and a slot **426** that engages with the fin **422**. Additionally, opposite of the slot **426**, the weight **404** includes a groove **428** that engages with a flange **430** of the cover **406**. The elastomeric material **416** may be coupled to the weight **404** so that the material **416** extends from the bottom surface **424** and also into the slot **426**. In one example, the elastomeric material **416** may be a unitary piece that extends through one or more holes within the weight **404**. In other examples, the elastomeric material **416** may be adhered to one or more external surfaces of the weight **404**. In still other examples, at least a portion of the elastomeric material **416** may form the weight **404** itself.

(153) In operation, when the cover **406** is in the locked configuration, the flange **430** engages with the groove **428** of the weight **404** and compresses the weight **404** into the channel **402**. As such, the elastomeric material **416** may engage with the track **410** and the fin **422** of the channel **402**. By engaging the elastomeric material **416** in more than one location, securement of the weight **404** within the channel **402** increases. This reduces undesirable movement and rattling of the weight **404** within the channel **402**. In some examples, the elastomeric material **416** may deform when compressed within the channel **402**. Since the cover **406** engages with only a portion of the weight **404**, when the cover **406** is lifted **432** for the unlocked configuration (not shown), the weight **404** can rotate **434** within the channel **402** so that the elastomeric material **416** may disengage from the

track **410** and the fin **422**. This rotational movement **434** enables the weight **404** to be more easily slidable within the channel **402** while in the unlocked configuration because the elastomeric material **416** is at least partially positioned away from the channel surfaces. In some examples, the elastomeric material **416** extending from the bottom surface **424** may be only proximate the groove **428** so as to increase rotational movement **434** of the weight **404**.

(154) The cover **406** is substantially L-shaped in cross-section (see FIG. **12**) and receives at least a portion of the weight **404** therein. The cover includes a first leg **436** that has the flange **430** and a second leg **438** that is adjacent to the sidewall **418** of the channel **402**. The flange **430** may be substantially parallel to the second leg **438** so as to increase the structural rigidity of the cover **406** in the lengthwise direction. The second leg **438** may extend at least partially within a depression **440** of the track **410** so as to decrease bending of the cover **406** while in the locked configuration. Additionally, in the example, a projection **442** of the cover **406** may be substantially cylindrical in shape. The projection **442** is received within a corresponding cylindrical chamber **444**. This projection **442** and chamber **444** structure increases the engagement of the cover **406** with the body **106** in the locked configuration (as illustrated in FIG. **11**). In some examples, a projection axis **446** of the projection **442** may be substantially parallel to a fastener axis **448**. This orientation guides the movement of the cover **406** between the locked configuration and the unlocked configuration. In some examples, the projection **442** may include a tapered nose. In this example, the weight **404** and the channel **402** may include complementary features **450** that index the location of the weight **404** to the channel **402**.

(155) FIG. **13** is a cross-sectional view of another weight assembly **500**. Certain components are described above, and thus, are not necessarily described further. Similar to the example described in FIGS. **9-12**, in this example, the weight assembly **500** includes a recessed channel **502** defined within the body **106** of the club head. The weight assembly **500** also includes a slidable weight **504** and a cover **506**. The cover **506** is shown in a locked configuration and a slot **508** of the weight **504** is engaged with a fin **510** of the channel **502**. However, in this example, a bottom surface **512** of the weight **504** is positioned directly against a track **514** of the channel **502**. Additionally, in this example, the bottom surface **512** of the weight **504** includes a hollow **516**. The hollow **516** reduces frictional sliding forces on the weight **504**, when the weight assembly **500** is in the unlocked configuration (not shown). The hollow **516** also enables for the size and shape of the weight **504** to be formed while maintaining the required or desired mass and/or density of the weight **504**. In some examples, an elastomeric material (not shown) may be disposed at least partially within the hollow **516**.

(156) FIG. **14** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **600**. Certain components are described above, and thus, are not necessarily described further. The weight assembly **600** includes a recessed channel **602** defined within the sole **102** of the body **106** of the club head **100**. The channel **602** has a substantially curved shape in the toe **114**-heel **116** direction so that the CG and the MOI of the club head **100** can be adjustable for fade-drawn bias. In some examples, the curve of the channel **602** matches the rear perimeter of the body **106**, where the sole **102** and the crown **118** are coupled together. The weight assembly **600** also includes a slidable weight **604**, a cover **606**, and a fastener **608**.

(157) In this example, the fastener **608** is positioned in the concave area of the curved channel **602** and towards the striking face **108** of the body **106**. This position enables the weight **604** to be positioned adjacent to the rear perimeter of the body **106** and increase the adjustability of the CG and MOI of the club head **100**, when compared to having the fastener **608** positioned in the convex area of the curved channel **602** and the weight **604** being closer to the striking face **108**.

Additionally, the weight **604** may slide completely from the toe **114** side to the heel **116** side and be located at any position of the channel **602** even adjacent to the fastener **608**. In other examples, the fastener **608** may be positioned in the convex area of the curved channel **602** as required or desired. The fastener **608** is also positioned at approximately the midpoint of the channel **602**. In other

examples, the fastener **608** may be offset from the midpoint of the channel **602**, or two or more fastener **608** may be used to couple the cover **606** to the body **106** (e.g., at each end of the channel **602**).

(158) FIG. **15** is a cross-sectional view of the club head **100** taken along line **15-15** in FIG. **14** and showing the weight assembly **600**. FIG. **16** is a cross-sectional view of the weight assembly **600** taken along line **16-16** in FIG. **14**. FIG. **17** is a cross-sectional view of the weight assembly **600** taken along line **17-17** in FIG. **14**. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **15-17**, the weight assembly **600** is illustrated in a locked configuration and the weight **604** includes a bottom surface **610** and a groove **612**. A tab **614** is disposed adjacent to the groove **612**. Additionally, the weight **604** includes an elastomeric material **614**. In this example, the elastomeric material **614** is coupled to the weight **604** and extends from the bottom surface **610** and also into the groove **612**. The elastomeric material **614** is oversized relative to the channel **602** (e.g., between a 0.1 millimeter and 1.0 millimeter overlap) so that the material **614** may deform while being compressed within the channel **602**. In other examples, the elastomeric material **614** may be adhered to the exterior surface of the weight **604**. In yet other examples, the elastomeric material **614** may at least partially form the weight **604** itself.

(159) The cover **606** is substantially C-shaped with a flange **616** that engages with the groove **612** of the weight **604**. Additionally, the cover **606** includes a top leg **618** and a side leg **620** that is opposite of the flange **616**. The top leg **618** has a thickness that is greater than the flange **616** and the side leg **620** so as to increase the structural rigidity of the cover **606** in a lengthwise direction. The fastener **608** is coupled to the cover **606** by a lock washer **186** that enables the fastener **608** to rotate relative to the cover **606** while allowing the cover **606** to move along a fastener axis **622** to raise and lower the cover **606** relative to the channel **602**.

(160) In operation, when the cover **606** is in the locked configuration, the flange **616** of the cover **606** is engaged within the groove **612** of the weight **604**. This compresses the weight **604** between the cover **606** and a bottom track **624** of the channel **602**. In the locked configuration, the elastomeric material **614** engages with both the cover **606** and the channel **602** to increase the securement of the weight **604** to one or more portion of the club head **100**. In some examples, a plurality of grooves **626** are defined within the track **624** that the elastomeric material **614** deforms into the grooves **626** to facilitate securement of the weight **604** within the channel **602**.

Additionally, the tab **614** of the weight **604** may be positioned proximate the outer surface **120** of the body **106** so that the position of the weight **604** may be visible. When the weight assembly **600** is in the unlocked configuration (not shown), the cover **606** is lifted at least partially out of the channel **602** so that the weight **604** may be selectively slidable therein, for example, via the tab **614**.

(161) Each end of the cover **606** may include a substantially cylindrical projection **628** that is received within a corresponding cylindrical chamber **630** of the channel **602**. The projections **628** extend along a projection axis **632** that is substantially parallel to the fastener axis **622**. This orientation guides the movement of the cover **606** between the locked configuration and the unlocked configuration. In some examples, the projections **628** may include a tapered nose. Additionally, the chamber **630** may be open into the interior cavity **122** of the body **106** as illustrated in FIGS. **15** and **16**. In other examples, the chamber **630** may be closed off from the interior cavity **122**. One or more support ribs **634** may also extend from the track **624** and within the interior cavity **122** as required or desired.

(162) FIG. **18** is an exploded perspective view of the golf club head **100** with another weight assembly **700**. Certain components are described above, and thus, are not necessarily described further. Similar to the example described in FIGS. **14-17**, in this example, the weight assembly **700** includes a recessed channel **702** defined within the body **106** of the club head **100** and the channel **702** has a substantially curved shape in the toe **114**-heel **116** direction. In some examples, the curve

of the channel **702** matches the rear perimeter of the body **106**, where the sole **102** and the crown **118** are coupled together. The weight assembly **700** also includes a slidable weight **704**, a cover **706**, and a fastener **708**. At each end of the cover **706**, projections **710** may extend for engagement within the channel **702**.

(163) FIG. **19** is a cross-sectional view of the weight assembly **700** taken along line **19-19** in FIG. **18**. Certain components are described above, and thus, are not necessarily described further. The weight assembly **700** is illustrated in the locked configuration in FIG. **19** and a bottom surface **712** of the weight **704** is positioned directly against a track **714** of the channel **702**. Additionally, in this example, the bottom surface **712** of the weight **704** includes a hollow **716**. The hollow **716** reduces frictional sliding forces on the weight **704**, when the weight assembly **700** is in the unlocked configuration (not shown). The hollow **716** also enables for the size and shape of the weight **704** to be formed while maintaining the required or desired mass and/or density of the weight **704**. In some examples, an elastomeric material (not shown) may be disposed at least partially within the hollow **716**.

(164) Additionally, the cover **706** includes an angled surface **718** that abuts the weight **704**. As such, when the weight assembly **700** is in a locked configuration (e.g., FIG. **19**), the cover **706** generates a compressive force **720** along the angled surface **718** that acts in both a downward direction and a transverse direction to secure the weight **704** between the cover **706** and an undercut sidewall **722** of the channel **702**. As such, the weight **704** is frictionally secured by the weight assembly **700** to one or more portions of the club head **100**.

(165) FIG. **20** is a partial cross-sectional perspective view of another weight assembly **800**. FIG. **21** is another cross-sectional view of the weight assembly **800**. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **20** and **21**, the cross-sectional views are substantially along a front-rear direction of the golf club head and, for example, similar to the examples described above in reference to FIGS. **16** and **17**. The weight assembly **800** includes a recessed channel **802** defined within the body **106**. The weight assembly **800** also includes a slidable weight (not shown), a cover **804**, and a fastener **806**. In this example, the channel **802** is defined by a bottom track **808** and two opposing sidewalls **810**, **812**. The bottom track **808** includes an elastomeric material **814** coupled thereto and that extends at least partially into the channel **802**. The elastomeric material **814** engages with the weight and further increases securement of the weight within the channel **802** in the locked configuration. Additionally, the elastomeric material **814** decreases rattling of the weight during the swing of the club head. Additionally or alternatively, the elastomeric material **814** may be coupled to one or more of the sidewalls **810**, **812** as required or desired. In still other examples, the elastomeric material **814** can be coupled to the cover **804**.

(166) In this example, the elastomeric material **814** extends along the longitudinal length of the channel **802**. At each end **816** of the elastomeric material **814**, a portion of the material may extend into an undercut area **818** within the channel **802** so as to secure the elastomeric material **814** within the channel **802**. In other examples, the elastomeric material **814** may be adhered within the channel **802** or the cover **804** as required or desired. The end **816** of the elastomeric material **814** may be offset **820** from a projection **822** of the cover **804** so that the elastomeric material **814** does not interfere with the movement of the cover **804** between the locked and unlocked configurations as described herein.

(167) FIG. **22** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **900** in a locked configuration. FIG. **23** is a cross-sectional view of the weight assembly **900** taken along line **23-23** in FIG. **22**. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **22** and **23**, the weight assembly **900** is illustrated in a locked configuration and includes a recessed channel **902** defined within the sole **102** of the body **106** of the club head **100**. The channel **902** has a substantially curved shape in the toe **114**-heel **116** direction so that the CG and the MOI of the club head **100** can be adjustable for

fade-drawn bias. In some examples, the curve of the channel **902** matches the rear perimeter of the body **106**, where the sole **102** and the crown **118** are coupled together. The weight assembly **900** also includes a toe-side slidable weight **904**, a heel-side slidable weight **906**, a toe side cover **908**, a heel side cover **910**, and a fastener **912**.

(168) In this example, the fastener **912** is disposed within the channel **902** and divides the weight assembly **900** approximately in half. By positioning the fastener **912** within the channel **902** the size of the weight assembly **900** on the club head **100** is reduced. Additionally, the mass of the fastener **912** is moved further rearward from the striking face **108** than those examples described above. The weights **904**, **906** extend from the inner convex side of the covers **908**, **910** as illustrated in FIG. 22. In other examples, the weights **904**, **906** may extend from the outer concave side of the covers **908**, **910** as required or desired. In this example, two slidable weights **904**, **906** are described since the fastener **912** prevents a weight from sliding completely from the toe side to the head side of the channel **902** and back. In some examples, the weight assembly **900** may include only one slidable weight and the fastener **912** and the covers **908**, **910** are configured to allow the weight to pass between the toe **114** side and the heel **116** side. In other examples, the weight assembly **900** may include only one slidable weight that requires the assembly to be completely disassembled so as to move the weight from the toe side to the head side and back. In still other examples, the weights **904**, **906** may be completely removable from the channel **902** as required or desired.

(169) One end of each cover **908**, **910** is engaged with the channel **902**, for example, with the projection/channel interface as described herein, while the other opposite end of each cover **908**, **910** is engaged with the fastener **912**. In the example, the fastener **912** includes a washer **914** that is disposed below the head. The washer **914** is a substantially cylindrical flange extending from the threaded shaft that engages with both corresponding groove **916** within the covers **908**, **910**. When the weight assembly **900** is in the locked configuration the covers **908**, **910** are disposed within the channel **902** and secured in place with the fastener **912**, via the grooves **916**, so that the weights **904**, **906** cannot slide within the channel **902** and are locked in place. Additionally, the covers **908**, **910** are flush with the outer surface **120** of the body **106**. In some examples, the portion of the covers **908**, **910** that define the grooves **916** may extend all the way to a bottom track **918** of the channel **902** so that overtightening of the fastener **912** is reduced or prevented.

(170) FIG. 24 is a perspective view of the sole **102** of the golf club head **100** with the weight assembly shown **900** in an unlocked configuration. FIG. 25 is a cross-sectional view of the weight assembly **900** taken along line 25-25 in FIG. 24. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. 24 and 25, the weight assembly **900** is illustrated in an unlocked configuration. When the weight assembly **900** moves from the locked configuration (shown in FIGS. 22 and 23), the fastener **912** is rotated so as to lift at least partially out of the channel **902**. This movement of the fastener **912** also lifts the ends of the covers **908**, **910** that are engaged with the washer **914** at least partially out of the channel **902** so as to enable the weights **904**, **906** to slide within the channel **902**. In some examples, the weights **904**, **906** may be engaged with the respective cover **908**, **910** so as to lift away from the track **918** for ease of movement.

(171) In some examples, the covers **908**, **910** and the fastener **912** may be completely removed from the body **106** as required or desired so as to completely remove the weights **904**, **906** from the channel **902**. However, moving the weight assembly **900** between the locked configuration) and the unlocked configuration does not require that the weight assembly **900** be uncoupled from the body **106**. As such, in the unlocked configuration, the covers **908**, **910** remain coupled to the body **106** so that it is less likely that the components become lost or misplaced.

(172) In this example, when the covers **908**, **910** are in the unlocked configuration, the ends of the covers **908**, **910** that are opposite of the fastener **912** and engaged with the channel **902** (e.g., with the projection/channel interface) remain engaged with the channel **902** and may form a pivot point

that the covers **908**, **910** rotate about. In other examples, the ends of the covers **908**, **910** that are opposite of the fastener **912** may lift at least partially out of the channel **902** as described herein. For example, through a cam and cutout interface as described above.

(173) FIG. **26** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **1000**. FIG. **27** is a cross-sectional view of the weight assembly **1000** taken along line **27-27** in FIG. **26**. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **26** and **27**, the weight assembly **1000** includes a substantially linear recessed channel **1002** defined within the sole **102**. The weight assembly **1000** also includes a slidable weight **1004**, a cover **1006**, and a fastener **1008**. In this example, the fastener **1008** may be positioned at approximately the midpoint of the channel **1002** and offset towards the rear of the club head **100**. By positioning the fastener **1008** at a midpoint location, the distance between the fastener **1008** and the far end(s) of the cover **1006** is reduced so that the engagement between the cover **1006** and the channel **1002** is increased for securement of the weight **1004**.

(174) Similar to the example described above in reference to FIGS. **6** and **7**, the channel **1002** is formed by two opposing sidewalls, a cover sidewall **1010** and an undercut sidewall **1012**, and a bottom track **1014** offset from the outer surface **120** of the body **106**. A partial wall **1016** also extends from the bottom track **1014**. The cover **1006** is located adjacent to the cover sidewall **1010** and includes an angled surface **1018**. As such, when the weight assembly **1000** is in a locked configuration (e.g., FIG. **27**), the cover **1006** generates a compressive force along the angled surface **1018** that acts in both a downward direction and a transverse direction to secure the weight **1004** between the cover **1006** and the undercut sidewall **1012**. Accordingly, the weight **1004** is frictionally secured by the weight assembly **1000** and at least partially underneath the angled surface **1018** and the undercut sidewall **1012**. Additionally, the cover **1006** completely engages with the partial wall **1016** via a groove **1020** so that the portion of the cover **1006** away from the fastener **1008** is restricted from moving within the channel **1002** (e.g., bending or flexing) towards the undercut sidewall **1012**. Furthermore, the partial wall **1016** is substantially parallel to the fastener axis (not shown) of the fastener **1008** so that the cover **1006** guides the movement between the locked and unlocked configuration.

(175) FIG. **28** is an exploded perspective view of the sole **102** of the golf club head **100** with another weight assembly **1100**. FIG. **29** is a cross-sectional view of the weight assembly **1100**. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **28** and **29**, the weight assembly **1100** includes a substantially linear recessed channel **1102** defined within the sole **102**. The weight assembly **1100** also includes a slidable weight **1104**, a cover assembly **1106**, and a fastener **1108**. In this example, the fastener **1108** may be positioned at approximately the midpoint of the channel **1102** and offset towards the rear of the club head **100**. As described above, when the cover assembly **1106** is in the locked configuration, the cover assembly **1106** is coupled to the body **106** so that the weight **1104** is secured within the weight assembly **1100** without movement or rattling. In this example, the cover assembly **1106** is a four piece assembly including a fastener member **1110**, two opposing longitudinal members **1112**, and a transverse member **1114**.

(176) When the cover assembly **1106** is moved towards the locked configuration (e.g., FIG. **29**), the fastener **1108** is tightened to the body **106**. The fastener **1108** engages with the fastener member **1110** and moves the fastener member **1110** along the fastener axis (not shown) and into the channel **1102**. The fastener member **1110** has a tapered surface that engages with both of the longitudinal members **1112** so that as the fastener member **1110** is pulled down within the channel **1102**, the longitudinal members **1112** are also pulled down within the channel **1102** and generate a compressive force **1116** along an angled surface **1118**. The compressive force **1116** acts in both a downward direction and a transverse direction on the transverse member **1114** to position the transverse member **1114** within the channel **1102** and compress the weight **1104** between the

transverse member **1114** and a sidewall **1120** of the channel.

(177) Additionally, to reduce or prevent pull-out of the weight assembly **1100** from the body **106**, the transverse member **1114** may engage with an undercut **1122** of the channel **1102**. The compressive force **1116** from the longitudinal members **1112** lock the transverse member against the undercut **1122** so as to prevent movement. Additionally or alternatively, a portion of the weight assembly **1100** may engage with the sidewall **1120** of the channel **1102** so as to reduce pull out of the weight assembly **1100** from the body **106**. Additionally, the fastener member **1110** also pushes the longitudinal members **1112** away from the fastener **1108** (e.g., arrows **1124**) so that ends **1126** of the members **1112** can engage with a corresponding chamber **1128** in the channel **1102** and also reduce pull out of the weight assembly **1100** from the body **106**.

(178) FIG. **30** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **1200**. FIG. **31** is a cross-sectional view of the weight assembly **1200** taken along line **31-31** in FIG. **30**. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **30** and **31**, the weight assembly **1200** is illustrated in an unlocked configuration and includes a recessed channel **1202**, a slidable weight **1204**, a cover **1206**, and a fastener **1208**. The structure, size, shape, and orientation of the channel **1202**, the weight **1204**, and the fastener **1208** may be similar to any of the examples described above. In this example, however, a width **1210** of the cover **1206** is extended towards the striking face **108** so that the cover **1206** forms a greater portion of the sole **102** and does not only cover a portion of the channel **1202**.

(179) In some examples, the cover **1206** may form greater than or equal to 75% of the surface area of the sole **102**. In other examples, the cover **1206** may form greater than or equal to 50% of the surface area of the sole **102**. In still other examples, the cover **1206** may form greater than or equal to 25% of the surface area of the sole **102**. In still further examples, the cover **1206** may be between about 10% and 90% of the surface area of the sole **102**. In other examples, the cover **1206** may be between about 25% and 75% of the surface area of the sole **102**.

(180) By enlarging the cover **1206** of the weight assembly **1200**, the golf club head structure that forms the sole **102** of the body **106** can be reduced. In some examples, the cover **1206** can be manufactured from a lighter weight material (e.g., composite materials, plastics, etc.) than the material that the body **106** is manufactured from. As such, the weight saved by the configuration of the sole construction can be used at other locations on the club head **100** as required or desired and further enable adjustment of the CG and MOI of the club head **100** for improving golf ball flight characteristics. In some examples, the weight saved by the sole construction can be included back into the slidable weight **1204**. For example, the cover **1206** may reduce the weight of the sole construction by 11 grams or more, some or all of which mass that can then be included at least partially into the weight **1204**.

(181) The cover **1206** can include a projection **1212** extending therefrom that is configured to engage with a corresponding chamber **1214** within each end of the channel **1202** for increasing the structural rigidity of the cover **1206** connection as described in the examples above. In one example, the projection **1212** may be substantially cylindrical and parallel to a fastener axis **1216**. At the opposite side of the cover **1206** from the fastener **1208**, the cover **1206** includes a brace **1218** adjacent to an extended edge **1220** that frictionally engages with the remaining sole **102** of the club head **100** to secure the edge **1220** to the body **106**. In some examples, the brace **1218** may extend at an angle that is substantially parallel to the fastener axis **1216** so as to guide the movement of the cover **1206** between the locked and unlocked configurations as described herein. The brace **1218** may include one or more brackets **1222** for increasing the structural rigidity of the brace **1218**.

(182) FIG. **32** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **1300**. Certain components are described above, and thus, are not necessarily described further. Similar to the example described in FIGS. **30** and **31**, the weight assembly **1300** includes an

enlarged cover **1302** that selectively secures a slidable weight **1304** to one or more portions of the club head **100**. In this example, however, a fastener **1306** is positioned more towards the striking face **108** and adjacent to an extended edge **1308** of the cover **1302**. This example increases the securement of the edge **1308** to the body **106** of the golf club head **100**. In other examples, the fastener **1306** may be positioned at any other location on the cover **1302** as required or desired. For example, towards the toe side **114**, towards the heel side **116**, centered on the cover **1302**, etc.

(183) FIG. **33** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **1400**. Certain components are described above, and thus, are not necessarily described further. Similar to the example described in FIGS. **30-32**, the weight assembly **1400** includes an enlarged cover **1402** that selectively secures a slidable weight **1404** to one or more portions of the club head **100**. In this example, however, the cover **1402** has an extended edge **1406** that is substantially V-shaped. Additionally, the cover **1402** is symmetrical in the toe **114**-heel **116** direction. In other examples, the cover **1402** may be asymmetrical in the toe **114**-heel **116** direction as required or desired.

(184) FIG. **34** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **1500**. Certain components are described above, and thus, are not necessarily described further. Similar to the example described in FIGS. **30-33**, the weight assembly **1500** includes an enlarged cover **1502** that selectively secures a slidable weight **1504** to one or more portions of the club head **100**. In this example, however, the cover **1502** is asymmetrical in the toe side **114** direction. In other examples, the cover **1502** may be asymmetrical in the heel side **116** direction as required or desired.

(185) FIG. **35** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **1600**. Certain components are described above, and thus are not necessarily described further. Similar to the example described in FIGS. **30-34**, the weight assembly **1600** includes an enlarged cover **1602** that selectively secures a slidable weight **1604** to one or more portions of the club head **100**. In this example, however, the cover **1602** has an extended edge **1606** that is substantially C-shaped. Additionally, the cover **1602** is symmetrical in the toe **114**-heel **116** direction. In other examples, the cover **1602** may be asymmetrical in the toe **114**-heel **116** direction as required or desired.

(186) FIG. **36** is a perspective view of the sole **102** of the golf club head **100** with another weight assembly **1700**. FIG. **37** is an exploded perspective view of the weight assembly **1700**. Certain components are described above, and thus, are not necessarily described further. Referring concurrently to FIGS. **36** and **37**, a recessed channel **1702** is defined within the sole **102** of the body **106** of the club head **100**. The channel **1702** has a substantially curved shape in the toe **114**-heel **116** direction so that the CG and the MOI of the club head **100** can be adjusted for fade-draw bias (e.g., the “F” and “D” indicia on a cover **1706** of the weight assembly **1700**). In the example, the curve of the channel **1702** substantially corresponds to the rear outer perimeter of the body **106**, where the sole **102** and the crown **118** are coupled together, and opposite of the striking face **108**. The weight assembly **1700** includes a slidable weight **1704**, a cover **1706**, and a fastener **1708**.

(187) In this example, the cover **1706** is substantially U-shaped with a toe end **1710** and an opposite heel end **1712**. The fastener **1708** is coupled to the cover **1706** by a lock washer **1714** (e.g., a retainer clip) and it is positioned on the inner concave side of the cover **1706** at approximately a midpoint between the ends **1710**, **1712**. The fastener **1708** is a threaded bolt that threadingly engages with a nut **1716** formed within the sole **102** of the body **106**. The lock washer **1714** enables the cover **1706** to linearly move M (e.g., raise and lower) along a fastener axis **1718** (shown in FIG. **38**) with respect to the recessed channel **1702** upon rotation of the fastener **1708**. The fastener **1708** is offset from the recessed channel **1702** towards the front and the striking face **108** of the body **106**. By offsetting the fastener **1708** from the recessed channel **1702**, the length of the recessed channel **1702** can be extended in the toe-heel direction so that the weight **1704** can be positioned at a greater number of locations on the sole **102**. Additionally, by positioning the

fastener adjacent to the inner concave side of the cover **1706**, the weight **1704** is disposed closer to the outer perimeter of the body **106** so that the weight **1704** increases the adjustability of the CG and MOI of the club head **100**.

(188) Each end **1710**, **1712** of the cover **1706** includes a projection **1720** extending therefrom. The projections **1720** are sized and shaped to be received within a corresponding chamber **1722** defined at the ends of the recessed channel **1702** and within the sole **102** of the body **106**. The projection **1720** may be substantially cylindrical in shape and increases the engagement of the cover **1706** with the body **106** so that the weight **1704** is restricted from moving or rattling when secured within the recessed channel **1702** by the cover **1706**. A projection axis **1724** of the projection **1720** (shown in FIG. **38**) is substantially parallel to the fastener axis **1718** so as to facilitate guiding the movement of the cover **1706** relative to the recessed channel **1702**.

(189) In operation, the weight assembly **1700** is selectively moveable between at least three configurations to enable the CG and the MOI of the club head **100** to be adjustable. More specifically, in a first or locked configuration, the cover **1706** is at least partially disposed within the recessed channel **1702** so that the weight **1704** is secured within the channel **1702** and movement is restricted. This locked configuration is illustrated in FIG. **36**. When the weight assembly **1700** is in the locked configuration, the projection **1720** is received at least partially within the chamber **1722** and engaged therewith. By engaging the cover **1706** to the body **106** at its ends **1710**, **1712**, when the weight **1704** is positioned away from the fastener **1708**, the cover **1706** still enables securement of the weight **1704** within the channel **1702** and reduces or prevents movement of the weight **1704** in the locked configuration. The locked configuration is used when swinging the golf club head **100**.

(190) Additionally, the weight assembly **1700** can be positionable into at least two other configurations that enable the weight **1704** to be selectively slidable with the recessed channel **1702** and that enable the weight **1704** to be completely removable from the weight assembly **1700** and the club head **100**. In a second or weight moving configuration, the cover **1706** is partially raised out of the recessed channel **1702** so that the position of the weight **1704** can be adjusted. However, the weight **1704** is still retained within the weight assembly **1700** and cannot be completely removed from the club head **100**. This configuration is illustrated in FIG. **40** and described further below. In a third or weight removal configuration, the cover **1706** is positioned so that the weight **1704** can be completely removed, for example, so that a different weight (e.g., having a different mass) can be used with the club head **100** so as to adjust the swing weight. This configuration is illustrated in FIG. **41** and described further below. In each of the three configurations, however, the cover **1706** remains coupled to the body **106** so that the cover **1706** does not have to be completely removed. In some examples, however, the weight assembly **1700** may include a fourth configuration (not illustrated), whereby the cover **1706** is completely removable from the body **106** as required or desired.

(191) A partial wall **1726** is disposed within the recessed channel **1702**. The weight **1704** has a corresponding recess **1728** (shown in FIG. **38**) so that the weight **1704** can slide along the partial wall **1726**. The partial wall **1726** at least partially prevents the weight **1704** from being completely removed when the weight assembly **1700** is in the weight moving configuration. In some examples, the weight **1704** may include at least one locating feature **1730** (FIG. **37**). The feature **1730** is sized and shaped to engage with one or more of a plurality of locating lugs **1732** (shown in FIG. **39**) that extend from the cover **1706** when the weight assembly **1700** is in the locked configuration. The locating lugs **1732** and feature **1730** facilitate locating the weight **1704** at specific locations within the recessed channel **1702**. In the example, the locating lugs **1732** are substantially frustoconical in shape and the locating features **1730** have a corresponding recessed shape. In other examples, the lugs **1732** and features **1730** can have any other shape and/or size that enable the cover **1706** and the weight **1704** to function as described herein.

(192) FIG. **38** is a cross-sectional view of the weight assembly **1700** taken along line **38-38** in FIG.

36. Certain components are described above, and thus, are not necessarily described further. As described above, the weight **1704** is secured within the recessed channel **1702** by the cover **1706**. The fastener **1708** positions and secures the cover **1706** to the body **106** of the golf club head, and thus, the fastener **1708** only retains the weight **1704** indirectly. In some examples, a washer (not shown) may be positioned on the fastener **1708** and between the body **106** and the cover **1706**. The recessed channel **1702** is formed in cross-section by a bottom track wall **1734** and a side wall **1736** arranged in a substantially L-shape configuration with a corner **1738**. In the example, the corner **1738** has an angle that is equal to or less than 90° . In another aspect, the corner **1738** has an angle that less than 90° so that the side wall **1736** is undercut. As such, when the weight assembly **1700** is in the locked configuration, the cover **1706** wedges the weight **1704** into the corner **1738** and against the side wall **1736** to frictionally secure the weight **1704** within the recessed channel **1702** and at least partially underneath the side wall **1736**.

(193) Adjacent to the corner **1738** and on the bottom track wall **1734**, the partial wall **1726** extends in an upward direction and has a height H.sub.1. The weight **1704** has a corresponding recess **1728** that receives at least a portion of the partial wall **1726**. The partial wall **1726** at least partially contains the weight **1704** within the weight assembly **1700** when in the locked and weight sliding configurations. The weight **1704** also includes a tail **1740** that projects from the recessed channel **1702** and out from underneath the cover **1706**. The tail **1740** of the weight **1704** provides structure for a user to grasp and slide the weight **1704** as required or desired. The tail **1740** is also visible on the outer surface of the club head so that its position is easily determined by visual inspection. In this example, the tail **1740** is at least partially corresponds to the shape of the bottom track wall **1734** of the recessed channel **1702**. In other examples, the tail **1740** can have any other size and/or shape as required or desired.

(194) FIG. 39 is an inside surface **1742** view of the cover **1706** of the weight assembly **1700** (shown in FIGS. 36-38). The cover **1706** is substantially U-shaped with a concave side that receives the fastener at an aperture **1744**. Proximate the convex side, the inside surface **1742** has the plurality of locating lugs **1732** that are configured to engage with the locating feature **1730** within the weight **1704** (shown in FIG. 37). When engaged (e.g., in the first, locked configuration), the cover **1706** wedges the weight in the corner of the recessed channel and against the side wall. However, when the cover **1706** raised out of the recessed channel, the locating lugs **1732** disengage from the weight so that the weight can be moved (e.g., in the weight sliding configuration) or so that the weight can be completely removed (e.g., in the weight removal configuration).

(195) Each end **1710**, **1712** of the cover **1706** includes the projection **1720** that, in addition to the fastener, secures the cover **1706** to the body of the club head. In the example, the projection **1720** engages with the chamber **1722** (shown in FIG. 37) in all three configurations (e.g., locked, weight moving, and weight removal) of the weight assembly. Furthermore, the projection **1720** also at least partially defines each of the three configurations. The projection **1720** is substantially cylindrical in shape and is configured to extend through the sole of the body and into the interior cavity of the club head via the chamber **1722**. In the example, the projection **1720** includes a plurality of flexible arms **1746** circumferentially spaced to form the substantially cylindrical projection **1720**. As illustrated, the projection **1720** includes three discrete flexible arms **1746**. In other examples, the projection **1720** may include any other number of flexible arms **1746** (e.g., 2, 4, 5, etc.) as required or desired.

(196) FIG. 40 is a cross-sectional view of the weight assembly **1700** taken along line 40-40 in FIG. 36 and in a weight sliding configuration. The projection **1720** has a distal end **1748** (relative to the inside surface **1742** of the cover **1706**) that is formed as a tapered nose so that the cover **1706** can be press fit into the body **106** of the golf club head and extend all the way into the interior cavity **122** and through the chamber **1722**. For example, the flexible arms **1746** can radially deflect so as to extend through the chamber **1722** and snap into place. This connection allows the cover **1706** to be secured to the body **106** and completely removed as required or desired. The cover **1706**,

however, does not need to be removed to adjust the weight **1704**. The distal end **1748** includes at least one stop **1750** that extends radially outward on the projection **1720**. In the example, the stop **1750** is formed as part of the tapered nose. The projection **1720** also has a proximal end **1752** (relative to the inside surface **1742** of the cover **1706**) that is formed as a substantially cylindrical post. The proximal end **1752** frictionally engages with the chamber **1722** when the cover **1706** is in the locked configuration. This engagement facilitates the cover **1706** securing the weight **1704** within the recessed channel **1702** (in addition to the fastener). Additionally, at least one rib **1754** extends radially on the projection **1720**. The rib **1754** is positioned between the distal end **1748** and the proximal end **1752**, and offset a distance D.sub.1 from the stop **1750**.

(197) To move the weight assembly **1700** from the locked configuration (shown in FIG. **36**), when the cover **1706** secures the weight **1704** within the recessed channel **1702**, to the weight moving configuration (shown in FIG. **40**) that enables the weight **1704** to slide within the recessed channel **1702**, the threaded fastener **1708** is rotated so that the cover **1706** raises out of the recessed channel **1702**. When the rib **1754** engages with an end wall **1756** of the chamber **1722**, further movement of the cover **1706** is restricted and the cover **1706** is raised to a height H.sub.2 relative to the body **106** of the club head. As such, the cover **1706** indicates that the weight assembly **1700** is in the weight moving configuration. To move the weight assembly **1700** past the weight moving configuration to the weight removal position, additional force can be induced into the weight assembly **1700** (e.g., via rotation of the fastener **1708**) to overcome the engagement between the rib **1754** and the chamber **1722** until the flexible arm **1746** flexes and the cover **1706** can further raise out of the recessed channel **1702**.

(198) FIG. **41** is a cross-sectional view of the weight assembly **1700** taken along line **40-40** in FIG. **36** and in a weight removal configuration. Once the engagement of the rib **1754** and the end wall **1756** is forcefully overcome (e.g., via rotation of the fastener **1708** driving movement of the cover **1706**), the weight assembly **1700** can move from the weight moving configuration (shown in FIG. **40**) to the weight removal configuration. In the weight removal configuration, the weight **1704** can be completely removed from the recessed channel **1702** because the cover **1706** is raised even further out of the recessed channel **1702**. When the stop **1750** engages with the end wall **1756** of the chamber **1722**, further movement of the cover **1706** is restricted and the cover **1706** is raised to a height H.sub.3. The height H.sub.3 is greater than the height H.sub.2 of the prior weight sliding configuration (shown in FIG. **40**). As such, the cover **1706** indicates that the weight assembly **1700** is in the weight removal configuration and the weight **1704** can be completely removed. In some examples, the weight assembly **1700** can be moved past the weight removal configuration and allow the cover **1706** to be completely removed. If this is the case, additional force is induced into the weight assembly **1700** to overcome the engagement between the stop **1750** and the chamber **1722** until the flexible arm **1746** flexes and the cover **1706** can be completely removed. In the example, the stop **1750** is radially larger than the rib **1754**, so that the force required to completely remove the cover **1706** is greater than the force required to move between the weight moving configuration and the weight removal configuration.

(199) Because the rib **1754** at least partially defines the weight moving configuration and the stop **1750** at least partially defines the weight removal configuration, the distance D.sub.1 (shown in FIG. **40**) between the rib **1754** and the stop **1750** defines the height that the cover **1706** rises between the two different configurations H.sub.2 and H.sub.3. In an aspect, the distance D.sub.1 may be about five millimeters. Additionally, in an example, the distance D.sub.1 may be at least equal to the height H.sub.1 of the partial wall **1726** (shown in FIGS. **37** and **38**), so that in the weight removal configuration, the weight **1704** can be lifted off of the partial wall **1726** and removed from the weight assembly **1700**. In other examples, either the rib **1754** or the stop **1750** may be completely removed from the cover **1706** so that the cover **1706** moves between only two configurations as required or desired.

(200) FIG. **42** is a perspective view of a sole **1802** of another golf club head **1800** with another

weight assembly **1804** in a locked configuration. The golf club head **1800** is a fairway-metal type golf club head having a body **1806** that includes a striking face **1808** with a lower edge **1810** and an upper edge **1812** (shown in FIG. 43), each extending between a toe **1814** and a heel **1816**. The sole **1802** extends from the lower edge **1810** on the bottom side of the club head **1800** and a crown **1818** extends from the upper edge **1812** on the top of the club head **1800**. The sole **1802**, the striking face **1808**, and the crown **1818** are coupled together so as to define an outer surface **1820** of the body **1806** with an interior cavity **1822** (shown in FIG. 44) formed within. A hosel **1824** is disposed at the heel **1816** and is configured to couple to a shaft (not shown). The functions of the components (e.g., sole, striking face, crown, hosel, etc.) of the fairway-metal type club head **1800** are similar to the component functions described above in the metalwood-type golf club head examples of FIGS. 1-41. However, fairway-metal type golf club heads **1800** may strike golf balls directly off the ground surface, thereby requiring or desiring a substantially smooth outer surface **1820** of the sole **1802** without any protruding portions. As illustrated in FIG. 42, the club head **1800** is a fairway-metal type club head, however, the body **1806** may form any type club head, such as an iron-type club head, hybrid-type club head, or metalwood-type club head (e.g., examples illustrated in FIGS. 1-41), as required or desired. Furthermore, the features of the weight assembly **1804** described below can additionally or alternatively be utilized in any type club head described herein as required or desired.

(201) In this example, a recessed channel **1826** is defined within the sole **1802** of the body **1806** of the club head **1800**. The channel **1826** extends in the toe **1814**-heel **1816** direction so that the CG and the MOI of the club head **1800** can be adjusted for fade-draw bias (e.g., the “F” and “D” indicia on a cover **1830** of the weight assembly **1804**). The weight assembly **1804** includes a slidable weight **1828** disposed at least partially within the channel **1826**, a cover **1830** that extends at least partially over the channel **1826**, and a fastener **1832** configured to couple the cover **1830** to the body **1806**. The fastener **1832** retains the weight **1828** in the recessed channel **1826** indirectly via the cover **1830** so that the weight **1828** can be used to adjust the CG and the MOI of the club head **1800**. In this example, the weight assembly **1804** and the recessed channel **1826** are located at a frontal section of the golf club head **1800**. By “frontal section,” it is meant that the weight **1828** is closer to the striking face **1808** than the rearmost outer perimeter of the body **1806**, where the sole **1802** and the crown **1818** are coupled together farthest from the striking face **1808**.

(202) As illustrated in FIG. 42, the weight assembly **1804** is in a locked configuration with the cover **1830** at least partially disposed within the recessed channel **1826**, and the weight **1828** secured within the channel **1826** and movement is restricted. When the cover **1830** and weight assembly **1804** are in the locked configuration, the weight **1828** is completely disposed within the channel **1826** and no portion of the weight **1828** extends above the outer surface **1820** of the body **1806**. Additionally, the cover **1830** has an exterior surface **1834** that when the cover **1830** and weight assembly **1804** are in the locked configuration, the exterior surface **1834** of the cover **1830** aligns with the outer surface **1820** of the body **1806** and no portion of the cover **1830** extends above the outer surface **1820** of the body **1806**. Because the weight assembly **1804** is completely disposed within the recessed channel **1826** and at least aligned with, or below, the outer surface **1820** of the body **1806**, the smoothness of the outer surface **1820** of the club head **1800** is maintained so as to promote good ground interaction.

(203) FIG. 43 is a perspective view of the sole **1802** of the golf club head **1800** with the weight assembly **1804** in an unlocked configuration. Certain components are described above, and thus, are not necessarily described further. Via rotation of the fastener **1832**, the cover **1830** can be raised at least partially out of the recessed channel **1826** and into the unlocked configuration. In the unlocked configuration, the weight **1828** is selectively slidable within the channel **1826** so as to adjust the CG and the MOI as required or desired. In this example, the weight **1828** is engaged with the cover **1830** so that the weight **1828** moves with the cover **1830** between the unlocked configuration and the locked configuration and raises at least partially out of the recessed channel

1826 when in the unlocked configuration. It should be appreciated, that while FIGS. **42** and **43** illustrate and describe the weight assembly **1804** in two different configurations, a locked configuration and an unlocked configuration, the weight assembly **1804** could be moveable between more than two configurations as required or desired. For example, the weight assembly **1804** can move between at least three configurations, a locked configuration, a weight moving configuration, and a weight removal configuration, as described above in reference to FIGS. **36-41**. (204) FIG. **44** is a cross-sectional view of the golf club head **1800** with the weight assembly **1804** taken along line **44-44** in FIG. **42**. FIG. **45** is a partial perspective cross-sectional view of the weight assembly **1804** taken along line **44-44** in FIG. **42**. Referring concurrently to FIGS. **44** and **45**, certain components are described above, and thus, are not necessarily described further. The fastener **1832** is a threaded bolt that threadingly engages with a nut **1836** formed within the sole **1802** of the body **1806**. The fastener **1832** is coupled to the cover **1830** by a lock washer **1838** so that linear movement (e.g., via rotation of the fastener **1832**) is transferred to the cover **1830** and the cover **1830** can move in and out of the recessed channel **1826** as described herein.

(205) In this example, the channel **1826** is defined by a bottom track **1840** and two opposing sidewalls **1842**, **1844**. A first sidewall **1842** is adjacent the striking face **1808** and a second sidewall **1844** is adjacent to the rear of the sole **1802**. The cover **1830** is substantially L-shaped with a long leg **1846** and a short leg **1848**. The short leg **1848** includes a portion that couples to the fastener **1832** and both the short leg **1848** and the fastener **1832** are positioned adjacent the second sidewall **1844**. The short leg **1848** also includes a flange **1850**. The weight **1828** includes a groove **1852** that is sized and shaped to receive the flange **1850**. The weight **1828** is slidably engaged with the cover **1830** and with the flange **1850** received at least partially within the groove **1852**. This engagement between the cover **1830** and the weight **1828** enables the weight **1828** to move (e.g., raise out and lower back into the channel **1826**) with the cover **1830** between the locked configuration (shown in FIG. **42**) and the unlocked configuration (shown in FIG. **43**), while also enabling the weight **1828** to slide relative to the cover **1830** in the toe-heel direction when the weight assembly is in the unlocked configuration. When the cover **1830** is in the locked configuration, the long leg **1846** also substantially covers the weight **1828** so as to increase the smoothness of the outer surface **1820** of the club head **1800**.

(206) The bottom track **1840** includes a plurality of bosses **1854** extending into the channel **1826**. In this example, there are three bosses **1854**, each which corresponds respectively to a fade bias position of the weight **1828**, a draw bias position of the weight **1828**, and a center-neutral position of the weight **1828**. Additionally, the first sidewall **1842** includes a plurality of dimples **1856** that correspond to the plurality of bosses **1854**. The weight **1828** includes a hollow **1858** that is sized and shaped to receive the boss **1854** and a position indicator **1860** that is sized and shaped to be received within the dimple **1856**. In operation, when the weight assembly **1804** is in the unlocked configuration (shown in FIG. **43**), the weight **1828** is raised above the bosses **1854** so that it can be selectively moved between the bosses **1854** and the dimples **1856** of the channel **1826**. Once the weight **1828** is positioned, the weight assembly **1804** can be moved to the locked configuration (shown in FIG. **42**) and the selected boss **1854** is received at least partially within the hollow **1858** of the weight **1828**, and the position indicator **1860** is received at least partially within the selected dimple **1856**.

(207) In this example, at least a portion of the position indicator **1860** of the weight **1828** is visible on the outer surface **1820** of the club head **1800**, when the weight assembly **1804** is in the locked configuration. This allows the user to easily visually verify the position of the weight **1828** on the club head **1800**. It should be appreciated that while three bosses **1854** and dimples **1856** are illustrated and described, any other number of bosses and dimples locating features may be provided to define the position of the weight **1828** within the recessed channel **1826**. For example, five sets of bosses and dimples may be provided. Additionally, the position indicator **1860** has a cutout so that when the cover **1830** is raised to the unlocked configuration (shown in FIG. **43**), the

position indicator **1860** can slide completely out of the dimple **1856** and move above the first sidewall **1842** to adjust the position of the weight **1828**.

(208) The cover **1830** can also include one or more projections **1862** that are sized and shaped to be received within a corresponding chamber **1864** of the recessed channel **1826**. The projections **1862** are configured to increase the engagement of the cover **1830** with the body **1806** so that the weight **1828** is restricted from moving or rattling when secured within the recessed channel **1826** by the cover **1830**. In some examples, the projections **1862** may be similar to the projections described above in reference to FIGS. **36-41** and include one or more flexible arms, a tapered nose, a stop, and at least one rib.

(209) FIG. **46** is a bottom view of the golf club head **1800** with another weight assembly **1900**. FIG. **47** is a perspective cross-section view of the golf club head **1800** with the weight assembly **1900** taken along line **47-47** in FIG. **46**. Referring concurrently to FIGS. **46** and **47**, certain components are described above, and thus, are not necessarily described further. Similar to the example described in FIGS. **42-45**, the weight assembly **1900** includes a cover **1902** that selectively secures a slidable weight **1904** within a recessed channel **1906**. The weight **1904** is engaged with the cover **1902** so that the weight **1904** moves with the cover **1902** between two or more configurations. In this example, however, the cover **1902** completely covers the weight **1904** within the channel **1906**, when the cover **1902** is in a locked configuration. The cover **1902** can be formed from an at least partially transparent material so that the position of the weight **1904** is visible to the user.

(210) FIG. **48** is a perspective view of another golf club head **2000**. FIG. **49** is a bottom view of the club head **2000** with another weight assembly **2002**. Referring concurrently to FIGS. **48** and **49**, the golf club head **2000** is an iron-type golf club head that includes a striking face **2004** configured to strike a golf ball. The striking face **2004** is connected to a top line portion **2006**, a toe portion **2008**, and a heel portion **2010**. The toe portion **2008** and the heel portion **2010** are also at least in part connected to the top line portion **2006**. The heel portion **2010** is connected to a hosel **2012** that is configured to couple to a shaft (not shown). The striking face **2004** is also connected to a sole **2014**. The golf club head **2000** also includes a back portion **2016** that is attached at least partially to the sole **2014**, the top line portion **2006**, the toe portion **2008**, and the heel portion **2010**.

(211) The components of the golf club head **2000**, such as the striking face **2004**, the top line portion **2006**, the toe portion **2008**, the heel portion **2010**, and the back portion **2016** may be of a metallic material, such as a steel. The components of the golf club head **2000** may be formed through a casting process. Some of the components may be cast as a single piece and the remainder of the components may be attached subsequent to the casting process. For instance, the sole **2014**, the top line portion **2006**, the toe portion **2008**, the heel portion **2010**, and the back portion **2016** may be cast as a single piece. The striking face **2004** may then be attached to that single piece via welding or any other suitable process for attaching two club head components to one another. In such an example, the striking face **2004** may be an insert.

(212) In operation, the sole **2014** generally provides the lower surface of the club head **2000** when the club head **2000** is placed in an address position. The club head **2000** defines a center of gravity (CG) and a moment of inertia (MOI) that impact flight characteristics of the golf ball when hit with the striking face **2004**. The weight assembly **2002** is coupled to the club head **2000** such that the CG and/or the MOI of the club head **2000** can be selectively adjusted as required or desired. In this example, the weight assembly **2002** includes a movable weight **2018**, a cover **2020** configured to secure the weight **2018** in place, and a fastener **2022** for coupling the weight assembly **2002** to one or more portions of the club head **2000**. A recessed elongated channel **2024** is formed in the sole **2014** of the club head **2000** and is sized and shaped to receive at least a portion of the weight **2018**. Similar to the examples described above, the fastener **2022** is adapted to retain the weight **2018** in the channel **2024** only indirectly by the cover **2020**. Additionally, the cover **2020** can be loosened or completely removed, via the fastener **2022**, to enable the weight **2018** to slide within the channel

2024 and selectively adjust the CG and the MOI as required or desired.

(213) In this example, the fastener **2022** is positioned at the toe end of the weight assembly **2002** and aligned with the channel **2024**. In other examples, the fastener **2022** may be positioned at the heel end of the weight assembly **2002** as required or desired.

(214) FIG. **50** is a cross-section view of another weight assembly **2100**. FIG. **51** is a schematic view of the weight assembly **2100**. Referring concurrently to FIGS. **50** and **51**, a recessed channel **2102** is defined within a body **2104** of a club head (e.g., club heads **100**, **1800**, and/or **2000** described above). The weight assembly **2100** includes a slidable weight **2106** and a cover **2108**. A fastener (not shown) is used to retain the weight **2106** within the channel **2102**. In this example, the cover **2108** includes an inside surface **2110** that engages with at least a portion of the weight **2106**, when the weight **2106** is secured within the channel **2102**. In this example, at least a portion of the inside surface **2110** of the cover **2108** includes a friction material liner **2112**. The friction material **2112** is configured to frictionally engage with the weight **2106** when the cover **2108** is in a locked configuration. By frictionally engaging the weight **2106** with the cover **2108**, the weight **2106** is secured within the channel **2102** while reducing or preventing the weight **2106** from rattling therein. In the example, the friction material can be a soft metal material, such as brass.

(215) The friction material **2112** may include a plurality of grooves **2114** on the mating surface with the weight **2106**. In this example, the grooves **2114** may be triangular in shape, although, other shapes are also contemplated herein. When a clamp load **2116** is applied to the friction material **2112**, the material yields to hold the weight **2106** in place (as shown in FIG. **51**) and match the particular surface combination of the channel **2102**, weight **2106**, and cover **2108**. Once the deformation takes place and contact stress is established, the friction material **2112** will not deform further. By frictionally engaging the weight **2106** with the cover **2108**, the weight **2106** can be positioned at any location within the channel **2102** and indexing features do not need to be included. Additionally, by removing the indexing features, the weight **2106** and channel **2102** have more substantially flat surfaces, which increases manufacturing efficiencies.

(216) In this example, the cover **2108** may also include one or more protruding notches **2118** that engage with a corresponding cavities **2120** within the body **2104**. The notches **2118** may be substantially circular in shape. The notches **2118** and cavities **2120** are described further below in reference to FIGS. **52** and **53**. It should be appreciated that while the friction material **2112** is illustrated as being coupled to the cover **2108**, the friction material **2112** can additionally or alternatively be coupled to the weight **2106**.

(217) FIG. **52** is a top view of the cover **2108** of the weight assembly **2100** (shown in FIG. **50**). FIG. **53** is a side view of the cover **2108**. Referring concurrently to FIGS. **52** and **53** certain components are described above, and thus, are not necessarily described further. The cover **2108** includes a plurality of protruding notches **2118** that engage with corresponding cavities **2120** within the body **2104**. By engaging the cover **2108** at a plurality of locations, the cover **2108** is restricted or prevented from bowing out of alignment with the outer surface of the body **2104** when securing the weight. As illustrated in FIG. **53**, the side cavities may be tapered so as to accept the cover **1206** sliding in at an angle.

(218) FIG. **54** is a bottom view of another golf club head **2200** with another weight assembly **2202** in a locked configuration. The golf club head **2200** includes a body **2204** having a sole **2206**, and with the weight assembly **2202** disposed on the sole **2206**. The body **2204** also includes a striking face and a crown (both not shown), such that the body **2204** has an outer surface **2208**. In an aspect, the golf club head **2200** can be a fairway-metal type golf club head, however, the body **2204** can form any type club head, such as an iron-type club head, hybrid-type club head, or driver or other metal-wood type club head (e.g., one or more of the examples illustrated in FIGS. **1-53**).

Additionally, the functions of the components (e.g., sole, striking face, crown, hosel, etc.) of the club head **2200** are similar to the component functions described above in FIGS. **1-53**.

Furthermore, the features of the weight assembly **2202** described below can additionally or

alternatively be utilized in any type club head described herein, and as required or desired.

(219) In this example, a recessed channel **2210** is defined within the sole **2206** of the body **2204** of the club head **2200**. The recessed channel **2210** extends in a toe-heel direction so that the CG and MOI of the club head **2200** can be adjusted (e.g., for fade-draw bias). The weight assembly **2202** includes a slidable weight **2212** disposed at least partially within the channel **2210**, a cover **2214** that extends at least partially over the channel **2210**, and a fastener **2216** configured to couple the cover **2214** to the body **2204**. The fastener **2216** retains the weight **2212** in the recessed channel **2210** indirectly via the cover **2214** so that the weight **2212** can be used to adjust the CG and MOI of the club head **2200**.

(220) As illustrated in FIG. **54**, the weight assembly **2202** is in a locked configuration with the cover **2214** at least partially disposed within the recessed channel **2210** and the weight **2212** secured within the channel **2210** so as to restrict movement. When the cover **2214** and the weight assembly **2202** are in the locked configuration, at least a portion of the weight **2212** is visible between the body **2204** and the cover **2214**. This configuration enables the user to more easily determine the placement of the weight **2212** within the recessed channel **2210**. The weight assembly **2202** can also be moved into an unlocked configuration as described herein. For example, via rotation of the fastener **2216**, the cover **2214** can be raised at least partially out of the recessed channel **2210** and enable the weight **2212** to be repositioned.

(221) In this example, the weight **2212** overlaps and engages with the cover **2214** so that both move together between the locked configuration and the unlocked configuration. Furthermore, this engagement is such that the weight **2212** is reduced or prevented from twisting and tilting relative to the cover **2214** when raising and lowering with respect to the recessed channel **2210**. As such, the weight **2212** is prevented from binding within the recessed channel **2210** during weight adjustment, and thereby, increasing performance of the weight assembly **2202**.

(222) FIG. **55** is a perspective, cross-sectional, view of the weight assembly **2202** taken along line **54-54** in FIG. **54**. FIG. **56** is a cross-sectional view of the cover **2214** taken along line **54-54** in FIG. **54**. Referring concurrently to FIGS. **55** and **56**, the cover **2214** has a shelf **2218** that is configured to slidably engage with the weight **2212**. In the example, the shelf **2218** is open in a direction that faces towards the striking face of the club head and away from the fastener **2216**. Additionally, the shelf **2218** extends within the cover **2214** in a toe-heel direction. It is appreciated, however, that the shelf **2218** can be defined within the cover **2214** in any other orientation and/or direction as required or desired to achieve the adjustable weight functionality as described herein. When the cover **2214** is in the unlocked position, the weight **2212** is raised relative to the club head such that the weight **2212** is selectively slidable within the shelf **2218** and the recessed channel **2210** (shown in FIG. **54**). Conversely, when the cover **2214** is in the locked position, the weight **2212** is disposed at least partially within the recessed channel **2210** and the shelf **2218**, and secured therein, so as to restrict or prevent movement of the weight **2212**. In the example, the shelf **2218** provides an overlap for the cover **2214** with the weight **2212** so as to reduce the weight **2212** from binding within the recessed channel.

(223) The shelf **2218** includes an outer wall **2220** and an opposite inner wall **2222**. As described herein, the outer wall and inner wall of the shelf **2218** are in reference to the interior cavity of the body **2204** of the club head (shown in FIG. **54**). As such, the outer wall **2220** is disposed proximate an exterior surface **2224** of the cover **2214**. The weight **2212** is configured to be slidably received at least partially between the outer wall **2220** and the inner wall **2222** of the shelf **2218** and against an inner wall **2226** of the shelf **2218**. The three walls of the shelf **2218** retain the weight **2212** within the cover **2214** so that the position of the weight **2212** is restricted or prevented from tilting relative to the cover **2214** when being moved between the locked configuration and unlocked configuration. This configuration restricts the weight **2212** from binding within the weight assembly **2202**, and thus, increases performance of the weight assembly **2202**.

(224) In the example, this position of the weight **2212** within the cover **2214** can be measured by a

tilt angle **2228** that is defined as an angular position of the weight **2212** relative to the outer wall **2220** of the shelf **2218**. In an aspect, the tilt angle **2228** is substantially the same in both the unlocked configuration and the locked configuration. In another aspect, the tilt angle **2228** is substantially parallel to the outer wall **2220** of the shelf **2218** in both the unlocked configuration and the locked configuration. The weight **2212** has an outer surface **2230** that is positioned directly against the outer wall **2220**, the inner wall **2222**, and the inner wall **2226** of the cover **2214** when received within the shelf **2218**. As such, the outer surface **2230** of the weight **2212** maintains its position directly against the walls of the shelf **2218** in both the unlocked configuration and locked configuration.

(225) The weight **2212** includes a position indicator **2232** that extends at least partially out of the shelf **2218**. The position indicator **2232** can be used to selectively slide the weight **2212** when the weight assembly **2202** is in the unlocked configuration. When in the locked configuration, the position indicator **2232** is visible between the cover **2214** and the body of club head so that the user can easily determine the weight characteristics of the club head. Additionally, the position indicator **2232** can be disposed within dimples (e.g., the dimples **1856** shown in FIG. 45) of the recessed channel. The weight **2212** also includes a hollow **2234** that is sized and shaped to receive a boss (e.g., the boss **1854** shown in FIG. 45) of the recessed channel. In the example, the hollow **2234** is disposed adjacent the inner wall **2222** of the shelf **2218**.

(226) The cover **2214** can also include one or more projections **2236** that are sized and shaped to be received within a corresponding chamber (not shown) of the recessed channel. The projection **2236** is configured to increase the engagement of the cover **2214** with the golf club head body so that the weight **2212** is restricted from moving or rattling when secured within the recessed channel by the cover **2214**. The projection **2236** can also be used to limit the extraction of the cover **2214** from the body **2204** (shown in FIG. 54) to create a soft stop before completely unscrewing and extracting the cover **2214** from the body. In some examples, the projection **2236** may be similar to the projections described above in reference to FIGS. 36-41 and include one or more flexible arms, a tapered nose, a stop, and at least one rib.

(227) In the example, a width **2238** of the outer wall **2220** relative to the inner wall **2226** is greater than a width **2240** of the inner wall **2222**. This configuration enables the weight **2212** to be retained within the shelf **2218** without tilting and binding up within the weight assembly **2202**. Additionally, the weight **2212** includes the hollow **2234** and the position indicator **2232** that can extend out from the shelf **2218** and enable the function of the weight assembly **2202** as described herein. For example, the inner wall **2226** enables the hollow **2234** of the weight **2212** to engage with corresponding structure within the recessed channel. In an aspect, the width of the outer wall is between approximately 2 to 4 times greater than the width of the inner wall. In another aspect, a ratio of the width **2238** of the outer wall **2220** to the width **2240** of the inner wall **2222** is greater than, or equal to, 2:1. In yet another aspect, the ratio of the width **2238** of the outer wall **2220** to the width **2240** of the inner wall **2222** is greater than, or equal to, 3:1. In still another aspect, the ratio of the width **2238** of the outer wall **2220** to the width **2240** of the inner wall **2222** is between approximately 2:1 and 4:1. It should be appreciated that other ratio values are also contemplated herein and may not be expressly listed above.

(228) FIG. 57 is a perspective view of the weight **2212**. FIG. 58 is a schematic top plan view of the weight **2212**. Referring concurrently to FIGS. 57 and 58, as well as FIG. 54, at least a portion of the weight **2212** is exposed and visible between the exterior surface of the cover **2214** and the outer surface **2208** of the body **2204**, when the weight assembly **2202** in the locked configuration. That is, a gap is formed at least partially between a portion of the cover **2214** and the body **2204**, and the weight **2212** at least partially fills this gap. For example, the position indicator **2232** may be exposed and visible on the golf club head **2200**. This configuration enables the position of the weight **2212** to be easily determined. However, the weight **2212** is not entirely exposed and visible. By reducing the portions of the weight **2212** exposed on the golf club head **2200**, the smoothness

between the outer surface **1820** of the club head **2200** and the cover **2214** is increased. As such, the golf club head **2200** has increased performance (e.g., striking golf balls directly off the ground surface, aerodynamic performance, etc.), while also including the weight adjustable function via the weight assembly **2202** as described herein.

(229) In the example, between approximately 0% and 30% of the weight **2212** is exposed and visible between the outer surface **2208** of the body **2204** and the exterior surface of the cover **2214** in the locked configuration. In an aspect, between approximately 10% and 20% of the weight **2212** is exposed between the outer surface **2208** of the body **2204** and the exterior surface of the cover **2214** in the locked configuration. In yet another aspect, approximately 16% of the weight **2212** is exposed. It should be appreciated that other percentage values are also contemplated herein and may not be expressly listed above. Although not shown in the figures, the weight **2212** can be completely invisible without departing from the scope and content of the present invention.

(230) With reference to FIG. **57**, the percentage of the weight **2212** visible and exposed (e.g., portion **2242**) may be based on the outer surface area **2230** of the weight **2212**. As used herein, the outer surface **2230** of the weight **2212** includes more than one side of the weight shape and the entire outer perimeter as illustrated in FIG. **57**. For example, in an aspect, between approximately 0% and 30% of the outer surface **2230** of the weight **2212** is exposed between the outer surface **2208** of the body **2204** and the exterior surface of the cover **2214** in the locked configuration. In another aspect, between approximately 10% and 20% of the outer surface **2230** of the weight **2212** is exposed between the outer surface **2208** of the body **2204** and the exterior surface of the cover **2214** in the locked configuration. In yet another aspect, approximately 16% of the outer surface **2230** the weight **2212** is exposed. It should be appreciated that other percentage values are also contemplated herein and may not be expressly listed above.

(231) With reference to FIG. **58**, the percentage of the weight **2212** visible and exposed (e.g., portion **2244**) may be based on a planar surface area **2246** of the weight **2212**. As used herein, the planar surface area **2246** is the surface area only on one projection side of the weight shape (e.g., top planar area). While the top planar area is illustrated in FIG. **58**, other weight sides (e.g., right planar area, left planar area, etc.) are also contemplated herein. For example, in an aspect, between approximately 0% and 30% of the planar surface area **2246** of the weight **2212** is exposed between the outer surface **2208** of the body **2204** and the exterior surface of the cover **2214** in the locked configuration. In another aspect, between approximately 10% and 20% of the planar surface area **2246** of the weight **2212** is exposed between the outer surface **2208** of the body **2204** and the exterior surface of the cover **2214** in the locked configuration. In yet another aspect, approximately 16% of the planar surface area **2246** the weight **2212** is exposed. It should be appreciated that other percentage values are also contemplated herein and may not be expressly listed above.

(232) FIG. **59** is a schematic perspective view of an exemplary test mule **2300** with another weight assembly **2302**. FIG. **60** is a cross-sectional view of the weight assembly **2302** taken along line **60-60** in FIG. **59**. FIG. **61** is another cross-sectional view of the weight assembly **2302** taken along line **61-61** in FIG. **59**. Referring concurrently to FIGS. **59-61**, the test mule **2300** represents a golf club head as described herein and can be utilized to test and develop features of the weight assembly **2302** as required or desired. The club head that the test mule **2300** represents can be any type of club head described herein as required or desired, such as, but not limited to, a metalwood-type golf club head, a fairway-metal type club head, an iron-type club head, or a hybrid-type club head. In an aspect, the club head that the test mule **2300** represents is a metalwood-type club head with a striking face, a sole extending from a lower edge of the striking face, and a crown extending from an upper edge of the striking face (all three components not shown in FIGS. **59-60**). In an aspect, a transition area where the sole and the crown couple together opposite the striking face is known as a skirt of the club head, and the weight assembly **2302** is disposed at least partially on the skirt. As such, the weight assembly **2302** is disposed at a rear perimeter of the club head and proximate where the sole and the crown couple together. In an aspect, the weight assembly **2302** is

disposed substantially at a rear portion of the club head opposite the striking face. One example of a weight assembly disposed on a skirt of a golf club head is shown in FIG. 8 and described above. In the example, the test mule **2300** includes a body **2304** having an outer surface **2306** that represents the body of the club head. Additionally, a bracket **2307** is coupled to the body **2304** to facilitate testing and development, and the bracket **2307** is not representative of the club head. In an aspect, the bracket **2307** is substantially triangular in shape.

(233) In this example, a recessed channel **2308** is defined in the outer surface **2306** of the body **2304**. The channel **2308** extends along a curve in a generally toe-heel direction so that the CG and the MOI of the club head can be adjusted via the weight assembly **2302**. The weight assembly **2302** includes a weight **2310** disposed at least partially within the channel **2308** and configured to slide therein, a cover **2312** that extends at least partially over the channel **2308**, and a fastener **2314** configured to couple the cover **2312** to the body **2304**. The fastener **2314** is configured to retain the weight **2310** in the recessed channel **2308** indirectly via the cover **2312** and so that the weight **2310** can be used to adjust the CG and the MOI of the club head.

(234) Similar to the examples described above, the weight assembly **2302** is configured to move between at least a locked configuration, shown in FIGS. 59 and 61, and an unlocked configuration, shown in FIG. 60. In the locked configuration, the cover **2312** is at least partially disposed within the recessed channel **2308**, and the weight **2310** is secured within the channel **2308** with movement restricted. In the unlocked configuration, the fastener **2314** enables the cover **2312** to move along a fastener axis so that the weight **2310** can slide relative to cover **2312** and the body **2304**. In an aspect, from the locked configuration, the fastener **2314** may be rotatable between about 2-3 turns to release the cover **2312** from the locked configuration and enable the weight **2310** to slide at least partially within the recessed channel **2308**. In another aspect, the fastener **2314** may rotate about 2½ turns to enable the weight **2310** to slide within the weight assembly **2302**.

(235) The cover **2312** has a first end **2316** and an opposite second end **2318**. The fastener **2314** is coupled to the first end **2316** (e.g., via a lock washer) of the cover **2312** and so that the fastener **2314** is used for attaching the cover **2312** to the body **2304**. The second end **2318** of the cover **2312** includes a projection **2320**. The projection **2320** of the second end **2318** is configured to engage with a corresponding chamber **2322** defined at the end of the recessed channel **2308**. In the locked configuration, the fastener **2314** secures the first end **2316** of the cover **2312** to the body **2304**, while the projection **2320** of the second end **2318** engages with the chamber **2322** of the channel **2308** so that a position of the weight **2310** within the recessed channel **2308** is retained between the first end **2316** and the second end **2318** of the cover **2312**. When the weight assembly **2302** is moved towards the unlocked configuration, the fastener **2314** is used to move the cover **2312** along the fastener axis and raise the cover **2312** at least partially out of the recessed channel **2308**. This configuration enables the weight **2310** to slide and be repositioned on the body **2304** of the golf club head. In this example, the projection **2320** extends in a direction that is substantially parallel to the fastener axis so that the cover **2312** can uniformly raise out of the recessed channel **2308**. In an aspect, the chamber **2322** is formed as an undercut in a sidewall of the recessed channel **2308**, and this undercut engages with the projection **2320** of the cover **2312**. The fastener **2314** is at the opposite end of the undercut.

(236) To assist in positioning the weight **2310** at preselected positions within the recessed channel **2308**, the cover **2312** includes at least one locating lug **2324** that extends from an inner surface of the cover **2312**. The weight **2310** includes a corresponding hollow **2326** shaped and sized to receive at least a portion of the locating lug **2324**. As illustrated in FIG. 60, the cover **2312** has three spaced apart locating lugs **2324** so that the hollow **2326** can selectively engage the lug **2324** at three discrete locations within the recessed channel **2308**. In an aspect, the locating lug **2324** may not be symmetrical, for example, one side of the lug **2324** can have a steeper angled side than an opposite more shallower angle side. Furthermore, in this example, the weight **2310** has a first inclined surface **2332** on one end and a second inclined surface **2334** on the opposite end proximate the side

of the hollow **2326**. The first and second inclined surfaces **2332**, **2334** are different and configured to engage with one of the steeper or shallower angled sides of the locating lug **2324**. This configuration allows for the weight **2310** to be positioned between two locating lugs **2324** and selectively engage therewith. As such and as illustrated in FIG. **60**, the weight **2310** can engage with the cover **2312** at two more discrete locations within the recessed channel **2308** and between pairs of locating lugs **2324**. When the locating lug(s) **2324** is engaged with the weight **2310**, the weight **2310** is retained more tightly within the cover **2312** to reduce or prevent rattling and further movement of the weight **2310** in the locked configuration. Additionally, one or more of the locating lugs **2324** can define a position of the weight **2310** on the golf club head. In other aspects, the cover **2312** can have five spaced apart locating lugs **2324**. Other numbers of locating lugs **2324** are also contemplated herein. It should also be appreciated that in other examples, a locating lug **2324** may be provided for every discrete location of the weight **2310** as required or desired.

(237) In this example, the weight **2310** is slidably engaged with the body **2304** within the recessed channel **2308**. The body **2304** includes a partial wall **2328** that is disposed within the recessed channel **2308**. The partial wall **2328** extends from a bottom of the recessed channel **2308**, and the weight **2310** includes a recess **2330** shaped and sized to receive at least a portion of the partial wall **2328**. By slidably engaging the partial wall **2328** and the recess **2330** of the weight **2310**, the weight **2310** does not move with the cover **2312** when the cover is moved towards the unlocked configuration. This retention of the weight **2310** within the recessed channel **2308** enables the locating lugs **2324** of the cover **2312** to disengage with the weight **2310** and allow the weight **2310** to slide and change positions. Additionally, the orientation of the weight **2310** within the recessed channel **2308** can be held by the partial wall **2328** when the cover **2312** is in the unlocked configuration so that the weight **2310** can slide more easily to different positions.

(238) FIG. **62** is a schematic perspective view of another test mule **2400** with another weight assembly **2402**. FIGS. **63A-E** are cross-sectional views of the weight assembly **2402** taken along line **63-63** in FIG. **62** and with a weight **2410** in a variety of different positions. FIG. **64** is another cross-sectional view of the weight assembly **2402** taken along line **64-64** in FIG. **62**. Referring concurrently to FIGS. **62-64**, and similar to the example described above in FIGS. **59-61**, the test mule **2400** represents a club head that can be any type of club head described herein as required or desired, and in an aspect, the weight assembly **2402** is disposed on a skirt of the club head and at a rear perimeter where the sole and the crown couple together. The test mule **2400** includes a body **2404** having an outer surface **2406** that represents the club head and a bracket **2407**. A recessed channel **2408** is defined in the outer surface **2406** of the body **2404**. The weight assembly **2402** includes the weight **2410** disposed at least partially within the channel **2408** and configured to slide therein, a cover **2412** that extends at least partially over the channel **2408**, and a fastener **2414** configured to couple the cover **2412** to the body **2404**. The cover **2412** has a first end **2416** and an opposite second end **2418**. The fastener **2414** is coupled to the first end **2416** (e.g., via a lock washer) of the cover **2412** and so that the fastener **2414** is used for attaching the cover **2412** to the body **2404**. The second end **2418** of the cover **2412** includes a projection **2420**. The projection **2420** of the second end **2418** is configured to engage with a corresponding chamber **2422** defined at the end of the recessed channel **2408** and to secure the second end **2418** to the body **2404**.

(239) In this example, to assist in positioning the weight **2410** at preselected positions within the recessed channel **2408**, the body **2404** includes at least one locating lug **2424** disposed within the recessed channel **2408**. In an aspect, the locating lug **2424** extends from a back wall of the recessed channel **2408** relative to the outer surface **2406**. In another aspect, the locating lug **2424** is substantially symmetrical with two similarly angled sides. The weight **2410** includes a corresponding hollow **2426** shaped and sized to receive at least a portion of the locating lug **2424**. Furthermore, in this example, the weight **2410** has inclined surfaces **2432** on each end proximate the side of the hollow **2426**. The inclined surfaces **2432** are similar to each other and configured to engage with the angled sides of the locating lugs **2424**. As illustrated in FIGS. **63A-E**, the cover

2412 has three spaced apart locating lugs **2424** so that the weight **2410** is selectively positionable at five discrete locations within the recessed channel **2408**. Other numbers of locating lugs **2424** are also contemplated herein. The locating lugs **2424** are not evenly spaced apart in the heel-toe direction of the club head and have two different spacing distances. In the example, the locating lug **2424** proximate the fastener **2414** is spaced further apart from the locating lug **2424** in the middle than the locating lug **2424** proximate the projection **2420**. In an aspect, the spacing between the fastener locating lug **2424** and the middle locating lug **2424** is approximately double the spacing between the projection locating lug **2424** and the middle locating lug **2424**. Additionally, in some examples, the locating lugs **2424** can be substantially cone-shaped.

(240) Starting with FIG. 63A, the weight **2410** is disposed adjacent to the fastener **2414** and the hollow **2426** is engaged with the locating lug **2424** proximate the fastener **2414**. As such, a portion of the weight **2410** is positioned on both sides of locating lug **2424**. The weight assembly **2402** is in a locked configuration so that the position of the weight **2410** relative to the body **2404** is secured. In some aspects, the far side of the weight **2410** can be positioned directly against a portion of an end wall of the recessed channel **2408** and/or a portion of the cover **2412** that couples to the fastener **2414**. Moving next to FIG. 63B, the weight assembly **2402** can be moved to an unlocked configuration (e.g., at least partially raising the cover **2412** out of the recessed channel **2408** to enable sliding movement of the weight **2410**) for repositioning the weight **2410** and adjusting the CG and/or MOI of the club head. Once the weight **2410** is repositioned, the weight assembly **2402** can be moved into the locked configuration (as shown) to secure the position of the weight **2410**. In this position, the weight **2410** is disposed between two locating lugs **2424** such that the hollow **2426** does not have a locating lug **2424** received therein. Rather, the inclined surfaces **2432** are engaged with a respective locating lug **2424**.

(241) In FIG. 63C, the hollow **2426** of the weight **2410** is engaged with the locating lug **2424** in the middle. In this position, one of the inclined surfaces **2432** is also engaged with the locating lug **2424** proximate the projection **2420**. In FIG. 63D, the hollow **2426** of the weight **2410** is engaged with the locating lug **2424** proximate the projection **2420**. In this position, one of the inclined surfaces **2432** is engaged with the locating lug **2424** in the middle. Lastly, in FIG. 63E, the weight **2410** is disposed adjacent to the projection **2420** and between the locating lug **2424** and an end wall of the recessed channel **2408**. The hollow **2426** does not have a locating lug **2424** received therein and one of the inclined surfaces **2432** is engaged with the locating lug **2424** proximate the projection **2420**. In the example, the far side of the weight **2410** can be positioned directly against a portion of an end wall of the recessed channel **2408** and/or a portion of the cover **2412** proximate the projection **2420**. When the locating lug(s) **2424** is engaged with the weight **2410**, the weight **2410** is retained more tightly within the recessed channel **2408** to reduce or prevent rattling and further movement of the weight **2410** in the locked configuration. Additionally, one or more of the locating lugs **2424** can define a position of the weight **2410** on the golf club head. In the example, by shifting the locating lug **2424** proximate the projection **2420** inward, the weight **2410** can more easily slide between all positions without binding. It should be appreciated, that the spacing of the locating lugs **2424** can take on any other configuration as required or desired. For example, the recessed channel **2408** can have five spaced apart locating lugs **2424** such that in each position the hollow **2426** of the weight **2410** engages with a locating lug **2424**. In other examples, three similarly spaced locating lugs **2424** can be used.

(242) Additionally, the weight **2410** is slidably engaged with the cover **2412**. The cover **2412** includes a flange **2428** that extends from an interior of the cover **2412** and the weight **2410** includes a groove **2430** shaped and sized to receive at least a portion of the flange **2428**. By slidably engaging the flange **2428** and the groove **2430** of the weight **2410**, the weight **2410** is configured to move with the cover **2412** when moved towards the unlocked configuration. This movement of the weight **2410** enables the weight **2410** to disengage with the locating lugs **2424** and so that the weight **2410** can slide and change positions.

(243) FIG. 65 is a partial perspective view of an exemplary recessed channel 2500 within a body 2502 of a test mule. FIG. 66 is another partial perspective view of the recessed channel 2500. Referring concurrently to FIGS. 65 and 66, the body 2502 is illustrated schematically and the test mule represents a club head that can be any type of club head described herein as required or desired. The recessed channel 2500 is configured to receive a slidable weight 2504 and a cover (not shown) is configured to selectively retain the weight 2504 in different positions. The cover is coupled to the body 2502 with a fastener (not shown) that defines a fastener axis. The recessed channel 2500 includes a chamber 2506 that is sized and shaped to receive a corresponding projection (not shown) of the cover. The chamber 2506 is defined on the opposite end of the recessed channel 2500 from the fastener location so that both ends of the cover are engaged with the body 2502 and increase the retention of the weight 2504.

(244) In this example, the recessed channel 2500 includes a lip 2508 proximate the chamber 2506. The lip 2508 extends into the recessed channel 2500 and is configured to engage with the cover at a corresponding duct (not shown). As such, when the cover is moved towards an unlocked configuration that allows the weight 2504 to slide within the recessed channel 2500, the end of the cover opposite the fastener remains at least partially engaged with the body 2502 to reduce or prevent the end of the cover from becoming loose relative to the body 2502. The lip 2508 is elongated in a direction that is substantially parallel to the fastener axis to enable movement of the cover as described herein. The lip 2508 can be positioned at a top wall of the recessed channel 2500, as illustrated in FIG. 65, at a bottom wall of the recessed channel 2500, as illustrated in FIG. 66, or both.

(245) FIG. 67 is a schematic perspective view of another test mule 2600 with another weight assembly 2602. FIG. 68 is a cross-sectional view of the weight assembly 2602 in a first configuration taken along line 67-67 in FIG. 67. FIG. 69 is a cross-sectional view of the weight assembly 2602 in a second configuration taken along line 67-67 in FIG. 67. Referring concurrently to FIGS. 67-69, and similar to the examples described above in FIGS. 59-64, the test mule 2600 represents a club head that can be any type of club head described herein as required or desired, and in an aspect, the weight assembly 2602 is disposed on a skirt of the club head and at a rear perimeter where the sole and the crown couple together. The test mule 2600 includes a body 2604 having an outer surface 2606 that represents the club head and a bracket 2607. A recessed channel 2608 is defined in the outer surface 2606 of the body 2604. In this example, however, the weight assembly 2602 includes a cover 2610 that is configured to be selectively oriented within the recessed channel 2608 and secured to the body 2604 to at least partially define a weight position of the club head and to adjust the CG and the MOI of the club head.

(246) In this example, the weight assembly 2602 includes the cover 2610 that is removably coupled to the body 2604 and at least partially within the recessed channel 2608. The cover 2610 has a first end 2612 and an opposite second end 2614. A fastener 2616 is mounted (e.g., via a lock washer) on the first end 2612 of the cover 2610 and is configured to couple to the body 2604. Additionally, a first weight 2618 is disposed at the first end 2612 of the cover 2610. In this example, the first weight 2618 defines the first end 2612 of the cover 2610 itself and is removable from the second end 2614 of the cover 2610 so that different mass weights 2618 are interchangeable and can form the cover 2610 as required or desired. The second end 2614 of the cover 2610 includes a projection 2620 extending therefrom.

(247) Each end of the recessed channel 2608 has a chamber 2622 and a fastener receiver 2624. The chamber 2622 is configured to engage with the projection 2620 of the cover 2610 and the fastener 2616 is configured to couple to the fastener receiver 2624. By having the recessed channel 2608 symmetrical at each end, the cover 2610 can be selectively coupled to the body 2604 so that the first weight 2618 can be oriented in either the first configuration (shown in FIG. 68) or the second configuration (shown in FIG. 69). In the first configuration, the first end 2612 of the cover 2610 is disposed on the heel side of the club head so that the first weight 2618 is positioned towards the

heel side of the body **2604**. In this configuration, the fastener **2616** is secured to the fastener receiver **2624** on the heel side and the projection **2620** of the cover **2610** engages with the chamber **2622** at the toe side. This leaves the fastener receiver **2624** on the toe side and the chamber **2622** on the heel side unused by the cover **2610**. Conversely, in the second configuration, the first end **2612** of the cover **2610** is disposed on the toe side of the club head so that the first weight **2618** is positioned towards the toe side of the body **2604**. In this configuration, the fastener **2616** is secured to the fastener receiver **2624** on the toe side and the projection **2620** of the cover **2610** engages with the chamber **2622** at the heel side. This leaves the fastener receiver **2624** on the heel side and the chamber **2622** on the toe side unused by the cover **2610**. In this example, the cover **2610** has a dog-bone type shape so that its position within the recessed channel **2608** can be switched as required or desired.

(248) Additionally or alternatively, a second weight **2626** may be coupled to a corresponding weight chamber **2628** defined in the body **2604** and within the recessed channel **2608**. The weight chamber **2628** is positioned at both ends of the recessed channel **2608** proximate the fastener receiver **2624** and is covered by the cover **2610** when coupled to the body **2604**. As such, the second weight **2626** is secured by the cover **2610** within the weight chamber **2628** and indirectly retained by the fastener **2616** of the weight assembly **2602**. In an aspect, the second weight **2626** may thread at least partially into the weight chamber **2628**. It should be appreciated that the position and use of the second weight **2626** does not necessarily need to correspond to the orientation of the cover **2610** and as illustrated in FIGS. **68** and **69**. For example, the second weight **2626** can be used opposite of the first weight **2618** and retained at least partially by the second end **2614** of the cover **2610**. In another example, only the first weight **2618** and no second weight **2626** can be utilized. In still another example, a pair of second weights **2626** may be used in the pair of weight chambers **2628**. By using more than one weight **2618**, **2626** the CG and the MOI of the club head can be more finely tuned as required or desired.

(249) FIG. **70** is a schematic perspective view of another test mule **2700** with another weight assembly **2702**. FIG. **71** is a partial cross-sectional view of the weight assembly **2702** in an unlocked configuration. FIG. **72** is a partial cross-sectional view of the weight assembly **2702** in a locked configuration. Referring concurrently to FIGS. **70-72**, and similar to the examples described above in FIGS. **59-64** and **67-69**, the test mule **2700** represents a club head that can be any type of club head described herein as required or desired, and in an aspect, the weight assembly **2702** is disposed on a skirt of the club head and at a rear perimeter where the sole and the crown couple together. The test mule **2700** includes a body **2704** having an outer surface **2706** that represents the club head. A recessed channel **2708** is defined in the outer surface **2706** of the body **2704**. In this example, however, the weight assembly **2702** includes a cover **2710** that is pivotably coupled to the body **2704** and a slidable weight **2712** to adjust the CG and the MOI of the club head. In an aspect, the cover **2710** is lighter in density than the weight **2712** so that a larger amount of mass can be used to manipulate the CG and the MOI.

(250) In this example, the cover **2710** has a first end **2714** and an opposite second end **2716**. A fastener **2718** is mounted on the first end **2714** of the cover **2710** (e.g., via a lock washer) and is configured to secure the first end **2714** to the body **2704** of the club head. The second end **2716** of the cover **2710** is pivotably coupled to the body **2704**. The weight **2712** is slidably coupled to the cover **2710** and is movable between the first end **2714** and the second end **2716**. In operation, the cover **2710** is pivotable about its second end **2716** between at least a locked configuration and an unlocked configuration (shown in FIG. **70**). In the locked configuration, the fastener **2718** secures the first end **2714** of the cover **2710** to the body **2704** and the weight **2712** is disposed at least partially within the recessed channel **2708** and retained therein by the cover **2710**. The position of the weight **2712** within the recessed channel **2708** between the first end **2714** and the second end **2716** of the cover **2710** is thereby retained indirectly by the fastener **2718**. In the unlocked configuration, the first end **2714** of the cover **2710** pivots out of the recessed channel **2708** to

enable the weight **2712** to be repositioned (e.g., slide along the cover **2710**) as required or desired. The unlocked configuration is illustrated in FIG. **70** and the weight **2712** moves with the cover **2710**.

(251) The second end **2716** of the cover **2710** can include a hook **2720** that pivotably engages with a post **2722** in the body **2704**. The hook **2720** includes a hard stop **2724** that is configured to engage with the body **2704** in the unlocked position so as to define the pivot limit of the cover **2710**. The hard stop **2724** can be tapered on one end so that the second end **2716** of the cover **2710** is more easily inserted into the body **2704** during assembly. In other example, the second end **2716** of the cover **2710** can be pivotably coupled to the body **2704** with a pin connection (not shown). The recessed channel **2708** can include one or more locating lugs **2726** to assist in positioning the weight **2712** as required or desired. In aspects, the weight **2712** can be positionable in two, four, or six discrete positions at least partially defined by the locating lugs **2726**. In this example, the weight **2712** is slidably engaged with the cover **2710** and pivots therewith. In other examples, the weight can be slidably engaged with the body so that it does not pivot with the cover. This example is described below in reference to FIG. **73**.

(252) FIG. **73** is a cross-sectional view of another weight assembly **2750** that can be used with the test mule **2700** (shown in FIG. **70**). In this example, the weight assembly **2750** includes a pivotable cover **2752** and a slidable weight **2754**. However, in this example, the weight **2754** is slidably engaged at least partially within the recessed channel **2708** of the body **2704** so that the weight **2754** does not pivot with the cover **2752**. The weight assembly **2750** includes a rail **2756** that secures the weight **2754** to the body **2704** while still enabling the weight **2754** to slide. In the locked configuration as illustrated in FIG. **73**, the cover **2752** secures the position of the weight **2754** relative to the body **2704** via the rail **2756**.

(253) FIG. **74** is an exploded perspective view of another test mule **2800** with another weight assembly **2802**. Similar to the examples described above in FIGS. **59-64** and **67-72**, the test mule **2800** represents a club head that can be any type of club head described herein as required or desired, and in an aspect, the weight assembly **2802** is disposed on a skirt of the club head and at a rear perimeter where the sole and the crown couple together. The test mule **2800** includes a body **2804** having an outer surface **2806** that represents the club head and a bracket **2807**. A recessed channel **2808** is defined in the outer surface **2806** of the body **2804**. In this example, however, the weight assembly **2802** includes a cover **2810** that is coupled to the body **2804** via a fastener **2812** that is disposed proximate a center of the cover **2810**. The cover **2810** is configured to secure a slidable weight **2814** while enabling a position of the weight **2814** to be selectively adjusted. The weight **2814** has an elongated U-shape so as to accommodate the center mounted fastener **2812** and a portion of the weight **2814** can be disposed on both sides of the fastener **2812**. The cover **2810** has projections **2816** at each end to engage with the recessed channel **2808**. The recessed channel **2808** includes locating lugs **2818** to assist in positioning the weight **2814** and dimples **2820** that receive at least a portion of a position indicator **2822** of the weight **2814**.

(254) FIG. **75** is a perspective view of a sole **2902** of another golf club head **2900** with another weight assembly **2904**. FIG. **76** is a cross-sectional view of the weight assembly **2904** taken along line **76-76** in FIG. **75**. FIG. **77** is a cross-sectional view of the weight assembly **2904** taken along line **77-77** in FIG. **75**. Referring concurrently to FIGS. **75-77**, the golf club head **2900** is a metalwood-type golf club head having a body **2906** that includes a striking face **2908** with a lower edge **2910** and an upper edge (not shown) extending between a toe **2912** and a heel **2914**. The sole **2902** extends from the lower edge **2910** on the bottom side of the club head **2900** and a crown **2916** extends from the upper edge on the top of the club head **2900**. The sole **2902**, the striking face **2908**, and the crown **2916** are coupled together so as to define an outer surface **2918** of the body **2906** with an interior cavity **2920** formed within. A hosel **2922** is disposed at the heel **2914** and is configured to couple to a shaft (not shown). The functions of the components (e.g., sole, striking face, crown, hosel, etc.) of the metalwood-type club head **2900** are similar to the component

functions described above. The body **2906** may form any type club head, such as a fairway-metal type club head, an iron-type club head, or a hybrid-type club head as required or desired. Furthermore, the features of the weight assembly **2904** described below can additionally or alternatively be utilized in any type club head described herein as required or desired.

(255) In this example, a recessed channel **2924** is defined within the sole **2902** of the body **2906** of the club head **2900**. The channel **2924** extends in the toe **2912**-heel **2914** direction so that the CG and the MOI of the club head **2900** can be adjusted for fade-draw bias. In an aspect, the recessed channel **2924** may be defined in a transition area where the sole **2902** and the crown **2916** couple together opposite the striking face **2908** and known as a skirt of the club head **2900**. As such, the recessed channel **2924** and the weight assembly **2904** are disposed at a rear perimeter of the club head **2900** and proximate where the sole **2902** and the crown **2916** couple together. In an aspect, the recessed channel **2924** and the weight assembly **2904** are disposed substantially at a rear perimeter portion of the club head **2900** opposite the striking face **2908**.

(256) The weight assembly **2904** includes a slidable weight **2926** disposed at least partially within the recessed channel **2924** and configured to slide therein, a cover **2928** that extends at least partially over the channel **2924** and adapted to releasably secure the weight **2926** within the recessed channel **2924**, and a fastener **2930** configured to couple the cover **2928** to the body **2906**. The fastener **2930** retains the weight **2926** in the recessed channel **2924** indirectly via the cover **2928** and so that the weight **2926** can be used to adjust the CG and the MOI of the club head. Similar to the examples described above, the weight assembly **2904** is configured to move between at least a locked configuration, shown in FIG. **79** and described further below, and an unlocked configuration, shown in FIG. **80** and described further below. In the locked configuration, the cover **2928** is at least partially disposed within the recessed channel **2924**, and the weight **2926** is secured within the channel **2924** with its movement restricted. In the unlocked configuration, the fastener **2930** enables the cover **2928** to move along a fastener axis **2932** so that the weight **2926** is released and can slide relative to cover **2928** and the body **2906**. Additionally, in this example, the cover **2928** can also at least partially rotate relative to the recessed channel **2924** and the body **2906** towards a weight removal configuration, shown in FIG. **81** and described further below.

(257) The body **2906** includes a partial wall **2934** that is disposed within the recessed channel **2924**. The partial wall **2934** extends from a bottom of the recessed channel **2924**, and the weight **2926** includes a recess **2936** shaped and sized to receive at least a portion of the partial wall **2934**. By slidably engaging the partial wall **2934** and the recess **2936** of the weight **2926**, the weight **2926** does not move with the cover **2928** when the cover is moved towards the unlocked configuration. To assist in positioning the weight **2926** at preselected positions within the recessed channel **2924**, the cover **2928** includes at least one locating lug **2938** that extends from an inner surface of the cover **2928**. The weight **2926** includes a corresponding hollow **2940** shaped and sized to receive at least a portion of the locating lug **2938**. As such, when the weight **2926** is engaged with the cover **2928**, the weight **2926** is retained more tightly within the cover **2928** to reduce or prevent rattling and further movement of the weight **2926** in the locked configuration. In the example, the inner surface of the cover **2928** includes an oblique surface **2942** that is configured to engage with a corresponding oblique surface **2944** on the weight **2926**. The oblique surfaces **2942**, **2944** taper in a direction such that their height above the bottom wall of the recessed channel **2924** is larger and increases along a direction that is away from the partial wall **2934**. This configuration urges the weight **2926** in a direction towards the top wall of the recessed channel **2924** and induces a compression force on the weight **2926** between the cover **2928** and the recessed channel **2924** for securing the weight **2926** therein.

(258) In this example, the cover **2928** is formed from a first portion **2946** and a second portion **2948**. The fastener **2930** engages with the first portion **2946** via a lock-washer (not shown) such that the entire cover **2928** is linearly moveable along the fastener axis **2932**. The second portion **2948** has a first end **2950** that is rotatably coupled to the first portion **2946** and an opposite second

end 2952 that has a projection 2954. Similar to the other examples described herein, the projection 2954 is configured to engage with a corresponding chamber 2956 defined in the body 2906 and within the recessed channel 2924, so that when the weight assembly 2904 is in the locked configuration, the second end 2952 more tightly secures the weight 2926 within the recessed channel 2924. By enabling the second portion 2948 of the cover 2928 to rotate relative to the first portion 2946 when the second end 2952 is not engaged with the recessed channel 2924, access to the weight 2926 is increased and allows for the weight 2926 to be completely removed from the club head 2900 as required or desired and as illustrated in FIG. 81.

(259) FIG. 78 is an exploded view of the cover 2928 of the weight assembly 2904 (shown in FIGS. 75-77). The cover 2928 includes the first portion 2946 that couples to the fastener 2930 and the second portion 2948. In the example, the first portion 2946 and the second portion 2948 may be discrete and separable from one another. In other examples, the first portion 2946 and the second portion 2948 may be fixed to each other, while still being rotatable relative to one another. The first portion 2946 has a first end 2958 with a bore that is shaped and sized to receive and couple to the fastener 2930. The bore extends in a direction along the fastener axis 2932. The first portion 2946 also has an opposite second end 2960 that is configured to rotatably couple to the second portion 2948. The second end 2960 has a cylinder 2962 that is spaced away from the first end 2958 and that extends in a direction that is substantially orthogonal to the fastener axis 2932. The cylinder 2962 rotatably engages the second portion 2948 and defines a rotation axis for the second portion 2948 to rotate relative to the first portion 2946.

(260) The second portion 2948 extends between the first end 2950 and the second end 2952. The first end 2950 has a hook 2964 that rotatably engages with the cylinder 2962 of the first portion 2946 such that the rotation axis of the second portion 2948 is substantially orthogonal to the fastener axis 2932. In an aspect, an outer surface 2966 of the hook 2964 is rounded so that the second portion 2948 can rotate around the cylinder 2962. The hook 2964 is formed at least partially by an arm 2968 that is elongated and engages with a sidewall of the bore of the first portion 2946 so that rotation of the second portion 2948 is partially limited. This configuration allows the second portion 2948 to move with the first portion 2946 when the first portion 2946 linearly moves along the fastener axis 2932. In some examples, the hook 2964 can snap-fit around the cylinder 2962 so that in order to separate the two portions 2946, 2948, a separation force is required. The second end 2952 of the second portion 2948 includes the projection 2954. Extending between the first end 2950 and the second end 2952 of the second portion 2948, a cutout 2970 is formed that is sized and shaped to at least partially receive the weight 2926 (shown in FIGS. 76 and 77). Within the cutout 2970, the locating lugs 2938 and the oblique surface 2942 of the cover 2928 are formed.

(261) FIG. 79 is a perspective view of the weight assembly 2904 in a locked configuration. In the locked configuration, the fastener 2930 is tightened to the body 2906 of the club head so that the cover 2928 is engaged to the body 2906 and at least partially within the recessed channel 2924 to secure a position of the weight 2926 within the recessed channel 2924. In the locked configuration, the fastener 2930 retains the weight 2926 in the recessed channel 2924 indirectly via the cover 2928. The first end of the first portion 2946 of the cover 2928 is secured to the body 2906 by the fastener 2930. The second end of the second portion 2948 of the cover 2928 is secured to the body 2906 via the projection 2954 (shown in FIG. 78) such that rotation of the second portion 2948 relative to the first portion 2946 is prevented. Between the first portion 2946 and the second portion 2948, the hook 2964 and cylinder 2962 (shown in FIG. 78) engagement restricts the portions 2946, 2948 from separating from one another in the locked configuration. In order to release the weight 2926 from its secured position, the fastener 2930 is used to move the weight assembly 2904 towards the unlocked configuration described below in reference to FIG. 80.

(262) FIG. 80 is a perspective view of the weight assembly 2904 in an unlocked configuration. In the unlocked configuration, the fastener 2930 is loosened with respect to the body 2906 of the club head. When the fastener 2930 is loosened, the cover 2928 linearly moves along the fastener axis

2932 (shown in FIG. **78**) and at least partially raises out of the recessed channel **2924**. In the unlocked configuration, the cover **2928** releases the weight **2926** so that the weight **2926** may slide within the recessed channel **2924**. The first end of the first portion **2946** of the cover **2928** remains secured to the body **2906** by the fastener **2930** in the unlocked configuration.

(263) In some examples, the second end of the second portion **2948** of the cover **2928** can remain partially engaged to the body **2906** via the projection **2954** (shown in FIG. **78**) in the unlocked configuration so that rotation of the second portion **2948** relative to the first portion **2946** remains restricted and the weight **2926** cannot be removed from the recessed channel **2924**. However, the weight **2926** is still enabled to slide and be repositioned as required or desired. In this example, to disengage the projection **2954** from the recessed channel **2924**, the fastener **2930** is used to further raise the cover **2928** along the fastener axis **2932** so as to position the cover **2928** in a weight removal configuration as described below in reference to FIG. **81**. In other examples, in the unlocked configuration the cover **2928** is raised such that the projection **2954** is disengaged from the recessed channel **2924** without any further movement along the fastener axis **2932**. In this example, the cover **2928** is positioned such that it can be moved towards a weight removal configuration without further movement via the fastener **2930** as described below in reference to FIG. **81**.

(264) FIG. **81** is a perspective view of the weight assembly **2904** in a weight removal configuration. In the weight removal configuration, the first portion **2946** of the cover **2928** is raised at least partially out of the recessed channel **2924** such that the projection **2954** of the second portion **2948** is disengaged from the chamber **2956** defined within the recessed channel **2924**. This disengagement enables the second portion **2948** of the cover **2928** to open towards the weight removal configuration by rotating relative to the first portion **2946** and in an outwards direction relative to the body **2906** of the club head. The fastener **2930** does not need to be used to open the second portion **2948** of the cover **2928**. The rotation of the second portion **2948** is around a rotation axis that is substantially orthogonal to the fastener axis **2932** (shown in FIG. **78**). By opening the cover **2928** the weight **2926** can be completely removed from the recessed channel **2924** as required or desired. Additionally, when the second portion **2948** is rotated relative to the first portion **2946** of the cover **2928**, the hook **2964** and cylinder **2962** engagement (shown in FIG. **78**) restricts the portions **2946**, **2948** from separating from one another in the weight removal configuration.

(265) FIG. **82** is a perspective view of a sole **3002** of another golf club head **3000** with another weight assembly **3004**. FIG. **83** is a cross-sectional view of the weight assembly **3004** taken along line **83-83** in FIG. **82**. Referring concurrently to FIGS. **82** and **83**, the golf club head **3000** includes a body **3006**, a striking face **3008**, a lower edge **3010**, an upper edge (not shown), a toe **3012**, a heel **3014**, a crown **3016**, an outer surface **3018**, an interior cavity **3020**, and a hosel **3022**, the functions of which are similar to the component functions described above. The body **3006** may form any type club head as described herein, such as, a metalwood-type club head, a fairway-metal type club head, an iron-type club head, or a hybrid-type club head as required or desired. Furthermore, the features of the weight assembly **3004** described below can additionally or alternatively be utilized in any type club head described herein as required or desired.

(266) In this example, a recessed channel **3024** is defined within the sole **3002**, and/or a transition area (e.g., skirt) where the sole **3002** and the crown **3016** couple together. The channel **3024** extends in the toe **3012**-heel **3014** direction so that the CG and the MOI of the club head **3000** can be adjusted for fade-draw bias. In an aspect, the recessed channel **3024** and the weight assembly **3004** are disposed substantially at a rear perimeter portion of the club head **3000** opposite the striking face **3008**.

(267) The weight assembly **3004** includes a slidable weight **3026** disposed at least partially within the recessed channel **3024** and configured to slide therein, a cover **3028** that extends at least partially over the channel **3024** and adapted to releasably secure the weight **3026** within the

recessed channel **3024**, and a fastener **3030** configured to couple the cover **3028** to the body **3006**. Similar to the examples described above, the weight assembly **3004** is configured to move between at least a locked configuration, shown in FIG. **85** and described further below, and an unlocked configuration, shown in FIG. **86** and described further below. In the locked configuration, the cover **3028** is at least partially disposed within the recessed channel **3024**, and the weight **3026** is secured within the channel **3024** with its movement restricted. In the unlocked configuration, the fastener **3030** enables the cover **3028** to move along a fastener axis **3032** so that the weight **3026** can slide relative to cover **3028** and the body **3006**. Additionally, in this example, the cover **3028** can also at least partially rotate relative to the recessed channel **3024** and the body **3006** towards a weight removal configuration, shown in FIG. **87** and described further below. Additionally, the body **3006** includes a partial wall **3034** that slidably engages the weight **3026** and the cover **3028** includes at least one locating lug **3036** to facilitate positioning of the weight **3026**.

(268) FIG. **84** is a perspective view of the cover **3028** of the weight assembly **3004** (shown in FIGS. **82** and **83**). With continued reference to FIG. **83**, the cover **3028** has a first end **3038** and an opposite second end **3040**. The second end **3040** has a projection **3042**, which similar to the other examples herein, the projection **3042** is configured to engage with a corresponding chamber **3044** defined in the body **3006** and within the recessed channel **3024**. As such, when the weight assembly **3004** is in the locked configuration, the second end **3040** more tightly secures the weight **3026** to the body **3006**. In some examples, a duct **3046** is defined in the cover **3028** that is configured to engage with a corresponding lip (not shown) that extends from the recessed channel **3024**. This duct and lip feature facilitates the sliding engagement of the second end **3040** of the cover **3028** with the recessed channel **3024** within the body **3006** and as described further above in reference to FIGS. **65** and **66**. The first end **3038** of the cover **3028** engages with an enlarged head **3048** of the fastener **3030** and the fastener **3030** is freely rotatable relative to the cover **3028**. In this example, the fastener **3030** is not coupled to the cover **3028** with a lock-washer, and the fastener **3030** is devoid of a lock-washer.

(269) The first end **3038** of the cover **3028** has a holder **3050** defined on an inner surface of the cover **3028** that is shaped and sized to receive the enlarged head **3048** of the fastener **3030**. The holder **3050** has an inner surface **3052** that is larger than the enlarged head **3048** so that the enlarged head **3048** is freely rotatable within the holder **3050**. In the example, the inner surface **3052** has a first sidewall **3054** that is substantially parallel to the fastener axis **3032** and an opposite second sidewall **3056** that is tapered relative to the fastener axis **3032**. In an aspect, the second sidewall **3056** is oriented so as to increase the gap between the first sidewall **3054** and the second sidewall **3056** in a direction that is towards an outer surface of the cover **3028**. When the fastener **3030** is tightened to or loosened from the body **3006** of the club head, the enlarged head **3048** is positioned against the first sidewall **3054** of the inner surface **3052**, as illustrated in FIG. **83**. This configuration aligns cover **3028** along the fastener axis **3032** so that the cover **3028** can linearly move along the fastener axis **3032** and engage or disengage the projection **3042** relative to the chamber **3044**. However, when the fastener **3030** is loosened from the body **3006** (e.g., the enlarged head **3048** raised from the body **3006** along the fastener axis **3032**) and the cover **3028** is disengaged from the body **3006**, the second end **3040** of the cover **3028** can also be angled away from the body **3006** via the orientation of the second sidewall **3056**. This movement of the cover **3028** enables the cover **3028** to at least partially rotate around the fastener axis **3032** towards a weight removal configuration as illustrated in FIG. **87** and described further below.

(270) The inner surface **3052** of the holder **3050** also includes an outer axial wall **3058** that is substantially orthogonal to the fastener axis **3032**. The outer axial wall **3058** confines the enlarged head **3048** within the holder **3050** in an axial direction along the fastener axis **3032** so that when the enlarged head **3048** is loosened and raised relative to the body **3006** of the club head, corresponding linear movement is induced on the cover **3028** even without use of a lock-washer. The outer axial wall **3058** can include an aperture **3060** so that a tool (not shown) can access the

enlarged head **3048**. In an aspect, the aperture **3060** has a diameter that is less than a diameter of the enlarged head **3048**. An opposite inner axial wall **3062** is configured to at least partially hook around the enlarged head **3048** so that when the enlarged head **3048** is tightened and lowered relative to the body **3006**, corresponding linear movement is induced on the cover **3028** even without use of a lock-washer. Additionally, the holder **3050** is a protruding component of the cover **3028** (e.g., via the inner axial wall **3062**) with an outer surface **3064** that extends at least partially circumferentially around the fastener axis **3032**. As such, the outer surface **3064** is curved and at least partially cylindrical in shape. In an aspect, the outer surface **3064** is curved and extends at least 180° around the fastener axis **3032**. The outer surface **3064** facilitates rotation of the cover **3028** relative to the body **3006**, when the cover **3028** is at least partially raised out of the recessed channel **3024**.

(271) In the example, the holder **3050** is accessible from either the top or bottom of the cover **3028** and allows the fastener **3030** to be at least partially inserted into the holder **3050** (e.g., the enlarged head **3048**). When the cover **3028** is coupled to the body **3006** via the fastener **3030** and at least partially inserted within the recessed channel **3024**, the holder **3050** is at least partially inserted within the recessed channel **3024** because it is a protruding feature so that the cover **3028** is restricted or prevented from being decoupled from the enlarged head **3048** without completely withdrawing the holder **3050** from the recessed channel **3024**.

(272) FIG. **85** is a perspective view of the weight assembly **3004** in a locked configuration. In the locked configuration, the fastener **3030** is tightened to the body **3006** of the club head so that the cover **3028** is engaged to the body **3006** and at least partially within the recessed channel **3024** to secure a position of the weight **3026** within the recessed channel **3024**. In the locked configuration, the fastener **3030** retains the weight **3026** in the recessed channel **3024** indirectly via the cover **3028**. The first end of the cover **3028** is secured to the body **3006** by the fastener **3030**. The second end of the cover **3028** is secured to the body **3006** via the projection **3042** (shown in FIG. **83**) such that rotation of the cover **3028** is prevented. In order to release the weight **3026** from its secured position, the fastener **3030** via access by the aperture **3060** is used to move the weight assembly **3004** towards the unlocked configuration described below in reference to FIG. **86**.

(273) FIG. **86** is a perspective view of the weight assembly **3004** in an unlocked configuration. In the unlocked configuration, the fastener **3030** is loosened with respect to the body **3006** of the club head. When the fastener **3030** is loosened, the cover **3028** linearly moves along the fastener axis **3032** (shown in FIG. **83**) and at least partially raises out of the recessed channel **3024** by the fastener head engaging with the outer axial wall of the holder **3050**. In the unlocked configuration, the cover **3028** releases the weight **3026** so that the weight **3026** may slide within the recessed channel **3024**. The first end of the cover **3028** remains secured to the body **3006** by the fastener **3030** in the unlocked configuration.

(274) In some examples, the second end of the cover **3028** can remain partially engaged to the body **3006** via the projection **3042** in the unlocked configuration so that rotation of the cover **3028** remains restricted and the weight **3026** cannot be removed from the recessed channel **3024**.

However, the weight **3026** is still enabled to slide and be repositioned as required or desired. In this example, to disengage the projection **3042** from the recessed channel **3024**, the fastener **3030** is used to further raise the cover **3028** along the fastener axis **3032** so as to position the cover **3028** in a weight removal configuration as described below in reference to FIG. **87**. In other examples, in the unlocked configuration the cover **3028** is raised such that the projection **3042** is disengaged from the recessed channel **3024** without any further movement along the fastener axis **3032**. In this example, the cover **3028** is positioned such that it can be moved towards a weight removal configuration without further movement via the fastener **3030** as described below in reference to FIG. **87**.

(275) FIG. **87** is a perspective view of the weight assembly **3004** in a weight removal configuration. In the weight removal configuration, the first end **3038** of the cover **3028** is raised at

least partially out of the recessed channel **3024** such that the projection **3042** of the second end **3040** is disengaged from the chamber **3044** defined within the recessed channel **3024**. This disengagement enables the second end **3040** of the cover **3028** to be opened towards the weight removal configuration by angling away from the body **3006** of the club head and rotating it around the fastener axis of the fastener **3030**. For example, the fastener head is positioned against the second sidewall **3056** (shown in FIG. **83**) to angle the second end **3040** away from the body **3006** and the second end **3040** can be rotated around the fastener **3030**. By opening the cover **3028**, the weight **3026** can be completely removed from the recessed channel **3024** as required or desired. Additionally, the holder **3050** (shown in FIG. **83**) of the fastener **3030** is still at least partially projecting into the recessed channel **3024** so that the cover **3028** cannot be completely removed from the body **3006** of the club head without further movement of the fastener **3030**. This configuration enables both the cover **3028** and the fastener **3030** to remain coupled to the body **3006** in the weight removal configuration.

(276) FIG. **88** is a perspective view of a sole **3102** of another golf club head **3100** with another weight assembly **3104**. The golf club head **3100** includes a body **3106**, a striking face **3108**, a lower edge **3110**, an upper edge (not shown), a toe (not shown), a heel **3112**, a crown **3114**, an outer surface **3116**, an interior cavity (not shown), and a hosel **3118**, the functions of which are similar to the component functions described above. The body **3106** may form any type club head as described herein, such as, a metalwood-type club head, a fairway-metal type club head, an iron-type club head, or a hybrid-type club head as required or desired. Furthermore, the features of the weight assembly **3104** described below can additionally or alternatively be utilized in any type club head described herein as required or desired.

(277) In this example, a channel **3120** is defined by the body **3106** and the channel **3120** is a through-opening that extends through the body **3106** between a toe-side opening **3122** and a heel-side opening **3124**. In some examples, the channel **3120** may be separated from the interior cavity of the body **3106** by a channel wall. In other examples, the channel **3120** may be at least partially open into the interior cavity of the body **3106**. The channel **3120** may be disposed within the sole **3102**, and/or a transition area where the sole **3102** and the crown **3114** couple together (e.g., skirt). The channel **3120** extends in the toe-heel direction so that the CG and the MOI of the club head **3100** can be adjusted for fade-draw bias. In an aspect, the channel openings **3122**, **3124** are disposed substantially at a rear perimeter portion of the club head **3100** opposite the striking face **3108** and a front-rear centerline of the club head **3100**.

(278) The weight assembly **3104** includes a slidable weight **3126** slidably engaged with a cover **3128** and a fastener **3130** configured to couple the cover **3128** to the body **3106**. The cover **3128** has a first end **3132** and an opposite second end **3134**. The fastener **3130** is coupled to the first end **3132** (e.g., via a lock-washer) and the cover **3128** extends in a direction that is along the fastener axis. Both the first end **3132** and the second end **3134** of the cover **3128** have an outer surface that is shaped and sized to align with and not extend from the outer surface **3116** of the club head **3100** when secured thereto. Similar to the examples described above, the weight assembly **3104** is configured to move between at least a locked configuration (not illustrated) and an unlocked configuration (shown in FIG. **88**). In the locked configuration, the cover **3128** is inserted within the channel **3120** so that the weight **3126** is completely disposed within the channel **3120** and within the body **3106** of the club head **3100**. The position of the weight **3126** on the cover **3128** is secured within the channel **3120** when the weight assembly **3104** is in the locked configuration. As such, the fastener **3130** retains the weight **3126** in the channel **3120** indirectly via the cover **3128**. In the unlocked configuration, the fastener **3130** enables the cover **3128** to be at least partially withdrawn from the channel **3120** and along the fastener axis so that the weight **3126** is at least partially extracted from the channel **3120**. When the weight **3126** is extracted from the body **3106**, the weight **3126** can slide relative to cover **3128** for removal and/or repositioning on the cover **3128**. As such, the position of the weight **3126** on the cover **3128** is adjustable so that the weight

assembly **3104** is used to adjust the GC and MOI of the club head **3100**.

(279) In this example, an inner surface of the cover **3128** includes at least one locating lug **3136** spaced along the fastener axis. The weight **3126** includes one or more corresponding hollows **3138** shaped and sized to receive at least a portion of the locating lug **3136**. Additionally, when the locating lug **3136** is engaged with the hollow **3138**, the weight **3126** is retained more tightly by the cover **3128** to reduce or prevent rattling and further movement of the weight **3126** in the locked configuration. In an aspect, the cover **3128** can include a shelf **3140** that the weight **3126** is supported by. The shelf **3140** allows the weight **3126** to be extracted from within the channel **3120** when the weight assembly **3104** is moved into the unlocked configuration.

(280) In some examples, the first end **3132** of the cover **3128** may be engaged at least partially with the channel **3120** so that the cover **3128** is coupled to the body **3106** of the club head **3100** in the unlocked configuration. In other example, the cover **3128** can be completely removable from the body **3106** of the club head **3100** as required or desired. By securing the slidable weight **3126** within the body **3106** of the club head **3100** aerodynamic performance of the outer surface **3116** of the club head **3100** can be increased.

(281) FIG. **89** is a perspective view of a sole **3202** of another golf club head **3200** with another weight assembly **3204** in a locked configuration. The golf club head **3200** includes a body **3206**, a striking face **3208**, a lower edge **3210**, an upper edge (not shown), a toe (not shown), a heel **3212**, a crown **3214**, an outer surface **3216**, an interior cavity (not shown), and a hosel **3218**, the functions of which are similar to the component functions described above. The body **3206** may form any type club head as described herein, such as, a metalwood-type club head, a fairway-metal type club head, an iron-type club head, or a hybrid-type club head as required or desired. Furthermore, the features of the weight assembly **3204** described below can additionally or alternatively be utilized in any type club head described herein as required or desired.

(282) In this example, a channel **3220** is defined by the body **3206** for supporting the weight assembly **3204**. The channel **3220** can include a first channel **3222** and a second channel **3224** that are in communication with one another. The first channel **3222** is defined in the sole **3202** of the body **3206** and extends in a front-rear direction of the club head **3200** with the striking face **3208** being the front of the club head **3200**. The second channel **3224** is defined in the sole **3202**, and/or a transition area (e.g., skirt) where the sole **3202** and the crown **3214** couple together. The second channel **3224** extends in the toe-heel direction so that the CG and the MOI of the club head **3200** can be adjusted for fade-draw bias. In an aspect, the second channel **3224** is disposed substantially at a rear perimeter portion of the club head **3200** opposite the striking face **3208**. As such, the first channel **3222** and the second channel **3224** are oriented in substantially orthogonal directions and on different planes of the club head **3200**.

(283) The weight assembly **3204** includes a slidable weight **3226** (shown in FIG. **91**) slidably engaged with a cover **3228** and a fastener **3230** configured to couple the cover **3228** to the body **3206**. In this example, the cover **3228** includes a fastener tab **3232** and a weight tray **3234**. The fastener tab **3232** is sized and shaped to be received within the first channel **3222** and the weight tray **3234** is sized and shaped to be received within the second channel **3224**. The fastener **3230** is coupled to the fastener tab **3232** (e.g., via a lock washer) and the weight tray **3234**. The fastener **3230** is configured to cooperate with both the fastener tab **3232** and the weight tray **3234** to retain and secure the weight **3226** within the body **3206** of the club head **3200**. Similar to the examples described above, the weight assembly **3204** is configured to move between at least a locked configuration, shown in FIG. **89**, and an unlocked configuration, shown in FIG. **90** and described further below. In the locked configuration, the cover **3228** is at least partially disposed within the recessed channel **3220**, and the weight **3226** is secured within the body **3206** of the club head **3200** with its movement restricted. In the unlocked configuration, the fastener **3230** releases the cover **3228** from the body **3206** so that the weight assembly **3204** can move towards a weight adjustment configuration, shown in FIG. **91** and described further below. In the weight adjustment

configuration, the position of the weight **3226** within the weight tray **3234** can be adjusted as required or desired.

(284) In the locked configuration, both the fastener tab **3232** and the weight tray **3234** are positioned within the body **3206** of the club head **3200** so that they are not protruding and aligned with the outer surface **3216** of the club head **3200**. Additionally, the weight **3226** is completely disposed within the body **3206** of the club head **3200**, and secured therein, in the locked configuration. The fastener **3230** retains the weight **3226** in the recessed channel **3220** indirectly via the cover **3228** and so that the weight **3226** can be used to adjust the CG and the MOI of the club head **3200**.

(285) FIG. **90** is a perspective view of the weight assembly **3204** in an unlocked configuration. In the unlocked configuration, the fastener **3230** is loosened such that it at least partially raises out of the first channel **3222** along the fastener axis. Because the fastener tab **3232** is coupled to the fastener **3230**, the fastener tab **3232** also raises at least partially out of the first channel **3222**. In the example, the fastener **3230** is positioned towards a front section of the first channel **3222**. This movement of the fastener **3230** and the fastener tab **3232**, however, does not correspond to the weight tray **3234** moving within the second channel **3224**. Rather, once the weight assembly **3204** is in the unlocked configuration, the user can utilize the fastener tab **3232** to pull the weight tray **3234** at least partially out of the second channel **3224** and access the weight **3226** as illustrated in FIG. **91**. In the unlocked configuration, the weight **3226** may still be disposed completely within the body **3206** of the club head.

(286) FIG. **91** is a perspective view of the weight assembly **3204** in a weight adjustment configuration. FIG. **92** is a cross-sectional view of the weight assembly **3204** taken along line **92-92** in FIG. **91**. Referring concurrently to FIGS. **91** and **92**, the weight tray **3234** has a first end **3236** with a pool **3238** that is sized and shaped to receive the weight **3226** and allow the weight to be selectively positionable (e.g., via sliding) therein. A second end **3240** of the weight tray **3234** threadably engages with the fastener **3230** so that the fastener **3230** can rotate relative thereto. A slot **3242** is defined between the first channel **3222** and the second channel **3224** so that the fastener **3230** can extend between the two and couple to both the fastener tab **3232** and the weight tray **3234**. The slot **3242** extends in a similar front-rear direction to the first channel **3222** so that the weight assembly **3204** can be linearly movable in a rearwards direction towards the weight adjustment configuration when the weight assembly **3204** is unlocked (e.g., disengage the fastener tab **3232** from the first channel **3222**).

(287) In operation, when the weight assembly **3204** is unlocked and then moved towards the weight adjustment configuration, the first end **3236** of the weight tray **3234** projects from the body **3206** so that the weight **3226** is accessible within the pool **3238** and its position can be adjusted. In some examples, the weight **3226** can be completely removable from the weight assembly **3204** as required or desired in the weight adjustment configuration.

(288) FIG. **93** is a perspective view of a sole **3302** of another golf club head **3300** with another weight assembly **3304**. FIG. **94** is a perspective view of the weight assembly **3304**. Referring concurrently to FIGS. **93** and **94**, the golf club head **3300** includes a body **3306**, a striking face **3308**, a lower edge **3310**, an upper edge (not shown), a toe (not shown), a heel **3312**, a crown **3314**, an outer surface **3316**, an interior cavity (not shown), and a hosel **3318**, the functions of which are similar to the component functions described above. The body **3306** may form any type club head as described herein, such as, a metalwood-type club head, a fairway-metal type club head, an iron-type club head, or a hybrid-type club head as required or desired. Furthermore, the features of the weight assembly **3304** described below can additionally or alternatively be utilized in any type club head described herein as required or desired.

(289) In this example, a recessed channel **3320** is defined within the sole **3302**, and/or a transition area (e.g., skirt) where the sole **3302** and the crown **3314** couple together. The channel **3320** extends in the toe-heel direction so that the CG and the MOI of the club head **3300** can be adjusted

for fade-draw bias. In an aspect, the recessed channel **3320** and the weight assembly **3304** are disposed substantially at a rear perimeter portion of the club head **3300** opposite the striking face **3308**.

(290) The weight assembly **3304** includes an insert **3322** that is configured to be inserted into the recessed channel **3320** and coupled to the body **3306** of the club head **3300**. The insert **3322** has one or more weights coupled thereto. In this example, a first weight **3324** and a second weight **3326** are coupled to the insert **3322**. The first weight **3324** may be a different mass than the second weight **3326**. An actuator **3328** is also coupled to the insert **3322** and disposed between the weights **3324**, **3326**. In the example, the actuator **3328** is rotatable relative to the insert **3322** with an enlarged head **3330** and a leadscrew **3332**. The enlarged head **3330** is captured within the insert **3322** and the leadscrew **3332** extends in a direction away from the insert **3322**. The actuator **3328**, however, is not used to couple the weight assembly **3304** to the body **3306** of the club head **3300**. Rather, the insert **3322** includes a locking member **3334** configured to selectively engage with the body **3306** of the club head **3300** and secure the weight assembly **3304** within the recessed channel **3320**. The locking member **3334** is engaged with the leadscrew **3332** and has a pair of opposing ends **3336** that project from the insert **3322**.

(291) In operation, the weight assembly **3304** is configured to move between a locked configuration, shown in FIG. **95** and described further below, and an unlocked configuration, shown in FIG. **96** and described further below. In the locked configuration, the insert **3322** is secured within the recessed channel **3320** via the ends **3336** of the locking member **3334** so that the weights **3324**, **3326** are coupled to the golf club head **3300**. In the unlocked configuration, the ends **3336** of the locking member **3334** disengage with the body **3306** of the club head **3300** so that the insert **3322** can be removed from the recessed channel **3320**. Once the insert **3322** is removed, one or more of the weights **3324**, **3326** can be changed out and replaced to adjust the weight in the weight assembly **3304**. In other examples, the insert **3322** can be flipped around and inserted back into the recessed channel **3320** so as to adjust the position of the weights **3324**, **3326** within the golf club head **3300**.

(292) FIG. **95** is a cross-sectional view of the weight assembly **3304** taken along line **93-93** in FIG. **93** in a locked configuration. In the locked configuration, the actuator **3328** is rotated such that the locking member **3334** is positioned substantially orthogonal to the rotation axis and in a linear orientation. This position of the locking member **3334** as illustrated in FIG. **95**, results in the ends **3336** projecting from the insert **3322** and engaging with corresponding chambers **3338** defined in the body **3306** of the club head and within the recessed channel **3320**. The locking member **3334** engaging with the body **3306** couples the weight assembly **3304** to the club head and secure the position and orientation of the weights **3324**, **3326**. The actuator **3328** is rotatable so as to move the locking member **3334** towards an unlocked configuration as described below to enable the insert **3322** to be removed and decoupled from the body **3306** of the club head.

(293) FIG. **96** is a cross-sectional view of the weight assembly **3304** taken along line **93-93** in FIG. **93** in an unlocked configuration. The locking member **3334** is a flexible member such that when the actuator **3328** is rotated, the rotational movement of the leadscrew **3332** induces corresponding linear movement on the center of the locking member **3334**. As such, the locking member **3334** can curve so as to retract the ends **3336** into the insert **3322**. This retraction of the ends **3336** of the locking member **3334** disengages the ends **3336** from the chambers **3338** and allows the weight assembly **3304** to be removed from the recessed channel **3320** and the body **3306** of the club head. In the example, the ends **3336** of the locking member **3334** are stiffer relative to the middle section so that the ends **3336** are able to engage and secure to the recessed channel **3320**.

(294) The unlocked configuration allows the weights **3324**, **3326** to be replaced or for the insert **3322** to be reinserted into the recessed channel **3320** in a flipped position and adjust the GC and MOI of the club head. The insert **3322** can include a pair of stops **3340** that engage with the ends **3336** of the locking member **3334** so as to help impart the curve into the locking member **3334** in

the unlocked configuration. In this example, the weight assembly **3304** can be substantially systematical in both the toe-heel direction and sole-crown direction so that the insert **3322** can be used to reposition the weights **3324**, **3326** within the body **3306**. Additionally, the shape and size of the recessed channel **3320** enables the locking member **3334** to move as described herein.

(295) FIG. **97** is a bottom view of a sole **3402** of a golf club head **3400** with another weight assembly **3404**. FIG. **98** is a perspective cross-sectional view of the golf club head **3400** taken along line **97-97** in FIG. **97** and in an locked configuration. FIG. **99** is another perspective cross-sectional view of the golf club head **3400** taken along line **97-97** in FIG. **97** and in an unlocked configuration. Referring concurrently to FIGS. **97-99**, the golf club head **3400** is a fairway-metal type golf club head having a body **3406** that includes a striking face **3408** with a lower edge **3410** and an upper edge **3412** extending between a toe **3414** and a heel **3416**. The sole **3402** extends from the lower edge **3410** on the bottom side of the club head **3400** and a crown **3418** extends from the upper edge **3412** on the top of the club head **3400**. The sole **3402**, the striking face **3408**, and the crown **3418** are coupled together so as to define an outer surface **3420** of the body **3406** with an interior cavity **3422** formed within. A hosel **3424** is disposed at the heel **3416** and is configured to couple to a shaft (not shown). The functions of the components (e.g., sole, striking face, crown, hosel, etc.) of the fairway-metal type golf club head **3400** are similar to the component functions described above. The body **3406** may form any type club head, such as a metalwood-type club head, an iron-type club head, or a hybrid-type club head as required or desired. Furthermore, the features of the weight assembly **3404** described below can additionally or alternatively be utilized in any type club head described herein as required or desired.

(296) In this example, a recessed channel **3426** is defined within the sole **3402** of the body **3406** of the club head **3400**. The channel **3426** extends in the toe **3414**-heel **3416** direction so that that the CG and the MOI of the club head **3400** can be adjusted for fade-draw bias. The weight assembly **3404** includes a slidable weight **3428** disposed at least partially within the channel **3426** and configured to slide therein, a cover **3430** that extends at least partially over the channel **3426** and adapted to releasably secure the weight **3428** within the channel **3426**, and a fastener **3432** configured to couple the cover **3430** to the body **3406**. The fastener **3432** retains the weight **3428** in the recessed channel **3426** indirectly via the cover **3430** so that the weight **3428** can be used to adjust the CG and the MOI of the club head **3400**. In this example, the weight assembly **3404** and the recessed channel **3426** are located at a frontal section of the golf club head **3400**.

(297) Similar to the examples described above, the weight assembly **3404** is configured to move between at least two positions via the fastener **3432**, for example, a locked configuration, shown in FIG. **98**, and an unlocked configuration, shown in FIG. **99**. In the locked configuration, the cover **3430** is at least partially disposed within the recessed channel **3426** in a secured position, and the weight **3428** is secured within the channel **3426** with its movement restricted. When the weight assembly **3404** is in the locked configuration, the weight **3428** is completely disposed within the channel **3426** and no portion of the weight **3428** extends above the outer surface **3420** of the body **3406**. Because the weight **3428** is completely disposed within the recessed channel **3426** and at least aligned with, or below, the outer surface **3420** of the body **3406**, the smoothness of the outer surface **3420** of the club head **3400** is maintained so as to promote good ground interaction. In the unlocked configuration, the fastener **3432** enables the cover **3430** to move along the fastener axis towards a raised position so that the weight **3428** is released and can slide relative to the cover **3430** and the body **3406**. The fastener **3432** is coupled to cover **3430** with a retaining clip **3433**. The retaining clip **3433** is configured to couple the fastener **3432** to the cover **3430** so that the fastener **3432** can rotate around the fastener axis and relative to the cover **3430**, however, the fastener **3432** and cover **3430** are coupled together for corresponding linear movement along the fastener axis. In an aspect, the retaining clip **3433** includes a lock washer. In this example, the cover **3430** includes a retention rib **3434** that is configured to slidably engage with a slit **3435** defined in the weight **3428**. The retention rib **3434** improves retention of the weight **3428** with the cover **3430** and as the

weight assembly **3404** moves between configurations. The retention rib **3434** is described in further detail below.

(298) FIG. **100** is another bottom view of the sole **3402** of the golf club head **3400**. The weight assembly **3404** (shown in FIGS. **97-99**) is not shown for clarity. The recessed channel **3426** is defined by a bottom track **3436** and two opposing sidewalls **3438**, **3440**. A first sidewall **3438** is adjacent the striking face **3408** and a second sidewall **3440** is adjacent to the rear of the sole **3402**. A nut **3442** is formed within the sole **3402** of the body **3406** proximate the second sidewall **3440**. The nut **3442** is configured to engage the fastener **3432** so that the cover **3430** (both shown in FIGS. **97-99**) is directly coupled to the body **3406**. Additionally, the recessed channel **3426** is defined by two opposing end walls **3444**, **3446**. A toe end wall **3444** is located on the toe side of the club head **3400** and a heel end wall **3446** is located on the heel side of the club head **3400**. In this example, no chambers or openings are defined at the ends of the recessed channel **3426** and the cover **3430** does not have any corresponding projections. In an aspect, the fastener **3432** is the only component of the weight assembly **3404** that extends into the interior cavity of the body **3406**, for example, via the nut **3442**. This configuration of the club head **3400** reduces dirt and debris from accumulating within the channel **3426** and within the body **3406**.

(299) The bottom track **3436** includes a plurality of bosses **3448** projecting into the channel **3426**. In this example, there are five bosses **3448** equally spaced in the toe-heel direction. The bosses **3448** are configured to selectively engage with the weight **3428** when in the locked configuration (shown in FIG. **98**) and so as to assist with positioning and retaining the weight **3428** within the recessed channel **3426** as described herein. In the example, the bosses **3448** can have a substantially frustoconical shape. Additionally, the first sidewall **3438** includes a plurality of dimples **3450** that correspond to the plurality of bosses **3448**. The dimples **3450** are configured to selectively engage with the weight **3428** when in the locked configuration and so as to assist with positioning and retaining the weight **3428** within the recessed channel **3426** as described herein. In the example, there are five bosses **3448** and five dimples **3450** so that the weight **3428** is selectively positionable at five predefined positions. It should be appreciated that any other number of predefined positions, (e.g., three) can be utilized as required or desired.

(300) Between the first sidewall **3438** and the end walls **3444**, **3446**, the recessed channel **3426** is defined by an oblique wall **3452**. The oblique walls **3452** are formed at the terminal end of the channel **3426** in the toe-heel direction. Because the projections and chambers have been eliminated from the weight assembly **3404** when compared to prior examples, the oblique walls **3452** are configured to engage with the cover **3430** and secure the ends of the cover **3430** when in the locked configuration.

(301) FIG. **101** is an inside surface view of the cover **3430** and the weight **3428** of the weight assembly **3404** (shown in FIGS. **97-99**). FIG. **102** is a side view of the cover **3430** and the weight **3428**. Referring concurrently to FIGS. **101** and **102**, the cover **3430** has a first end **3454** and an opposite second end **3456**. A rabbet **3458** is formed on one side of the cover **3430** and extends at least partially between the first end **3454** and the second end **3456**. The rabbet **3458** is defined by at least an outer wall **3460** that forms an exterior surface of the cover **3430** and a sidewall **3462** of the cover **3430**. The rabbet **3458** is sized and shaped to at least partially receive the weight **3428** and allow the weight **3428** to slide therein. An end wall **3464** is defined at both the first end **3454** and the second end **3456** and define a terminal end of the rabbet **3458**. A fastener receiver **3466** is disposed opposite of the rabbet **3458** on the cover **3430** and is configured to support the fastener **3432** (shown in FIGS. **97-99**) at an aperture **3468**. The fastener **3432** couples the cover **3430** to the body of the club head at the fastener receiver **3466**.

(302) The retention rib **3434** projects from the outer wall **3460** within the rabbet **3458** and is elongated extended in a direction between the first end **3454** and the second end **3456** of the cover **3430**. The retention rib **3434** is configured to engage the weight **3428** so as to improve the retention of the weight **3428** to the cover **3430**. The retention rib **3434** is shaped and sized to be at least

partially received within the slit **3435** (shown in FIGS. **98-99**) of the weight **3428**. This engagement between the weight **3428** and the cover **3430** via the retention rib **3434** enables the weight **3428** to move with the cover **3430** between the locked and unlocked configuration and so that the weight **3428** can be slidably repositioned when in the unlocked configuration. In this example, the retention rib **3434** is discontinuous in the elongated direction so that the weight **3428** can be completely removed from the weight assembly as required or desired.

(303) In the example, the retention rib **3434** can include two discrete sections, a first rib **3470** and a second rib **3472** separated by a gap **3474** that is configured to allow the weight **3428** to be at least partially removably received within the rabbet **3458**. As shown in FIG. **101**, the retention rib **3434** is disposed proximate a distal end of the outer wall **3460** and on the opposite side of the cover **3430** from the fastener receiver **3466**. As such, the retention rib **3434** is spaced **3476** from the sidewall **3462** of the cover **3430**. The retention rib **3434** extends from each of the end walls **3464** and the gap **3474** is disposed proximate the second end **3456** of the cover **3430**. In another aspect, the gap **3474** can be disposed proximate the first end **3454** of the cover **3430**, or proximate a midpoint of the cover **3430**. In other aspects, the gap **3474** can be disposed at one of the end walls **3464** such that the retention rib **3434** only extends from one end wall **3464**, or more than one gap **3474** can be present within the retention rib **3434**. In still another example, two or more parallel retention ribs **3434** can be used as required or desired.

(304) The cover **3430** has an outside surface **3478** that is configured to align with the outer surface of the club head when in the locked configuration, and an opposite inside surface **3480** that faces the recessed channel **3426** (shown in FIG. **100**) of the club head. A thickness **3482** of the cover **3430** is defined between the outside surface **3478** and the inside surface **3480** in a direction that is substantially parallel to a fastener axis of the fastener **3432** (shown in FIGS. **97-99**). The thickness **3482** of the cover **3430** at the end walls **3464** is substantially equal to the thickness **3482** of the cover **3430** at a midpoint of the rabbet **3458** between the first end **3454** and the second end **3456**. As such, the end walls **3464** of the cover **3430** are devoid of any projections and engage directly with the recessed channel **3426** of the club head.

(305) At least one hollow **3484** is defined in the weight **3428** and in a surface that is opposite of the slit **3435** (shown in FIGS. **98-99**). The hollow **3484** is sized and shaped to engage with the boss **3448** disposed within the channel **3426** (both shown in FIG. **100**). In the example, the hollow **3484** includes a fully defined hollow **3484** and two partially defined hollows **3484** that flank the fully defined hollow **3484**. This configuration enables for use of a more elongated weight **3428** while accommodating an increase in the amount of weight positioning locations, e.g., five and as illustrated in the depicted example.

(306) FIG. **103** is another inside surface view of the cover **3430**. FIG. **104** is a cross-sectional view of the cover **3430** taken along line **104-104** in FIG. **103**. Referring concurrently to FIGS. **103** and **104**, certain components are described above, and thus, are not necessarily described further. The rabbet **3458** is formed by the outer wall **3460** and the sidewall **3462**. The retention rib **3434** is disposed proximate the distal end of the outer wall **3460**. A shelf **3486** extends from the distal end of the sidewall **3462** and is configured to at least partially support the weight **3428** (shown in FIGS. **101-102**). The retention rib **3434** and the shelf **3486** extend in substantially orthogonal directions. In the example, the retention rib **3434** has a substantially square-shaped cross-section. It should be appreciated that the retention rib **3434** can have any other cross-sectional shape as required or desired.

(307) In this example, the rabbet **3458** has a length **3488** that is defined between the end walls **3464** at the first and second ends of the cover **3430**. The gap **3474** of the retention rib **3434** also has a length **3490** that is defined between the two sections of the rib. In an example, the length **3488** of the rabbet **3458** is between about two to four times the length **3490** of the gap **3474**. In an aspect, the length **3488** of the rabbet **3458** is about three times the length **3490** of the gap **3474**. As shown in FIG. **103**, the retention rib **3434** has an arcuate shape in plan view that corresponds to the

elongated shape of the rabbet **3458** defined within the cover **3430**.

(308) The end walls **3464** at each end of the cover **3430** have their inner surface aligned with the inner surface of the sidewall **3462** such that the end walls **3464** directly engage with the recessed channel **3426** (shown in FIG. **100**) without the use of projections. Each end of the cover **3430** also includes a chamfer **3492** that corresponds to the oblique walls **3452** (shown in FIG. **100**) of the channel **3426** which enables the ends of the cover to securely engage with the club head in the locked configuration and reduce weight rattling therein.

(309) FIG. **105** is a partially exploded, perspective view of an iron-type golf club head **3500** with another weight assembly **3504**. In particular, the golf club head **3500** is a wedge type golf club head. FIG. **106** is another partially exploded, perspective view of the golf club head **3500** of FIG. **105**. FIG. **107** is a back view of the golf club head **3500** of FIG. **105** with the weight assembly **3504** in a locked configuration. FIG. **108** is another back view of the golf club head **3500** of FIG. **105** with the weight assembly **3504** in an unlocked configuration. FIG. **109** is another partially exploded, perspective view of the golf club head **3500** of FIG. **105**. FIG. **110** is a top view of the golf club head **3500** of FIG. **105** without the weight assembly **3504**. FIG. **111** is a cross-sectional view in the toe-to-heel direction of the golf club head **3500** of FIG. **105** taken along line **110a-110a** in FIG. **110** and with the weight assembly **3504** in the locked configuration. FIG. **112** is a cross-sectional view in the toe-to-heel direction of the golf club head **3500** of FIG. **105** taken along line **110a-110a** in FIG. **110** and with the weight assembly **3504** in the unlocked configuration. In FIG. **111** and FIG. **112**, the golf club head is shown with the loft being perpendicular to the ground plane, which is different from the setup at an address position that usually has the shaft that is vertical to the ground plane. The address position, as defined by the current application, sets up the golf club head at an orientation that has a lie angle of 60 degrees similar to the requirements of the USGA. Once the lie angle is set at 60 degrees, the face angle of the golf club head is set to be square, which is defined as having a face angle of 0 degrees.

(310) Referring concurrently to FIGS. **105-112**, the golf club head **3500** is an iron-type golf club head. The iron-type golf club head may be, for example, a wedge-type golf club head. The golf club head **3500** includes a body **3506** having a striking face **3508** with a lower leading edge **3510** and an opposite upper topline edge **3512** extending between a toe **3514** and a heel **3516**. The striking face **3508** may have multiple grooves **3519** formed therein to help impart spin on a golf ball when struck by the striking face. The body **3506** has a sole **3502** extending from the lower leading edge **3510** on the bottom side of the golf club head **3500** to a rearward portion **3522** of the sole **3502**. The rearward portion **3522** may be defined as an edge or corner on the body **3506**. A back portion **3550** is between the rearward portion **3522** of the sole **3502** and the top portion **3503**. The back portion **3550** comprises all surfaces of an outer surface **3520** of the body **3506** of the golf club head **3500** that are both between the rearward portion **3522** of the sole **3502** and the top portion **3503** and also between the toe **3514** and the heel **3516**. Specific examples of the back portion **3550** will be described in more detail below. A top portion **3503** of the body **3506** is coupled between the upper topline edge **3512** and the back portion **3550**. In some embodiments, the top portion **3503** is a thin surface. In some other embodiments, the top portion **3503** is not present, and the upper topline edge **3512** may be directly connected to the back portion **3550**. The striking face **3508**, the sole **3502**, the back portion **3550**, and the top portion **3503** are coupled together so as to define at least part of the outer surface **3520** of the body **3506**. A hosel **3501** is disposed at the heel **3516** and is configured to couple to a shaft (not shown).

(311) A recessed channel **3526** is defined in the outer surface **3520** of the body **3506**, and in particular, in the back portion **3550**. The channel **3526** extends along the toe **3514**-heel **3516** direction so that the CG and the MOI of the golf club head **3500** can be adjusted for fade-draw bias. The weight assembly **3504** is adapted to be coupled to the body **3506** at the channel **3526**. The weight assembly **3504** includes a weight **3528** disposed at least partially within the channel **3526** and configured to be movable along the toe **3514**-heel **3516** direction within the channel **3526**, a

cover **3530** that extends at least partially over the channel **3526** and is adapted to releasably secure the weight **3528** in the channel **3526**, and a fastener **3532** configured to couple the cover **3530** to the body **3506**. The fastener **3532** retains the weight **3528** in the recessed channel **3526** only indirectly via the cover **3530**. The fastener **3532** can therefore be used to secure the weight **3528** in the channel **3526** or to release the weight **3528** so that the weight **3528** can be moved at least along the toe **3514**-heel **3516** direction in the channel **3526**. As used herein, references to movement along the toe-heel direction includes along the heel-toe direction as well.

(312) The weight assembly **3504** is configured to move between at least two positions via the fastener **3532**, for example, a locked configuration as shown in FIGS. **107** and **111**, and an unlocked configuration as shown in FIGS. **108** and **112**. In some embodiments, the weight assembly **3504** may also be configured to move to a weight removable configuration in which the weight assembly **3504** may be removed entirely from channel **3526**. In other embodiments, however, the fastener **3532** may be configured to not allow the weight assembly **3504** to be removed entirely from channel **3526**, so as to prevent the weight **3528** from being accidentally lost, among other reasons. In the locked configuration, the cover **3530** is at least partially disposed within the recessed channel **3526** so that its movement is restricted. When the weight assembly **3504** is in the locked configuration, the weight **3528** is disposed in the channel **3526**, and the position of the weight **3528** is secured by the cover **3530** so that it cannot move. The locked configuration may be used when a golf club including the golf club head **3500** is being swung so that the CG and the MOI of the golf club head **3500** is set during the swing. The fastener **3532** is selectively moveable along a fastener axis **3570** (see FIGS. **111** and **112**). The fastener **3532** may be coupled to the cover **3530** with a retaining clip **3533** so that the fastener **3532** can rotate around the fastener axis **3570** and relative to the cover **3530**. Accordingly, the fastener **3532** and the cover **3530** are coupled together so that the cover **3530** moves with the fastener **3532** along the fastener axis **3570**.

(313) In other embodiments, the retaining clip **3533** may not be present so that the cover **3530** does not necessarily move with the fastener **3532** when the fastener **3532** moves. However, in such embodiments, when the fastener **3532** is partially moved to a raised position to at least partially release the cover **3530**, the cover **3530** may be moveable along the fastener axis **3570** to the extent that the fastener **3532** has been moved.

(314) The weight assembly **3504** is moved from the locked configuration to the unlocked configuration by adjusting the position of the fastener **3532** to the raised position so that the weight **3528** is released and can move inside the channel **3526** at least along the toe **3514**-heel **3516** direction and relative to the body **3506** and cover **3530**. The unlocked configuration may be used to adjust the CG and MOI of the golf club head **3500** when a golf club including the golf club head **3500** is not being swung.

(315) The weight assembly **3504** may be moved from the unlocked configuration to the weight removable configuration by further adjusting the position of the fastener **3532** to a position high enough so that the weight **3528** can be removed from the channel **3526** and/or decoupled from the cover **3530**. In some embodiments, the weight assembly **3504** may be entirely decoupled from the body **3506** in the weight removable configuration. The weight removable configuration may be used to allow the weight **3528** to be replaced. For example, a plurality of weights having different masses may be usable with the weight assembly **3504**, and the weights having different masses may be interchanged for greater control of the CG and MOI of the golf club head **3500**.

(316) In this nonexclusive example, and as shown for example in FIG. **107**, the back portion **3550** includes a lower back surface **3552** coupled between the rearward portion **3522** of the sole **3502** and the top portion **3503**, a first back surface **3554** coupled between the lower back surface **3552** and the top portion **3503**, an intermediate surface **3555** coupled between the first back surface **3554** and the top portion **3503**, and a second back surface **3556** coupled between the intermediate surface **3555** and the top portion **3503**.

(317) The lower back surface **3552** may be substantially flat in some embodiments. In some other embodiments, the lower back surface **3552** may be substantially flat except for a curve along the toe **3514**-heel **3516** direction. A height (when viewed in the orientation of FIG. **105**) of the lower back surface **3552** may be the lowest among heights of the lower back surface **3552**, the first back surface **3554**, the intermediate back surface **3555**, and the second back surface **3556**. In some embodiments, the lower back surface **3552** may not be present, and the first back surface **3554** may extend from the rearward portion **3522** of the sole **3502**.

(318) The first back surface **3554** may be substantially flat and may extend from an edge of the lower back surface **3552** distal to the rearward portion **3522** toward the intermediate surface **3555** along a direction such that at least part of a virtual extension of the first back surface **3554** intersects the striking face **3508**. The first back surface **3554** may also extend towards (e.g., may extend along a direction to be closer to) the upper topline edge **3512** of the golf head club **3500**.

(319) The intermediate back surface **3555** may be a curved or substantially flat surface extending from an edge of the first back surface **3554** distal to the lower back surface **3552** to the second back surface **3556**. In some embodiments, the intermediate back surface **3555** is not present, and the second back surface **3556** extends from the first back surface **3554**.

(320) The second back surface **3556** may be substantially flat and parallel to the striking face **3508**. The second back surface **3556** extends from an edge of the intermediate back surface **3555** distal to the first back surface **3554** to the top portion **3503**. The first and second surfaces **3554** and **3556** generally form an obtuse angle. For example, an obtuse angle may be formed where virtual extensions of the first and second surfaces **3554** and **3556** meet.

(321) Although the back portion **3550** has been described as including the lower back surface **3552**, the first back surface **3554**, the intermediate back surface **3555**, and the second back surface **3556**, and the recessed channel **3526** has been described as being formed in the first back surface **3554**, embodiments of the present disclosure are not limited thereto. The back portion **3550** may include any one or more of the lower back surface **3552**, the first back surface **3554**, the intermediate back surface **3555**, and the second back surface **3556**. Although lower back surface **3552**, the first back surface **3554**, the intermediate back surface **3555**, and the second back surface **3556** have been illustrated and described as having certain features, the lower back surface **3552**, the first back surface **3554**, the intermediate back surface **3555**, and the second back surface **3556** are not limited thereto. These features are provided for purpose of describing example embodiments, not for purpose of limitation. Furthermore, the recessed channel **3526** may be formed in any one or more surfaces included in the back portion **3550**.

(322) In this example, and as shown for example in FIG. **110**, the recessed channel **3526** is formed in the back portion **3550**, for example, in the first back surface **3554**. A fastener receiver **3572** adapted to receive and engage with the fastener **3532** is also defined in the back portion **3550**, for example, at a junction between the first and second surfaces **3554** and **3556** or at the intermediate back surface **3555**. A fastener cutout **3571** is formed around the fastener receiver **3572** and is shallower than the fastener receiver **3572**. The fastener cutout **3571** forms a hollow in the second back surface **3556**. The cover **3530** includes a positioner protrusion **3557** having a shape and size corresponding to the fastener cutout **3571** so that the positioner protrusion **3557** can engage with the fastener cutout **3571** to at least partially define the position of the cover **3530** in the locked configuration and, in some embodiments, the unlocked configuration.

(323) The channel **3526** has first and second opposing sidewalls **3538** and **3540** that extend along the toe **3514**-heel **3516** direction. The second sidewall **3540** is adjacent to the fastener receiver **3572** and the first sidewall **3538** is adjacent to the rearward portion **3522** of the sole **3502**. The channel **3526** also has toe and heel opposing end walls **3544** and **3546**, where the toe end wall **3544** is adjacent to the toe **3514** and the heel end wall **3546** is adjacent to the heel **3516**. The channel **3526** has a bottom track **3536** offset from the outer surface **3520** of the body **3506** and disposed both between the first and second sidewalls **3538** and **3540** and also between the toe and heel end

walls **3544** and **3546**.

(324) The body **3506** includes multiple locating lugs **3534** on the bottom track **3536**. The locating lugs **3534** protrude from the bottom track **3536** towards an opening of the recessed channel **3526**. As shown in FIG. **109**, the weight **3528** has a bottom indent **3562** shaped and sized to allow the weight **3528** to selectively engage with the locating lugs **3534** to thereby at least partially define the position of the weight **3528** in the recessed channel **3526**. The first sidewall **3538** has multiple dimples **3524** at a top of the first sidewall **3524** at the opening of the recessed channel **3526**. The dimples **3524** are cutouts or hollows in the first sidewall **3538** and have positions arranged along the toe **3514**-heel **3516** direction that respectively correspond to the locating lugs **3534**. The weight **3528** comprises a main body **3560** and a position indicator **3568** that protrudes from the main body **3560**. The position indicator **3568** is shaped and sized to selectively engage with the dimples **3524** to visually indicate the position of the weight **3528** inside the channel **3526**. In the locked configuration, the cover **3530** may cover and conceal the main body **3560** of the weight **3528**, but the position indicator **3568** may extend from under the cover **3568** and be positioned at least partially in one of the dimples **3524**. In the locked configuration, between 0% and 30% of an outer surface of the weight **3528** is visible.

(325) As shown in FIGS. **111** and **112**, the cover **3530** includes a flange **3558**, and the weight **3528** has a groove **3566** shaped and sized to receive at least part of the flange **3558** to slidably engage the weight **3528** with the cover **3530**. Accordingly, when the weight assembly **3504** is in the unlocked configuration, the weight **3528** is able to slide back and forth inside the recessed channel **3526**.

(326) As shown in FIGS. **111** and **112**, the striking face **3508** has an outermost surface **3518** and a plurality of grooves **3519**, each having a longitudinal axis extending along the toe **3514**-heel **3516** direction. The outermost surface **3518** may be substantially flat except for the grooves **3519**. A striking face plane **3580** is tangential to the outermost surface **3518** and extends beyond the bounds of the striking face **3508** itself. In the example toe-to-heel cross-sectional views of FIG. **111** and FIG. **112**, a Y coordinate is defined as extending up and down (generally top to bottom), a Z coordinate is defined as extending left and right (generally front to back), and an X coordinate is defined as extending into and out of the page (generally toe-to-heel). In the examples of FIG. **111** and FIG. **112**, with the club head **3500** oriented as shown, the striking face plane **3580** is within the X-Y plane, and the Z axis is orthogonal to the striking face plane (the X-Y plane, as depicted). In examples, the fastener axis **3570**, which includes a virtual extension of an axis along which the fastener **3532** moves between a locked and unlocked position, intersects the striking face plane **3580**. In some examples, the intersection of the fastener axis **3570** and the striking face plane **3580** occurs at a point closer to the lower leading edge **3510** of the striking face **3508** than to the upper topline edge **3512** of the striking face **3508**. In some embodiments, the fastener axis **3570** intersects the striking face **3508** itself. In some other embodiments, the fastener axis **3570** does not intersect the striking face **3508** but still intersects the striking face plane **3580** closer to the lower leading edge **3510** than the upper topline edge **3512**, e.g., at a point below the lower leading edge **3510**.

(327) A fastener angle **3584** of the fastener axis **3570** may be measured in a cross-sectional plane. For example, as used herein, the cross-sectional plane is a plane that is orthogonal to the longitudinal axis of at least one groove **3519** (such as the groove **3519A** nearest to the lower leading edge **3510**, or the groove **3519B** that is second-nearest to the lower leading edge **3510**). That is, the longitudinal axis of the at least one groove **3519** may be normal to the cross-sectional plane. In the examples of FIGS. **111** and **112**, the cross-sectional plane is depicted as the Y-Z plane. The fastener axis **3570** forms the fastener angle **3584** with the striking face plane **3580** in the cross-sectional plane. It is recognized that the fastener axis **3570** may not lie entirely within the cross-sectional plane (here, the Y-Z plane), and the vector of the fastener axis **3570** may include some component in the X direction (e.g., the fastener axis **3570** may tilt in a toe **3514**-heel **3516** direction); however, as used herein, the fastener angle **3584** is measured only in the cross-sectional

plane, ignoring any toe-to-heel (or “X”) component of the vector defining the fastener axis **3570**. The fastener angle **3584** is defined in the cross-sectional plane and is measured in a counter-clockwise direction from the striking face plane **3580** to the fastener axis **3570** when the cross-sectional plane is viewed in a toe-to-heel direction, as shown in FIGS. **111** and **112**. The fastener angle **3584** may range from about 0 degrees to about 90 degrees. For example, the fastener angle **3584** of the fastener axis **3570** relative to the striking face plane **3580** may be from about 5 degrees to about 85 degrees, about 10 degrees to about 80 degrees, about 15 degrees to about 75 degrees, about 20 degrees to about 70 degrees, about 25 degrees to about 65 degrees, about 30 degrees to about 60 degrees, about 35 degrees to about 55 degrees, about 40 degrees to about 50 degrees, or about 45 degrees. In further examples, the fastener angle of the fastener axis **3570** relative to the striking face plane **3580** may be non-negative and may be less than about 80 degrees, less than about 70 degrees, less than about 60 degrees, less than about 50 degrees, less than about 45 degrees, less than about 40 degrees, less than about 30 degrees, less than about 20 degrees, less than about 10 degrees, less than about 5 degrees, or about zero degrees. In other examples, the fastener angle **3584** of the fastener axis **3570** relative to the striking face plane **3580** may even be negative, depending on the configuration of the back portion **3550** of the clubhead **3500**, for example, about 0 degrees to about -5 degrees, about 0 degrees to about -10 degrees, about 0 degrees to about -20 degrees, about 0 degrees to about -30 degrees, or about 0 degrees to about -45 degrees. The foregoing recited ranges of the fastener angle **3584** are non-limiting example ranges, and the fastener angle **3584** may be within any range subsumed within the range of -45 degrees to about 135 degrees, for example, about 28 degrees to about 63 degrees or about 85 degrees to about 115 degrees. By forming the recessed channel **3526**, and disposing the weight assembly **3504**, in the back portion **3550** of the iron-type golf club head **3500** instead of in the sole **3502**, interaction between the weight assembly **3504** and the ground may be reduced when a golf club including the golf club head **3500** is swung. Accordingly, dirt and other materials may be substantially prevented from entering into the recessed channel **3526** and interfering with the weight assembly **3504**.

(328) In examples, the fastener **3532** may have a height that can be measured in the cross-sectional plane along the direction from the lower leading edge **3510** to the upper leading edge **3512**. In examples, the height of the fastener **3532** increases when the weight assembly **3504** moves from the locked configuration to the unlocked configuration.

(329) FIG. **113** is a back view of a golf club head **3500** with another weight assembly **3604** in the locked configuration. The golf club head **3500** of the embodiment illustrated in FIG. **113** may include some features similar to, or the same as, features illustrated and described above with respect to the golf club head **3500** of FIGS. **105-112**. Certain components are described above, and thus, are not necessarily described further. Referring to the example of FIG. **113**, the golf club head **3500** has a body including a toe **3514**, a heel **3516**, a top portion **3503**, a striking face (not shown), a sole **3502**, and a back portion **3550**. The back portion **3550** includes a lower back surface **3552**, a first back surface **3554**, an intermediate back surface **3555**, and a second back surface **3556**. A recessed channel (not shown) is formed in the first back surface **3554**, and the weight assembly **3604** is couplable to the body of the golf club head **3500** at the recessed channel. The weight assembly **3604** includes a weight (not shown), a cover **3630**, and a fastener **3632**.

(330) In this example, the weight assembly **3604** is similar to weight assembly **3504**, except that when the weight assembly **3604** is in the locked configuration, the cover **3630** entirely covers the weight and the recessed channel, including a bottom track (not shown) of the recessed channel. In examples, the cover **3630** may comprise one or more see-through openings **3690** to provide an indication of where the weight is located within the recessed channel. For example, the body of the golf club head **3500** may have a plurality of locating lugs (similar to the locating lugs **3534** shown in FIG. **110**) at a bottom track of the recessed channel to engage with the weight and at least partially define the position of the weight when the weight assembly **3604** is in the locked

configuration, and the cover **3630** may include a plurality of see-through openings **3690** at positions respectively corresponding to the plurality of locating lugs. In examples, the weight may be adapted to slidably engage with a flange on the cover **3730** (e.g., similar to how the weight **3528** in the embodiment shown in FIG. **106** is adapted to slidably engage with the flange **3558** of the cover **3530**), may be adapted to slidably engage with a rail on the body (e.g., a rail protruding from the bottom track or from a sidewall of the recessed channel), or may not be adapted to slidably engage with either the cover or the body. The weight assembly **3604** may include one or more of the weight assembly features described herein to enable the CG and the MOI of the golf club head **3500** to be adjustable for fade-draw bias, while securing the weight in the locked configuration indirectly by the cover **3630**. In examples, faster **3632** forms a fastener angle measured in a cross-sectional plane with respect to a striking face and/or striking face plane of the golf club head **3500** in the manner described above with respect to weight assembly **3504**.

(331) FIG. **114** is a back view of a golf club head **3500** with another weight assembly **3704**, and FIG. **115** is a top view of the golf club head **3500** of FIG. **114**. The golf club head **3500** of the embodiment illustrated in FIGS. **114** and **115** may include some features similar to, or the same as, features illustrated and described above with respect to the golf club head **3500** of FIGS. **105-112**. Certain components are described above, and thus, are not necessarily described further. Referring to the example of FIGS. **114** and **115**, the golf club head **3500** has a body including a toe **3514**, a heel **3516**, a top portion **3503**, a striking face (not shown), a sole **3502**, and a back portion **3550**. The back portion **3550** includes a lower back surface **3552**, a first back surface **3554**, an intermediate back surface **3555**, and a second back surface **3556**. A recessed channel **3526** is formed in the first back surface **3554**, and the weight assembly **3604** is couplable to the body of the golf club head **3500** at the recessed channel **3526**. The weight assembly **3704** includes a weight **3728**, a cover **3730**, and a fastener **3732**.

(332) In this example, when the weight assembly **3704** is in the locked configuration, the cover **3730** exposes at least part of the weight **3728** and at least part of the recessed channel (e.g., at least part of a bottom track of the recessed channel **3526**). For example, the cover **3730** may have a cutout portion overlapping at least part of the recessed channel **3526** while still allowing the cover **3730** to releasably secure the weight **3728** in the recessed channel **3526**. Accordingly, a position of the weight **3728** in the recessed channel **3526** when the weight assembly **3704** is in the locked configuration may be visible through the cutout portion of the cover **3730**. In examples, the weight **3728** may be adapted to slidably engage with a flange on the cover **3730** (e.g., similar to how the weight **3528** in the embodiment shown in FIG. **106** is adapted to slidably engage with the flange **3558** of the cover **3530**), may be adapted to slidably engage with a rail on the body (e.g., a rail protruding from the bottom track or from a sidewall of the recessed channel), or may not be adapted to slidably engage with either the cover **3730** or the body. The weight assembly **3704** may include one or more of the weight assembly features described herein to enable the CG and the MOI of the golf club head **3500** to be adjustable for fade-draw bias, while securing the weight **3728** in the locked configuration. In examples, the faster **3732** forms a fastener angle measured in a cross-sectional plane with respect to a striking face and/or striking face plane of the golf club head **3500** in the manner described above with respect to weight assembly **3504**.

(333) FIG. **116** is a back view of the golf club head **3500** with another weight assembly **3804** in the locked configuration. FIG. **117** is a back view of the golf club head **3500** of FIG. **116** when the weight assembly **3804** is in the unlocked configuration. FIG. **118** is a partially exploded, back view of the golf club head **3500** of FIG. **116**. FIG. **119** is a cross-sectional view in the toe-to-heel direction of the golf club head **3500** of FIG. **116** taken along line **116a-116a** when the weight assembly **3804** is in the locked configuration. FIG. **120** is a cross-sectional view in the toe-to-heel direction of the golf club head **3500** of FIG. **116** taken along line **116a-116a** when the weight assembly **3804** is in the unlocked configuration. The golf club head **3500** of the embodiment illustrated in FIGS. **116-120** may include some features similar to, or the same as, features

illustrated and described above with respect to the golf club head **3500** of FIGS. **105-112**. Certain components are described above, and thus, are not necessarily described further. Referring to the example of FIGS. **116-120**, the golf club head **3500** has a body including a toe **3514**, a heel **3516**, a top portion **3503**, a striking face **3508** (see FIG. **119**), a sole **3502**, and a back portion **3550**. The back portion **3550** includes a lower back surface **3552**, a first back surface **3554**, an intermediate back surface **3555**, and a second back surface **3556**. A recessed channel **3526** is formed in the first back surface **3554**, and the weight assembly **3804** is coupled to the body of the golf club head **3500** at the recessed channel **3526**.

(334) The weight assembly **3804** includes a weight **3828**, a cover **3830**, and a fastener **3832**. The fastener **3832** is coupled to the cover **3830** via a retaining clip **3533**. In this example, the weight includes a main body **3860** and a protruding position indicator **3868** extending from an upper portion of the main body **3860**. The upper portion of the main body **3860** refers to a portion of the main body **3860** closest to the cover **3830** and/or to the opening of the recessed channel **3526** when the weight assembly **3804** is in the locked configuration. A lower portion of the main body **3860** refers to a portion of the main body **3860** closest to a bottom of the recessed channel **3526** when the weight assembly **3804** is in the locked configuration. The recessed channel **3526** extends along a toe **3514**-heel **3516** direction and the cover **3830** has two opposing long sides extending along the toe **3514**-heel **3516** direction—a first long side **3886** distal to the striking face **3508** and a second long side **3887** proximal to the striking face **3508**. The cover **3830** also has an interior surface **3888** facing the recessed channel **3526** when the weight assembly **3804** is in the locked configuration, and an exterior surface **3889** facing away from the recessed channel **3526** when the weight assembly **3804** is in the locked configuration. In some embodiments, when the weight assembly **3804** is in the locked configuration, at least part of the exterior surface **3889** may be level with the first back surface **3554**, and at least part of the interior surface **3888** is offset from the first back surface **3554** and is inside the recessed channel **3526**.

(335) In the locked configuration, the cover **3830** covers and conceals at least part of the main body **3860**, and at least part of the protruding position indicator **3868** extends from under the cover **3830** to be exposed. Accordingly, the protruding position indicator **3868** may indicate the position of the weight **3828** in the recessed channel **3526** when the weight assembly **3804** is in the locked configuration. The protruding position indicator **3868** may be on (e.g., in contact with) part of the outer surface (e.g., the first back surface **3554**) of the body of the golf club head **3500** in the locked configuration. In this example, the weight **3828** is not adapted to slidably engage with the cover **3830**. When the cover **3830** is moved from the locked configuration to the unlocked configuration, the weight **3828** remains in the recessed channel **3526**. In the unlocked configuration, the weight **3828** can be gripped by the protruding position indicator **3868** and moved along the toe **3514**-heel **3516** direction. In some embodiments where the body includes locating lugs (similar to the locating lugs **3534** shown in FIG. **10**) at a bottom track of the recessed channel **3526**, the weight **3828** can be lifted up slightly by the protruding position indicator **3868** prior to being moved along the toe **3514**-heel **3516** direction. The cover **3830** has at least one locating groove **3859** in the interior surface **3888** shaped and sized to receive at least part of the protruding position indicator **3868** and adapted to engage with the protruding position indicator **3868** to at least partially define the position of the weight **3828**. The at least one locating groove **3859** may include a plurality (e.g., five) locating grooves, and the protruding position indicator **3868** can selectively engage with the locating grooves **3859**. The locating grooves **3859** may extend from the first side **3886** toward (e.g., at least part way to) the second side **3887**. The weight assembly **3804** may include one or more of the weight assembly features described herein to enable the CG and the MOI of the golf club head **3500** to be adjustable for fade-draw bias, while securing the weight in the lock configuration. In examples, the fastener **3832** forms a fastener angle **3584** measured in a cross-sectional plane with respect to a striking face **3508** and/or striking face plane **3580** of the golf club head **3500** in the manner described above with respect to weight assembly **3504**.

(336) FIG. 121 is a back view of a golf club head 3900 with another weight assembly 3904 in the locked configuration. In this example, the golf club head 3900 is an iron-type golf club head, but not a wedge-type golf club head. In particular, the golf club head 3900 is illustrated as a 3-iron, Titleist® (Trademark of Acushnet) golf club head, but the embodiment is not limited thereto. FIG. 122 is a back view of the golf club head 3900 of FIG. 121 when the weight assembly 3904 is in the unlocked configuration. FIG. 123 is a perspective view of the golf club head 3900 of FIG. 121 when the weight assembly 3904 is in the unlocked configuration. FIG. 124 is a partially exploded, perspective view of the golf club head 3900 of FIG. 121. FIG. 125 is a cross-sectional view in the toe-to-heel direction of the golf club head 3900 of FIG. 121 along line 121a-121a when the weight configuration 3904 is in the locked configuration. FIG. 126 is a cross-sectional view of the golf club head 3900 of FIG. 121 along line 121a-121a when the weight configuration 3904 is in the unlocked configuration. The golf club head 3900 of the embodiment illustrated in FIGS. 121-126 may include some features similar to, or the same as, features illustrated and described above with respect to the golf club head 3500 of FIGS. 105-112.

(337) Referring to FIGS. 121-126, the golf club head 3900 has a body including a toe 3914, a heel 3916, a top portion 3903, a striking face 3908, a sole 3902, and a back portion 3950. A hosel 3901 is disposed at the heel 3916 and is configured to couple to a shaft (not shown). The striking face 3908 has an upper topline edge 3912 and an opposite lower leading edge 3910. As shown in FIGS. 125 and 126, the striking face 3908 also has an outermost surface 3918 and grooves 3919 extending along the toe 3914-heel 3916 direction. The sole 3902 extends from the lower leading edge 3910 and has a rearward portion 3922 distal to the lower leading edge 3910.

(338) The back portion 3950 is positioned between the rearward portion 3922 of the sole 3902 and the top portion 3903 and includes all surfaces of an outer surface of the body of the golf club head 3900 that are both between the rearward portion 3922 of the sole 3902 and the top portion 3903 and also between the toe 3914 and the heel 3916. In this example, the back portion 3950 includes a lower back surface 3952, a first back surface 3954, an intermediate back surface 3955, and a second back surface 3956. In examples, the second back surface 3956, along with the cover 3930, acts to conceal the cavity formed in the rear of the club head 3900; however, second back surface 3956 stays stationary, while the cover 3930 moves between a locked and unlocked position.

(339) A recessed channel 3926 is formed in the first back surface 3954, and the weight assembly 3904 is couplable to the body of the golf club head 3900 at the recessed channel 3926. The recessed channel 3926 has bottom track 3936 offset from the outer surface of the body of the golf club head 3900. The recessed channel 3926 also has opposing first and second sidewalls 3938 and 3940 extending along the toe 3914-heel 3916 direction, the first sidewall 3938 being proximal or adjacent to the lower back surface 3952 and the second sidewall 3940 being distal to the lower back surface 3952. At least one dimple 3924 may be formed in the first sidewall 3938 at an upper end of the first sidewall 3938 where an opening of the recessed channel 3926 is formed in the outer surface of the body.

(340) The weight assembly 3904 includes a weight 3928, a cover 3930, and a fastener 3932. A fastener receiver 3972 is formed in the body of the golf club head 3900 and is adapted to receive at least part of the fastener 3932 and to engage with the fastener 3932. The cover 3930 includes a flange 3958, and the weight 3928 has a groove 3966 shaped and sized to receive at least part of the flange 3958 and to slidably engage the weight 3928 with the cover 3930. The weight 3928 has a main body 3960 and a protruding position indicator 3968 protruding from the main body 3960. The at least one dimple 3924 are shaped and sized to receive at least part of the protruding position indicator 3968. In the locked configuration of the weight assembly 3904, the cover 3930 may cover and conceal the main body 3960 of the weight 3928, and the protruding position indicator 3968 may at least partially protrude from under the cover 3930 and engage with a dimple of the at least one dimple 3924 to visibly indicate the position of the weight 3928 in the locked configuration. The weight assembly 3904 may include one or more of the weight assembly features described

herein to enable the CG and the MOI of the golf club head **3900** to be adjustable for fade-draw bias, while securing the weight in the lock configuration.

(341) In examples, the fastener **3932** forms a fastener angle **3984** measured in a cross-sectional plane with respect to a striking face **3908** and/or a striking face plane **3980** of the golf club head **3900** in the manner described above with respect to weight assembly **3504**.

(342) Although specific embodiments and aspects were described herein and specific examples were provided, the scope of the technology is not limited to those specific embodiments and examples. For instance, while many of the present examples have been depicted particularly for use with a driver, a fairway metal, and an iron, any the present technology may be applied to any metal wood, fairway metal or wood, iron, or hybrid golf club. Further, each of the above examples may be combined with another and/or one or more features of some examples may be combined with other examples. One skilled in the art will recognize other embodiments or improvements that are within the scope and spirit of the present technology. Therefore, the specific structure, acts, or media are disclosed only as illustrative embodiments. In addition, if the limits of the terms “about,” “substantially,” or “approximately” as used in the following claims are unclear from the foregoing specification to one having skill in the art, those terms shall mean within ten percent of the value described. The scope of the technology is defined by the following claims and any equivalents therein.

Claims

1. An iron-type golf club head comprising: a body comprising: a striking face having a lower leading edge; a sole extending from the lower leading edge; and a back portion positioned rearward of the striking face and coupled to a rearward portion of the sole; a recessed channel formed in the back portion; and a weight assembly comprising: a weight at least partially disposed within the recessed channel and configured to move therein, a cover adapted to releasably secure the weight within the recessed channel, and a fastener coupling the cover to the body and selectively movable along a fastener axis forming a fastener angle of about 0 degrees to about 90 degrees to a striking face plane, wherein the striking face plane is tangential to an outermost surface of the striking face, the fastener angle is formed in a cross-sectional plane, and the fastener angle is measured from the striking face plane to the fastener axis along a counter-clockwise direction when the cross-sectional plane is viewed in a toe-to-heel direction, and wherein the recessed channel has a bottom track offset from the outer surface of the body, and wherein the cover extends partially over the recessed channel and exposes at least part of the bottom track.

2. The iron-type golf club head of claim 1, wherein the cover is positionable in at least an unlocked configuration whereby the cover is raised at least partially out of the recessed channel and the weight is selectively movable within the recessed channel, and a locked configuration whereby the cover is at least partially disposed within the recessed channel and the weight is secured within the recessed channel.

3. The iron-type golf club head of claim 2, wherein the weight is engaged with the cover so as to move with the cover between the locked configuration and the unlocked configuration.

4. The iron-type golf club head of claim 3, wherein the cover comprises a flange, and wherein the weight has a groove shaped and sized to receive at least a portion of the flange to slidably engage the weight with the cover.

5. The iron-type golf club head of claim 2, wherein the weight remains in the recessed channel when the cover moves between the locked configuration and the unlocked configuration.

6. The iron-type golf club head of claim 2, wherein between about 0% and about 30% of an outer surface of the weight is visible in the locked configuration.

7. The iron-type golf club head of claim 1, wherein the iron-type golf club head is a wedge-type golf club head.

8. An iron-type golf club head comprising: a body comprising: a striking face having an upper topline edge and lower leading edge; a sole extending from the lower leading edge and having a rearward portion distal to the lower leading edge; and a first back surface coupled to the rearward portion and extending towards the upper topline edge, wherein one or more of the striking face, the sole, and the first back surface define at least part of an outer surface of the body; a recessed channel defined in the first back surface; and a weight assembly comprising: a weight at least partially disposed within the recessed channel, a cover extending at least partially over the recessed channel, and a fastener coupling the cover to the body, the fastener being adapted to retain the weight in the recessed channel indirectly by the cover, and wherein the cover is positionable in at least an unlocked configuration, whereby the cover is raised at least partially out of the recessed channel and the weight is selectively moveable within the recessed channel, and a locked configuration whereby the cover is at least partially disposed within the recessed channel and the weight is secured within the recessed channel, and wherein a fastener axis defined by the fastener intersects the striking face.
9. The iron-type golf club head of claim 8, wherein a fastener axis defined by the fastener intersects a striking face plane tangential to an outermost surface of the striking face at a point closer to the lower leading edge than to the upper topline edge.
10. The iron-type golf club head of claim 8, wherein the fastener has a height defined along a direction from the lower leading edge to the upper topline edge, and wherein the height of the fastener increases as the position of the cover is changed from the locked configuration to the unlocked configuration.
11. The iron-type golf club head of claim 8, wherein the fastener defines a fastener axis forming a fastener angle of about 0 degrees to about 90 degrees to a striking face plane, wherein the striking face plane is tangential to an outermost surface of the striking face, the fastener angle is formed in a cross-sectional plane, and the fastener angle is measured from the striking face plane to the fastener axis along a counter-clockwise direction when the cross-sectional plane is viewed in a toe-to-heel direction.
12. The iron-type golf club head of claim 8, wherein the body further comprises a second back surface coupled between the first back surface and the upper topline edge of the striking face, and wherein the second back surface forms an obtuse angle with the first back surface at a junction between the first back surface and the second back surface.
13. The iron-type golf club head of claim 12, wherein the body has a fastener hollow at a position between the first and second back surfaces and adapted to receive at least part of the fastener, and wherein the recessed channel is spaced apart from the fastener hollow.
14. The iron-type golf club head of claim 8, wherein the body comprises at least one locating lug on a bottom track of the recessed channel offset from the outer surface of the body, and wherein the weight comprises an indent shaped and sized to receive at least a portion of the at least one locating lug to at least partially define the position of the weight within the recessed channel.
15. The iron-type golf club head of claim 8, wherein the recessed channel has a first sidewall along a toe-heel direction and adjacent to the rearward portion of the sole, the first sidewall comprising at least one dimple, and wherein the weight comprises a protruding position indicator configured to selectively engage with the at least one dimple.
16. The iron-type golf club head of claim 8, wherein the cover comprises at least one locating groove, and wherein the weight comprises a protruding position indicator configured to selectively engaged with the at least one locating groove.
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